

Notice that D_4 will be larger than D_3 in order to provide a shoulder for the bearing. Therefore, it will be safe. Its actual diameter will be specified after we have completed the stress analysis and selected the bearing at B . The bearing manufacturer's catalog will specify the minimum acceptable diameter to the right of the bearing to provide a suitable shoulder against which to seat the bearing.

3. **Point C:** Point C is the location of gear C with a well-rounded fillet to the left, a profile keyseat at the gear, and a retaining ring groove to the right. The use of a well-rounded fillet at this point is actually a design decision that requires that the design of the bore of the gear accommodate a large fillet. Usually this means that a chamfer is produced at the ends of the bore. The bending moment at C is

$$M_C = \sqrt{M_{Cx}^2 + M_{Cy}^2} = \sqrt{(12\,230)^2 + (16\,800)^2} = 20\,780 \text{ lb} \cdot \text{in}$$

To the left of C the torque of 21 000 lb·in exists with the profile keyseat giving $K_t = 2.0$. Then

$$D_5 = \left[\frac{32(2)}{\pi} \sqrt{\left[\frac{2.0(20\,780)}{25\,500} \right]^2 + \frac{3}{4} \left[\frac{21\,000}{83\,000} \right]^2} \right]^{1/3} = 3.22 \text{ in}$$

To the right of C there is no torque, but the ring groove suggests $K_t = 3.0$ for design, and there is reversed bending. We can use Equation (12-24) with $K_t = 3.0$, $M = 20\,780 \text{ lb} \cdot \text{in}$ and $T = 0$.

$$D_5 = \left[\frac{32(2)}{\pi} \sqrt{\left(\frac{(3.0)(20\,780)}{25\,500} \right)^2} \right]^{1/3} = 3.68 \text{ in}$$

Applying the ring groove factor of 1.06 raises the diameter to 3.90 in.

This value is higher than that computed for the left of C , so it governs the design at point C .

4. **Point D:** Point D is the seat for bearing D , and there is no torque or bending moment here. However, there is a vertical shearing force equal to the reaction at the bearing. Using the resultant of the x - and y -plane reactions, the shearing force is

$$V_D = \sqrt{(1223)^2 + (1680)^2} = 2078 \text{ lb}$$

We can use Equation (12-16) to compute the required diameter for the shaft at this point:

$$D = \sqrt{2.94 K_t(V)N/s_n} \quad (12-16)$$

Referring to Figure 12-2, we see a sharp fillet near this point on the shaft because of the small fillet radius necessary to permit the firm seating of the face of the bearing race against the shaft shoulder. Then a stress concentration factor of 2.5 should be used:

$$D_6 = \sqrt{\frac{2.94(2.5)(2078)(2)}{25\,500}} = 1.094 \text{ in}$$

This is very small compared to the other computed diameters, and it will usually be so. In reality, the diameter at D will probably be made much larger than this computed value because of the size of a reasonable bearing to carry the radial load of 2078 lb.

Summary The computed minimum required diameters for the various parts of the shaft in Figure 12-2 are as follows:

$$D_1 = 1.65 \text{ in}$$

$$D_2 = 3.30 \text{ in}$$

$$D_3 = 3.55 \text{ in}$$

$$D_5 = 3.90 \text{ in}$$

$$D_6 = 1.094 \text{ in}$$

Also, D_4 must be somewhat greater than 3.90 in in order to provide adequate shoulders for gear C and bearing B .

We now specify convenient decimal-inch dimensions for the six diameters. Choose the bearing seat dimensions from Table 14-3. Choose all other dimensions from Appendix 2. Table 12-1 shows one possible set of recommended diameters.

TABLE 12-1 Recommended Diameters

Mating part	Diameter number	Minimum diameter	Specified diameter (basic size)
	(from Design Example 12-1 and Figure 12-2)		
Gear	D_1	1.65 in	1.800 in
Nothing	D_2	3.30 in	3.400 in
Bearing	D_3	3.55 in	3.7402 in (95 mm)
Nothing	D_4	$> D_3$ or D_5	4.400 in
Gear	D_5	3.90 in	4.000 in
Bearing	D_6	1.094 in	1.7717 in (45 mm)

Diameters D_3 and D_6 are the decimal equivalents of the metric diameters of the inner races of bearings from Table 14-3. The procedures in Chapter 14 would have to be used to determine whether bearings having those diameters are suitable to carry the given radial loads. Also, D_4 would have to be checked to see whether it provides a sufficiently high shoulder against which to seat the bearing mounted at point B on the shaft. Then detailed specifications for fillet radii, lengths, keyseats, and retaining ring grooves would have to be defined, along with suitable tolerances. See Chapters 13–15 for guidance. The actual values for stress concentration factors and the size factor should then be determined. Finally, the stress analysis should be repeated to ensure that the resulting design factor is acceptable. Equation (12-23) can be solved for N and evaluated for actual conditions.

Design Example 12-2

The shaft shown in Figure 12-13 receives 110 hp from a water turbine through a chain sprocket at point C . The gear pair at E delivers 80 hp to an electrical generator. The V-belt sheave at A delivers 30 hp to a bucket elevator that carries grain to an elevated hopper. The shaft rotates at 1700 rpm. The sprocket, sheave, and gear are located axially by retaining rings. The sheave and gear are keyed with sled runner keyseats, and there is a profile keyseat at the sprocket. Use SAE 1040 cold-drawn steel for the shaft. Compute the minimum acceptable diameters D_1 through D_7 as defined in Figure 12-13.

Solution

First, the material properties for the SAE 1040 cold-drawn steel are found from Appendix 3:

$$s_y = 71\,000 \text{ psi} \quad s_u = 80\,000 \text{ psi}$$

Then from Figure 5-11, $s_n = 30\,000$ psi. Let's design for a reliability of 0.99 and use $C_R = 0.81$. The shaft size should be moderately large, so we can assume $C_s = 0.85$ as a reasonable estimate. Then the modified endurance strength is

$$s'_n = s_n C_s C_R = (30\,000)(0.85)(0.81) = 20\,650 \text{ psi}$$

This application is fairly smooth: a turbine drive and a generator and a conveyor at the output points. A design factor of $N = 2$ should be satisfactory.

Torque Distribution in the Shaft: Recalling that all of the power comes into the shaft at C , we can then observe that 30 hp is delivered down the shaft from C to the sheave at A . Also, 80 hp is delivered down the shaft from C to the gear at E . From these observations, the torque in the shaft can be computed:

$$T_A = T_{AC} = (63\,000)(30 \text{ hp})/1700 \text{ rpm} = 1112 \text{ lb} \cdot \text{in} \quad \text{from } A \text{ to } C \text{ in shaft}$$

$$T_E = T_{CE} = (63\,000)(80 \text{ hp})/1700 \text{ rpm} = 2965 \text{ lb} \cdot \text{in} \quad \text{from } C \text{ to } E \text{ in shaft}$$

Figure 12-14 shows a plot of the torque distribution *in the shaft* superimposed on the sketch of the shaft. When designing the shaft at C , we will use 2965 lb·in *at C and to the right*, but we can use 1112 lb·in *to the left of C*. Notice that no part of the shaft is subjected to the full 110 hp that comes into the sprocket at C . The power splits into two parts as it enters the shaft. When analyzing the sprocket itself, we must use the full 110 hp and the corresponding torque:

$$T_C = (63\,000)(110 \text{ hp})/1700 \text{ rpm} = 4076 \text{ lb} \cdot \text{in} \quad (\text{torque on the sprocket})$$

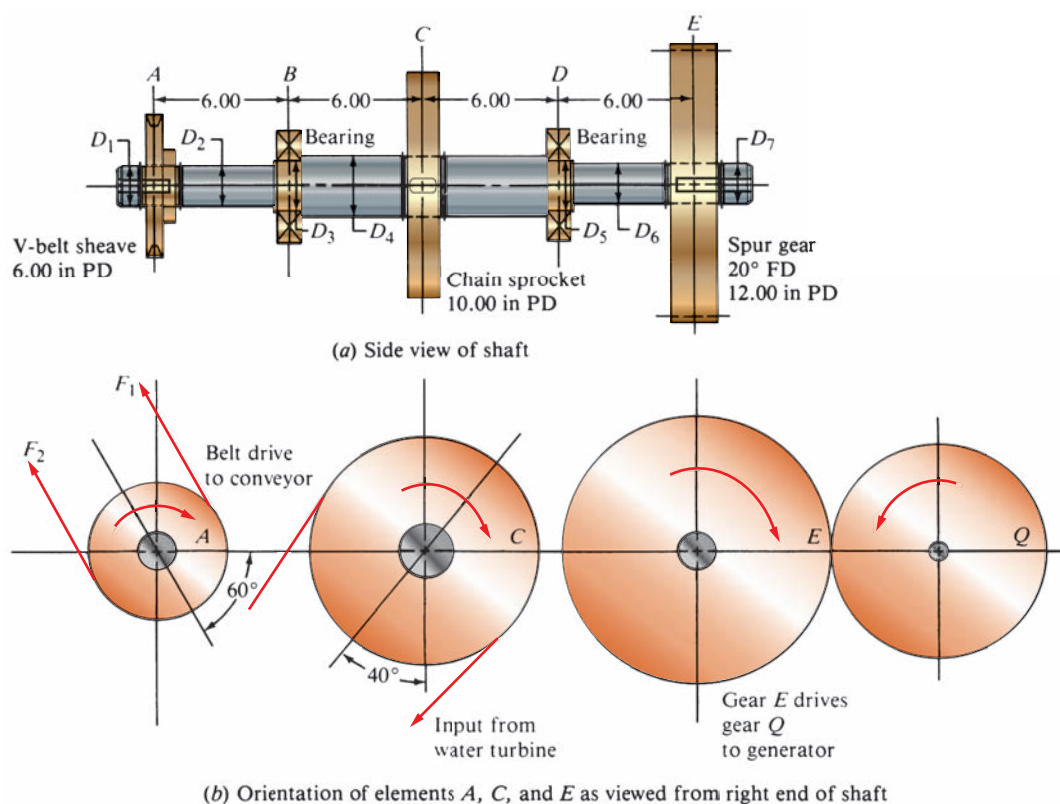


FIGURE 12-13 Shaft design for Example Problem 12-2

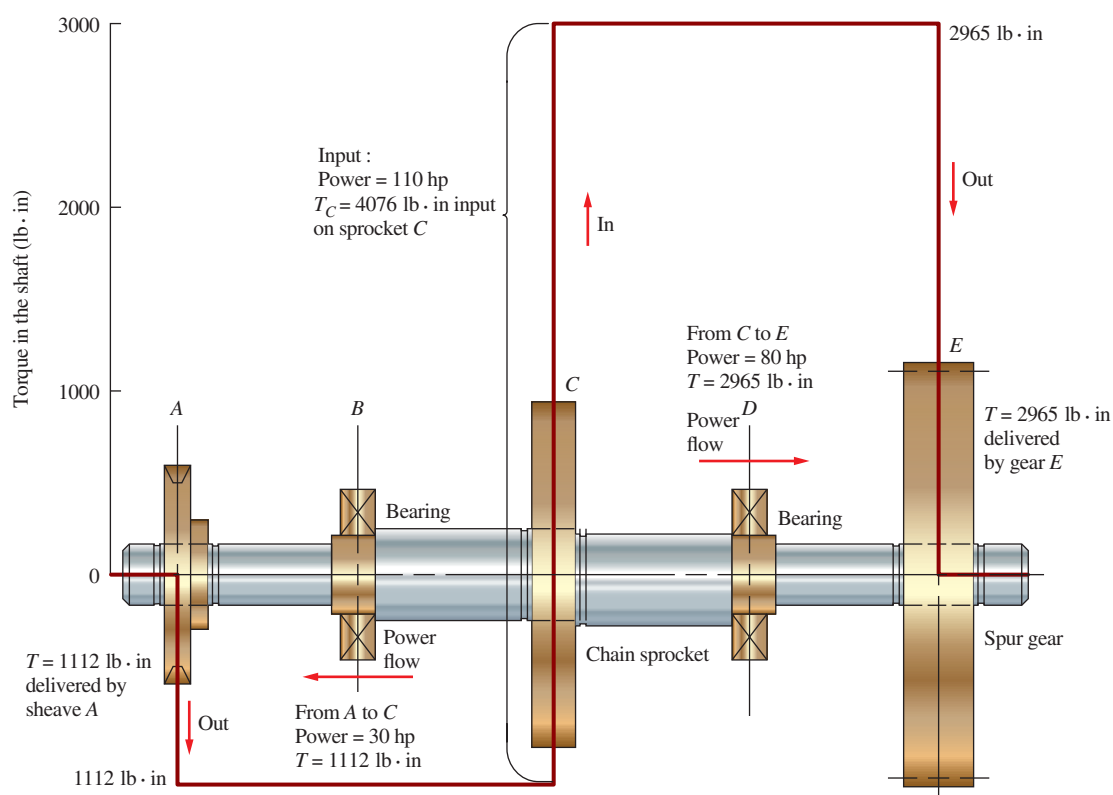


FIGURE 12-14 Torque distribution in the shaft

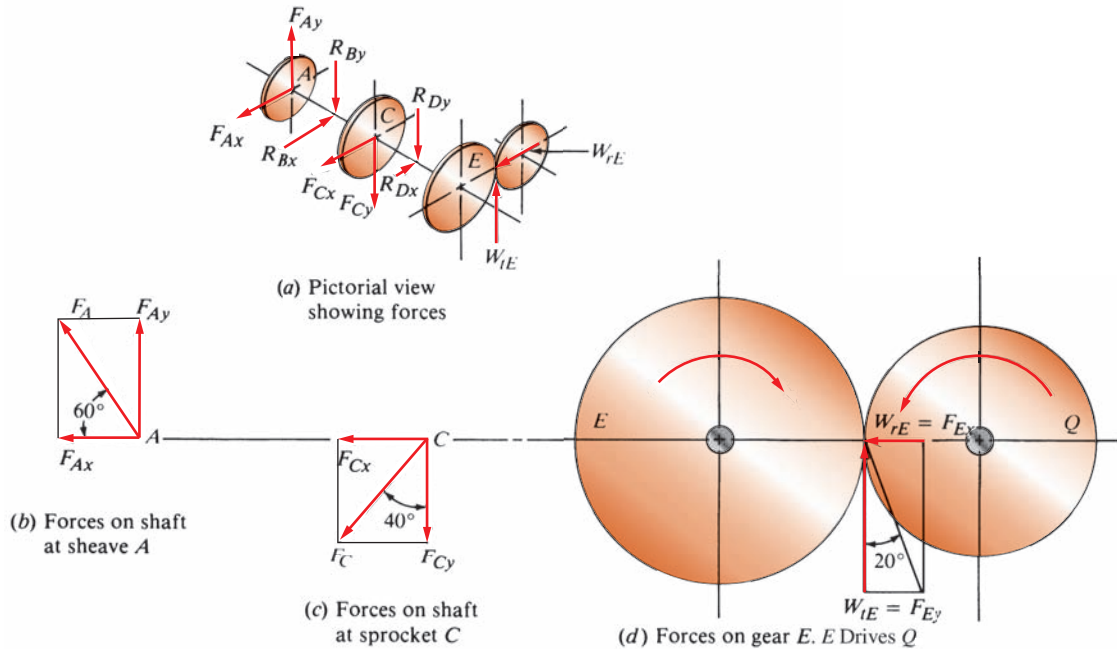


FIGURE 12-15 Forces resolved into x - and y -components

Forces: We will compute the forces at each element separately and show the component forces that act in the vertical and horizontal planes, as in Design Example 12-1. Figure 12-15 shows the directions of the applied forces and their components for each element.

1. **Forces on sheave A:** Use Equations (12-7), (12-8), and (12-12):

$$F_N = F_1 - F_2 = T_A / (D_A/2) = (1112 \text{ lb} \cdot \text{in}) / 3.0 \text{ in} = 371 \text{ lb} \quad (\text{net driving force})$$

$$F_A = 1.5 F_N = 1.5(371 \text{ lb}) = 556 \text{ lb} \quad (\text{bending force})$$

The bending force acts upward and to the left at an angle of 60° from the horizontal. As shown in Figure 12-15, the components of the bending force are

$$F_{Ax} = F_A \cos(60^\circ) = (556 \text{ lb}) \cos(60^\circ) = 278 \text{ lb} \leftarrow (\text{toward the left})$$

$$F_{Ay} = F_A \sin(60^\circ) = (556 \text{ lb}) \sin(60^\circ) = 482 \text{ lb} \uparrow (\text{upward})$$

2. **Forces on sprocket C:** Use Equation (12-6):

$$F_C = T_C / (D_C/2) = (4076 \text{ lb} \cdot \text{in}) / 5.0 \text{ in} = 815 \text{ lb}$$

This is the bending load on the shaft. The components are

$$F_{Cx} = F_C \sin(40^\circ) = (815 \text{ lb}) \sin(40^\circ) = 524 \text{ lb} \leftarrow (\text{to the left})$$

$$F_{Cy} = F_C \cos(40^\circ) = (815 \text{ lb}) \cos(40^\circ) = 624 \text{ lb} \downarrow (\text{downward})$$

3. **Forces on gear E:** The transmitted load is found from Equation (12-2), and the radial load from Equation (12-3). The directions are shown in Figure 12-15.

$$F_{Ey} = W_{tE} = T_E / (D_E/2) = (2965 \text{ lb} \cdot \text{in}) / 6.0 \text{ in} = 494 \text{ lb} \uparrow (\text{upward})$$

$$F_{Ex} = W_{rE} = W_{tE} \tan(\phi) = (494 \text{ lb}) \tan(20^\circ) = 180 \text{ lb} \leftarrow (\text{to the left})$$

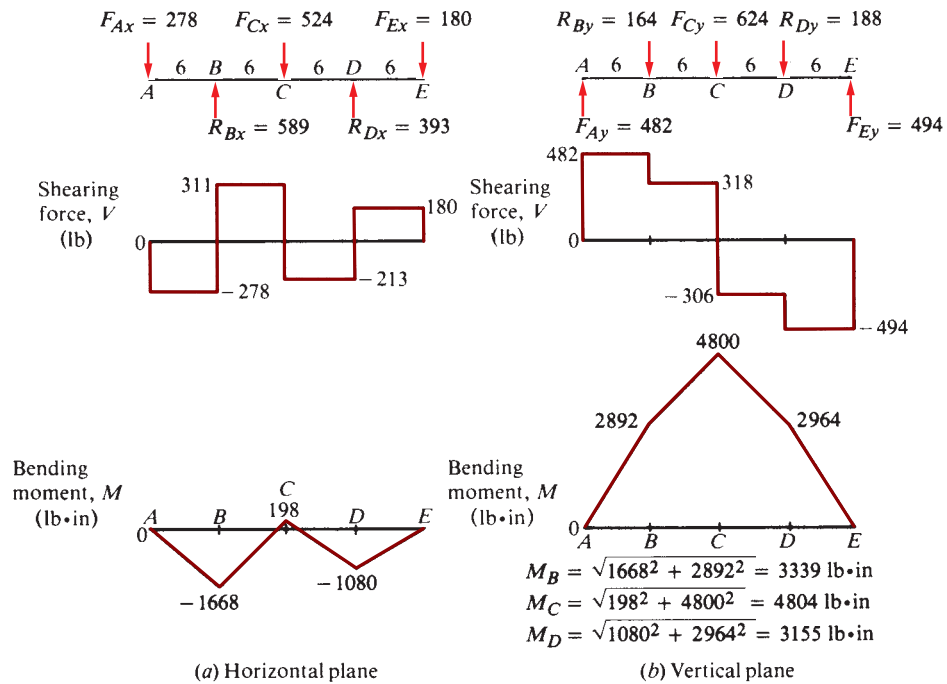


FIGURE 12-16 Load, shear, and moment diagrams

Load, Shear, and Moment Diagrams: Figure 12-16 shows the forces acting on the shaft at each element, the reactions at the bearings, and the shearing force and bending moment diagrams for both the horizontal (x -) and vertical (y -) planes. In the figure, the computations of the resultant bending moment at points B , C , and D are also shown.

Design of the Shaft: We will use Equation (12-24) to determine the minimum acceptable diameter of the shaft at each point of interest. Because the equation requires a fairly large number of individual operations, and because we will be using it at least seven times, it may be desirable to write a computer program just for that operation. Or the use of a spreadsheet would be nearly ideal. See Section 12-9. Note that we can use Equation (12-24) even though there is only torsion or only bending by entering zero for the missing value.

Equation (12-24) is repeated here for reference. In the solution below, the data used for each design point are listed. You may want to verify the calculations for the required minimum diameters. The design factor of $N = 2$ has been used.

$$D = \left[\frac{32N}{\pi} \sqrt{\left(\frac{K_t M}{s'_n} \right)^2 + \frac{3}{4} \left(\frac{T}{s_y} \right)^2} \right]^{1/3}$$

1. **Point A:** Torque = 1112 lb·in; moment = 0. The sheave is located with retaining rings. Because the torque is steady, we will not use a stress concentration factor in this calculation, as discussed in Section 12-4. But then we will find the nominal diameter at the groove to the right of A by increasing the computed result by about 6% to determine the nominal minimum full diameter for the shaft. The result should be conservative for typical groove geometries.

Using Equation (12-24), $D_1 = 0.65$ in. Increasing this by 6% gives $D_1 = 0.69$ in.

2. **To the left of point B:** This is the relief diameter leading up to the bearing seat. A well-rounded fillet radius will be specified for the place where D_2 joins D_3 . Thus,

$$\text{Torque} = 1112 \text{ lb}\cdot\text{in} \quad \text{Moment} = 3339 \text{ lb}\cdot\text{in} \quad K_t = 1.5$$

Then $D_2 = 1.70$ in.

3. **At point B and to the right:** This is the bearing seat with a shoulder fillet at the right, requiring a fairly sharp fillet:

$$\text{Torque} = 1112 \text{ lb}\cdot\text{in} \quad \text{Moment} = 3339 \text{ lb}\cdot\text{in} \quad K_t = 2.5$$

Then $D_3 = 2.02$ in.

TABLE 12-2 Specification of Values

Mating part	Diameter number	Minimum diameter	Specified diameter	
			Fraction	Decimal
Sheave	D_1	0.69	$1\frac{3}{4}$	1.750
Nothing	D_2	1.70	$1\frac{3}{4}$	1.750
Bearing	D_3	2.02	$2\frac{1}{4}$	2.250
Sprocket	D_4	2.57	$2\frac{3}{4}$	2.750
Bearing	D_5	1.98	2	2.000
Nothing	D_6	1.68	$1\frac{3}{4}$	1.750
Gear	D_7	0.96	$1\frac{3}{4}$	1.750

4. **At point C:** It is planned that the diameter be the same all the way from the right of bearing B to the left of bearing D . The worst condition is at the right of C , where there is a ring groove and the larger torque value is

$$\text{Torque} = 2965 \text{ lb} \cdot \text{in} \quad \text{Moment} = 4804 \text{ lb} \cdot \text{in} \quad K_t = 3.0$$

Then $D_4 = 2.57$ in after applying the ring groove factor of 1.06.

5. **At point D and to the left:** This is a bearing seat similar to that at B :

$$\text{Torque} = 2965 \text{ lb} \cdot \text{in} \quad \text{Moment} = 3155 \text{ lb} \cdot \text{in} \quad K_t = 2.5$$

Then $D_5 = 1.98$ in.

6. **To the right of point D:** This is a relief diameter similar to D_2 :

$$\text{Torque} = 2965 \text{ lb} \cdot \text{in} \quad \text{Moment} = 3155 \text{ lb} \cdot \text{in} \quad K_t = 1.5$$

Then $D_6 = 1.68$ in.

7. **At point E:** The gear is mounted with retaining rings on each side:

$$\text{Torque} = 2965 \text{ lb} \cdot \text{in} \quad \text{Moment} = 0 \quad K_t = 3.0$$

Then $D_7 = 0.96$ in. after applying the ring groove factor of 1.06.

Summary with
Convenient Values
Specified

Using Appendix 2, we specify convenient fractions at all places, including bearing seats (see Table 12-2). It is assumed that inch bearings of the pillow block type will be used.

We decided to make the diameters D_1 , D_2 , D_6 , and D_7 the same to minimize machining and to provide a little extra safety factor at the ring grooves. Again the bearing bore sizes would have to be checked against the load rating of the bearings. The size of D_4 would have to be checked to see that it provides a sufficient shoulder for the bearings at B and D .

The size factor and the stress concentration factors must also be checked.

12-8 SHAFT DESIGN EXAMPLE— BENDING AND TORSION WITH AXIAL FORCES

Example Problem 12-3 shows a shaft carrying a chain sprocket along with a wormgear. One of the components of the forces acting on the wormgear acts parallel to the axis of the shaft, creating a direct axial stress component in the shaft to the left of point D . We describe one

method for handling this situation within the development of the stress analysis for that part of the shaft in the problem solution.

Except for the presence of the axial stress in the shaft, the design procedure is the same as that for Design Examples 12-1 and 12-2. For this reason, much of the detailed manipulation of formulas is omitted to condense the presentation of the analysis.

Example Problem 12-3

A wormgear is mounted at the right end of the shaft as shown in Figure 12-17. The gear has the same design as that discussed in Example Problem 10-7 and delivers 6.68 hp to the shaft at a speed of 101 rpm. The magnitudes and directions of the forces on the gear are given in the figure. Notice that there is a system of three orthogonal forces acting on the gear. The power is transmitted by a chain sprocket at *B* to drive a conveyor removing cast iron chips from a machining system. Design the shaft.

Solution The torque on the shaft from the wormgear at point *D* to the chain sprocket at *B* is

$$T_B = T_D = T_{BD} = W_{tG}(D_G/2) = (962 \text{ lb})(4.333 \text{ in}) = 4168 \text{ lb} \cdot \text{in}$$

also, $T_{AB} = 0$. The torque diagram is drawn in Figure 12-17.

The force on the chain sprocket is

$$F_c = T_B(D_s/2) = (4168 \text{ lb} \cdot \text{in})/[(6.71 \text{ in})/2] = 1242 \text{ lb}$$

This force acts horizontally toward the right as viewed from the end of the shaft.

Bending Moment Diagrams: Figure 12-18 shows the forces acting on the shaft in both the vertical and the horizontal planes and the corresponding shearing force and bending moment diagrams. You should review these diagrams, especially that for the vertical plane, to grasp the effect of the axial force of 265 lb. Notice that because it acts above the shaft, it creates a bending moment at the end of the shaft of

$$M_D = W_{xG} \cdot D_{G/2} = (265 \text{ lb})[(8.667 \text{ in})/2]$$

$$M_D = 1148 \text{ lb} \cdot \text{in}$$

It also affects the reactions at the bearings. The resultant bending moments at *B*, *C*, and *D* are also shown in the figure.

In the design of the entire system, we must decide which bearing will resist the axial force. For this problem, let's specify that the bearing at *C* will transfer the axial thrust force to the housing. This decision places a compressive stress in the shaft from *C* to *D* and requires that means be provided to transmit the axial force from the wormgear to the bearing. The geometry proposed in Figure 12-17 accomplishes this, and it will be adopted for the following stress analysis. The procedures are the

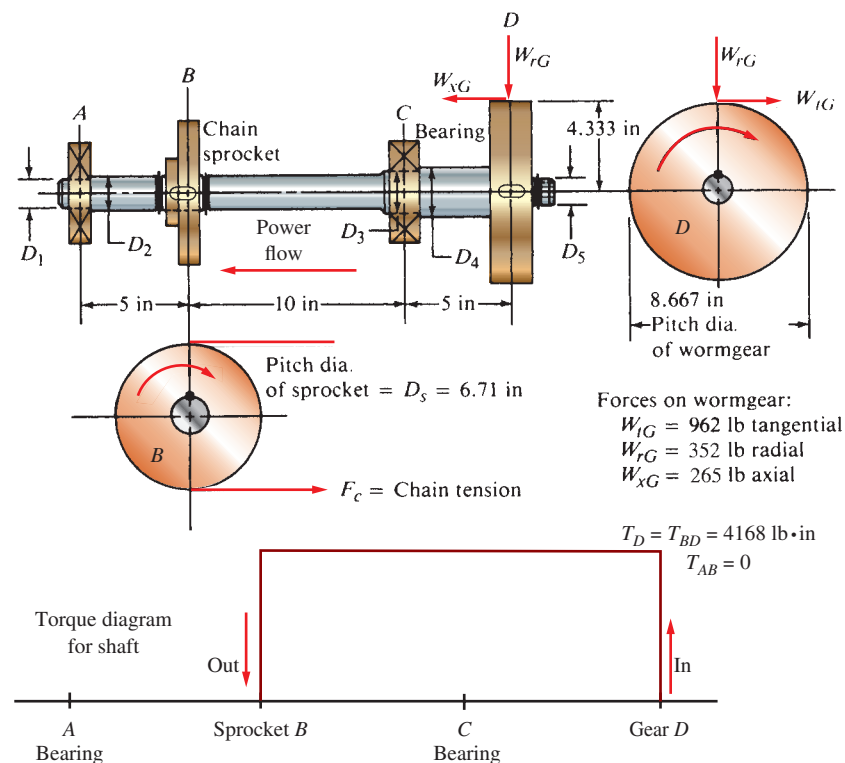


FIGURE 12-17 Shaft design

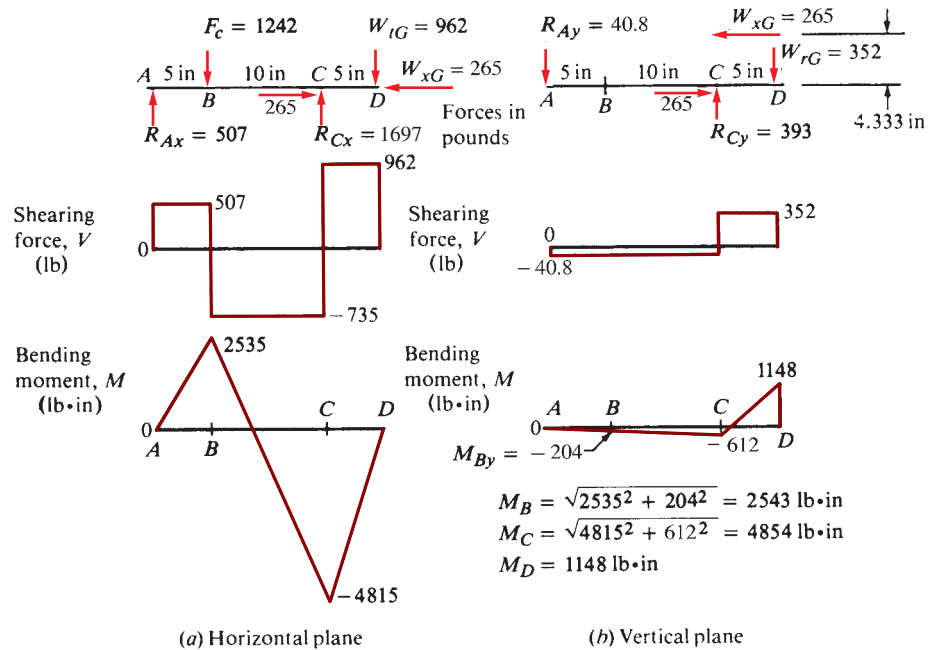


FIGURE 12-18 Load, shear, and moment diagrams for the shaft in Figure 12-16

same as those used in Design Examples 12-1 and 12-2 and only summary results will be shown. The consideration of the axial compressive stress is discussed, along with the computations at point C on the shaft.

Material Selection and Design Strengths: Medium-carbon steel with good ductility and a fairly high strength is desired for this demanding application. We will use SAE 1340 OQT 1000 (Appendix 3), having an ultimate strength of 144 000 psi, a yield strength of 132 000 psi, and a 17% elongation. From Figure 5-11 we estimate $s_n = 50\,000$ psi. Let's use an initial size factor of 0.80 and a reliability factor of 0.81 for a reliability of 0.99. Then

$$s'_n = (50\,000 \text{ psi})(0.80)(0.81) = 32\,400 \text{ psi}$$

Because the use of the conveyor is expected to be rough, we will use a design factor of $N = 3$, higher than average.

Except at point A, where only a vertical shear stress exists, the computation of the minimum required diameter is done using Equation (12-24).

- Point A:** The left bearing mounts at point A, carrying the radial reaction force only, which acts as a vertical shearing force in the shaft. There is no torque or bending moment here.

The vertical shearing force is

$$V = \sqrt{R_{Ax}^2 + R_{Ay}^2} = \sqrt{(507 \text{ lb})^2 + (40.8 \text{ lb})^2} = 509 \text{ lb}$$

We can use Equation (12-16) to compute the required diameter for the shaft at this point:

$$D = \sqrt{2.94 K_t(V)N/s'_n} \quad (12-16)$$

Referring to Figure 12-17, we see a sharp fillet near this point on the shaft. Then a stress concentration factor of 2.5 should be used:

$$D = \sqrt{\frac{2.94(2.5)(507 \text{ lb})(3)}{(32\,400 \text{ lb/in}^2)}} = 0.588 \text{ in}$$

As seen before, this is quite small, and the final specified diameter will probably be larger, depending on the bearing selected.

- Point B:** The chain sprocket mounts at point B, and it is located axially by retaining rings on both sides. The critical point is at the right of the sprocket at the ring groove, where $T = 4168 \text{ lb}\cdot\text{in}$, $M = 2543 \text{ lb}\cdot\text{in}$, and $K_t = 3.0$ for bending.

The computed minimum diameter required is $D_2 = 1.93$ in at the base of the groove. We should increase this by approximately 6%, as discussed in Section 12–3. Then

$$D_2 = 1.06(1.93 \text{ in}) = 2.05 \text{ in}$$

3. **To the left of point C:** This is the relief diameter for the bearing seat. The diameter here will be specified to be the same as that at B , but different conditions occur: Torque = 4168 lb·in, $M = 4854$ lb·in, and $K_t = 1.5$ for the well-rounded fillet for bending only. The required diameter is 1.91 in. Because this is smaller than that at B , the previous calculation will govern.
4. **At point C and to the right:** The bearing will seat here, and it is assumed that the fillet will be rather sharp. Thus, $T = 4168$ lb·in, $M = 4854$ lb·in, and $K_t = 2.5$ for bending only. The required diameter is $D_3 = 2.26$ in.

The axial thrust load acts between points C and D . The inclusion of this load in the computations would greatly complicate the solution for the required diameters. In most cases, the axial normal stress is relatively small compared with the bending stress. Also, the fact that the stress is compressive improves the fatigue performance of the shaft, because fatigue failures normally initiate at points of tensile stress. For these reasons, the axial stress is ignored in these calculations. The computed diameters are also interpreted as nominal minimum diameters, and the final selected diameter is larger than the minimum. This, too, tends to make the shaft safe even when there is an added axial load. When in doubt, or when a relatively high axial tensile stress is encountered, the methods of Chapter 5 should be applied. Long shafts in compression should also be checked for buckling.

5. **Point D:** The wormgear mounts at point D . We will specify that a well-rounded fillet will be placed to the left of D and that there will be a sled runner keyseat. Thus, $T = 4168$ lb·in, $M = 1148$ lb·in, and $K_t = 1.6$ for bending only. The computed required diameter is $D_5 = 1.24$ in. Notice that D_4 must be greater than either D_3 or D_5 because it provides the means to transfer the thrust load from the wormgear to the inner race of the bearing at C .

Summary and Selection of Convenient Diameters Table 12–3 presents a summary of the required diameters and the specified diameters for all parts of the shaft in this design example. See Figure 12–17 for the locations of the five diameters. For this application we have chosen to use fractional-inch dimensions from Appendix 2 except at the bearing seats, where the use of metric bearing bores from Table 14–3 are selected.

Comment As a partial check on the validity of the design decision to complete the calculations for shaft diameters while avoiding the axial stress between points C and D due to the axial force from the wormgear, let's now compute the magnitude of the axial compressive stress due to the 265-lb force. The shaft size between points C and D is $D_4 = 3.00$ in. Then,

$$\begin{aligned} \text{Axial stress} &= \sigma_{CD} = -W_{xG}/A_4 \\ A_4 &= \pi D_4^2/4 = \pi(3.00 \text{ in})^2/4 = 7.069 \text{ in}^2 \\ \sigma_{CD} &= -265 \text{ lb}/7.069 \text{ in}^2 = -37.5 \text{ psi Compression} \end{aligned}$$

Because of this rather low stress level, it seems reasonable to conclude that avoiding the axial compressive stress in the initial design analysis was warranted.

TABLE 12–3 Summary of Shaft Diameters

Mating part	Diameter number	Minimum diameter	Specified diameter	
			Fraction (metric)	Decimal
Bearing A	D_1	0.59 in	(35 mm)	1.3780 in
Sprocket B	D_2	2.05 in	$2\frac{1}{4}$ in	2.250 in
Bearing C	D_3	2.26 in	(65 mm)	2.5591 in
(Shoulder)	D_4	$>D_3$	3 in	3.000 in
Wormgear D	D_5	1.24 in	$1\frac{1}{2}$ in	1.500 in

12-9 SPREADSHEET AID FOR SHAFT DESIGN

A spreadsheet is useful to organize the data required to compute the minimum required shaft diameter at various points along a shaft and to complete the calculation using Equations (12-16) and (12-24). Note that Equation (12-24) can be used for bending only, torsion only, or the combination of bending and torsion.

Figure 12-19 shows a typical example, using U.S. Customary units, for data from Design Example 12-1. Describe the application in the upper panel for future reference. Then complete the following steps:

- Enter the shaft material specification along with its ultimate and yield strength properties found from tables in the appendices.
- Find the basic endurance strength from Figure 5-11 considering the ultimate tensile strength and the manner of production (ground, machined, etc.).

- Enter values for the size factor and the reliability factor. The spreadsheet then computes the modified endurance strength, s'_n .
- Enter the stress concentration factor for the point of interest.
- Enter the design factor.
- After analysis such as that shown in Design Example 12-1, enter the torque and the components of the bending moment in the x - and y -planes that exist at the point of interest along the shaft. The spreadsheet computes the combined bending moment.
- Enter the components of the vertical shearing force in the x - and y -planes. The spreadsheet computes the combined shearing force.

The minimum acceptable shaft diameters from both Equation (12-16) (vertical shear only) and Equation (12-24) (torsion and/or bending) are computed. You must observe which required diameter is larger.

DESIGN OF SHAFTS			
<i>Application:</i>	<i>Design Example 12–1, Drive for a Blower System</i> <i>Diameter D_3—To right of point B—Bending and torsion</i>		
This design aid computes the minimum acceptable diameter using Equation (12–24) for shafts subjected to steady torsion and/or rotating bending. Equation (12–16) is used when only vertical shear stress is present.			
<i>Input Data</i>		<i>(Insert values in italics.)</i>	
<i>Shaft material specification:</i>	<i>SAE 1144 OQT 1000 Steel</i>		
<i>Tensile strength:</i>	<i>$s_u =$</i>	<i>118 000 psi</i>	
<i>Yield strength:</i>	<i>$s_y =$</i>	<i>83 000 psi</i>	
<i>Basic endurance strength:</i>	<i>$s_n =$</i>	<i>42 000 psi</i>	From Figure 5–11
<i>Size factor:</i>	<i>$C_s =$</i>	<i>0.75</i>	From Figure 5–12
<i>Reliability factor:</i>	<i>$C_R =$</i>	<i>0.81</i>	From Table 5–3
Modified endurance strength:	<i>$s'_n =$</i>	25 515 psi	<i>Computed</i>
<i>Stress concentration factor:</i>	<i>$K_t =$</i>	<i>2.5 Sharp fillet</i>	
<i>Design factor:</i>	<i>$N =$</i>	<i>2 Nominal $N = 2$</i>	
<i>Shaft Loading Data: Bending and Torsion</i>			
<i>Bending moment components:</i>	<i>$M_x =$</i>	<i>21 000 lb · in</i>	<i>$M_y = 7640 \text{ lb} \cdot \text{in}$</i>
Combined bending moment:	<i>$M =$</i>	22 347 lb · in	<i>Computed</i>
<i>Torque:</i>	<i>$T =$</i>	<i>21 000 lb · in</i>	
Minimum shaft diameter:	<i>$D =$</i>	3.55 in	<i>Computed from Eq. (12–24)</i>
<i>Shaft Loading Data: Vertical Shearing Force Only</i>			
<i>Shearing force components:</i>	<i>$V_x =$</i>	<i>764 lb</i>	<i>$V_y = 2520 \text{ lb}$</i>
Combined shearing force:	<i>$V =$</i>	2633 lb	<i>Computed</i>
Minimum shaft diameter:	<i>$D =$</i>	1.232 in	<i>Computed from Eq. (12–16)</i>

FIGURE 12-19 Spreadsheet aid for shaft design

12-10 SHAFT RIGIDITY AND DYNAMIC CONSIDERATIONS

The design processes outlined in this chapter thus far have concentrated on stress analysis and ensuring that the shaft is safe with regard to the bending and torsional shear stresses imposed on the shaft by power-transmitting elements. Rigidity of the shaft is also a major concern for several reasons:

1. Excessive radial deflection of the shaft may cause active elements to become misaligned, resulting in poor performance or accelerated wear. For example, the center distance between shafts carrying precision gears should not vary more than approximately 0.005 in (0.13 mm) from the theoretical dimension. Improper meshing of the gear teeth would occur and actual bending and contact stresses could be significantly higher than those predicted in the design for the gears.
2. Guidelines are reported in Section 20-2 for the recommended limits for bending and torsional deflection of a shaft according to its intended precision.
3. Shaft deflection is also an important contributor to a tendency for shafts to vibrate as they rotate. A flexible shaft will oscillate in both bending and torsional modes, causing movements that are greater than the static deflections due just to gravity and applied loads and torques. A long slender shaft tends to whip and whirl with relatively large deformations from its theoretical straight axis.
4. The shaft itself and the elements mounted on it should be balanced. Any amount of unbalance causes centrifugal forces to be created that rotate with the shaft. Large unbalances and high rotating speeds may create unacceptable force levels and shaking of the rotating system. An example with which you may be familiar is an automobile wheel that is “out of balance.” You can actually feel the vibration through the steering wheel as you drive. Having the tire and wheel balanced reduces the vibration to acceptable levels.
5. The dynamic behavior of the shaft may become dangerously destructive if it is operated near its *critical speed*. At critical speed, the system is in resonance, the deflection of the shaft continues to increase virtually without bound, and it will eventually self-destruct.

Critical speeds of typical machinery shaft designs are several thousand revolutions per minute. However, many variables are involved. Long shafts used for vehicular driveshafts, power screws, or agitators may have much lower critical speeds and they must be checked. It is typical to keep operating speeds to at least 20% below or above the critical speed. When operating above the critical speed, it is necessary to accelerate the shaft rapidly

through the critical speed because it takes time for dangerous oscillations to develop.

Analysis to determine the critical speed is complex, and computer programs are available to assist in the calculations. See Internet site 1. The objective is to determine the natural frequency of the shaft while carrying the static weight of elements such as gears, sprockets, and pulleys. The stiffness of bearings is also a factor. One fundamental expression for natural frequency, ω_n , is,

$$\omega_n = \sqrt{k/m}$$

where k is the stiffness of the shaft and m is its mass. It is desirable to have a high critical speed, well above operating speed, so stiffness should be high and mass low. The primary variables over which a designer has control are the material and its modulus of elasticity, E , its density, ρ , the shaft diameter, D , and the shaft length, L . The following functional relationship can help to understand the influence of each of these variables:

$$\omega_n \propto (D/L^2)\sqrt{E/\rho}$$

where the symbol $\propto$ indicates proportionality among the variables. It is desirable to reduce the deflection of the shaft to produce a high critical speed. Other approaches are given here.

1. Making the shaft more rigid can prevent undesirable dynamic behavior.
2. Larger shaft diameters add rigidity.
3. Shorter shaft lengths reduce deflections and raise critical speeds.
4. Placing active elements on the shaft close to support bearings is recommended.
5. Reducing the weight of elements carried by the shaft reduces static deflection and raises critical speed.
6. Selection of a material for the shaft with a high ratio of E/ρ (modulus of elasticity/density) is desirable. While most metals have similar ratios, that for composite materials is typically high. For example, long driveshafts for vehicles that must operate at high speeds are frequently made from hollow tubular composite materials using carbon fibers. See Internet sites 2 and 3.
7. Bearings should have a high stiffness in terms of radial deflection as a function of load.
8. Mountings for bearings and housings should be designed with high rigidity.
9. References 3-5 and 7 provide additional information and analytical methods for estimating critical speeds. References 9-11 discuss machinery vibration.

Internet site 4 includes a section on critical speeds that can be used for calculating approximate values for several different support conditions and loading patterns. Internet sites 5 and 6 describe two software packages

that contain beam analysis programs that can be used to calculate beam deflections.

12-11 FLEXIBLE SHAFTS

At times it is desirable to transmit rotational motion and power between two points that are not aligned with each other. Flexible shafts can be used in such situations to couple a driver, such as a motor, to a driven device along a curved or dynamically moving path. The flexibility allows the driven point to be offset from the driving point in either parallel or angular fashion. Unidirectional shafts are used for power transmission in applications such as automation systems, industrial machinery, agricultural equipment, aircraft actuators, seat adjusters, medical devices, dentistry, speedometers, woodworking, and jeweler's tools. Bidirectional flexible shafts are used in remote control, valve actuation, and safety devices. They can often provide more design flexibility than other design options such as right-angle gear drives or universal joints. The offset between the driver and driven machine can be as much as 180° provided that the bend radius is above a specified minimum. Torque capacity increases with increasing bend radius. Efficiencies are in the range from 90% to 95%.

The construction consists of a flexible cable-like core connected to fittings on each end. The fittings facilitate connection with the driving and driven machines. The casing protects nearby equipment or people from contact with the rotating core. Designs are available for transmitting power in either direction, clockwise or counterclockwise. Some will operate bidirectionally.

See Internet sites 7–9 for additional information about commercially available flexible shafts. Table 12–4 gives a sampling of sizes and ratings from Internet site 7. Many more sizes are available.

TABLE 12–4 Sampling of Operating Torque Capacity for Flexible Shafts

Data for shafts of medium stiffness operating at 12-in (305-mm) radius			
Shaft diameter		Torque capacity	
(in)	(mm)	(lb · in)	(N · m)
0.127	3.23	1.7	0.19
0.183	4.65	4.3	0.49
0.245	6.22	20	2.26
0.304	7.72	33	3.73
0.495	12.57	86	9.72
0.740	18.80	265	29.95
0.990	25.15	386	43.62

Data adapted from Internet site 7.

REFERENCES

1. American Society of Mechanical Engineers. ANSI Standard B106.1M-1985. *Design of Transmission Shafting*. New York: American Society of Mechanical Engineers, 1985.
2. American Society of Mechanical Engineers. ANSI Standard B4.1-67(R2009). *Preferred Limits and Fits for Cylindrical Parts*. New York: American Society of Mechanical Engineers, 2009.
3. Sadegh, Ali, Theodore Baumeister, and Eugene Avalonne. *Marks' Standard Handbook for Mechanical Engineers*. 11th ed. New York: McGraw-Hill, 2007.
4. Paez, Thomas, and Allan Piersol. (Editors). *Harris' Shock and Vibration Handbook*. 6th ed. New York: McGraw-Hill, 2010.
5. Oberg, Erik et al. *Machinery's Handbook*. 30th ed. New York: Industrial Press, 2015.
6. Pilkey, Walter D., and Deborah F. Pilkey. *Peterson's Stress Concentration Factors*. 3rd ed. New York: John Wiley & Sons, 2008.
7. Budynas, R. G., and K. J. Nisbett. *Shigley's Mechanical Engineering Design*. 10th ed. New York: McGraw-Hill, 2015.
8. Soderberg, C. R. "Working Stresses." *Journal of Applied Mechanics* 57 (1935): A-106.
9. Wowk, Victor. *Machinery Vibration: Alignment*. New York: McGraw-Hill, 2000.
10. Wowk, Victor. *Machinery Vibration: Balancing*. New York: McGraw-Hill, 1998.
11. Wowk, Victor. *Machinery Vibration: Measurement and Analysis*. New York: McGraw-Hill, 1991.
12. American Society of Mechanical Engineers. ANSI Standard B4.2-1978(R2009). *Preferred Metric Limits and Fits*. New York: American Society of Mechanical Engineers, 2009.

INTERNET SITES FOR SHAFT DESIGN

1. **Hexagon Software.** Computer software called WL1+ for shaft calculations. Can handle up to 100 cylindrical or conical shaft segments and up to 50 individual forces. Computes shearing force and bending moment diagrams, slope, deflection, bending stresses, and shearing stresses. Critical speeds for flexure and torsional vibration are calculated.
2. **Spinning Composites.** Manufacturer of composite shafts for vehicular and industrial applications. Includes discussion of critical speeds.
3. **Advanced Composite Products and Technology, Inc.** Manufacturer of composite products for aerospace, military, commercial and industrial markets, including drive-shafts and high-speed rotors.
4. **RoyMech.** An Internet site for useful information, tables, and formulas related to mechanical engineering; developed by Roy Beardmore. One section deals with critical speeds of shafts.

5. **Orand Systems, Inc.** Developer and distributor of *Beam 2D Stress Analysis* software. Analyzes beams with numerous loads, cross sections, and support conditions. Computes load, shearing force, bending moment, slope, and deflection outputs in graphical and numerical data formats. Can be used for the bending calculations for shafts.
6. **MDSolids.** Educational software for stress analysis that includes several modules including beams, torsion, trusses, and columns.
7. **S. S. White Technologies, Inc.** Manufacturer of flexible shafts. Site includes rating charts for torque capacity versus the minimum radius of the shaft during operation.
8. **Elliott Manufacturing Co.** Manufacturer of flexible shafts.
9. **Otto Suhner AG.** Manufacturer of flexible shafts, spiral bevel gears, and small, high speed air motors and electric motors. Site includes rating charts for power and torque capacity of flexible shafts.

PROBLEMS

Problems 1 through 30 relate to one of Figures P12–1 through P12–17 showing shafts carrying a variety of combinations of gears, belt sheaves, chain sprockets, and a few other items such as a flywheel and a propeller-type fan. There are multiple ways in which the problems may be assigned, as some deal with only torque levels and radial forces in the shaft due to selected power-transmitting elements while others are comprehensive design problems for the shaft that carries several elements. The

following table may help instructors decide how to assign the problems for student solution and may help students comprehend how the sets of problems lead to the more general shaft design. Any combination of problems may be chosen.

Torques and Forces Acting Radial to Shaft	Comprehensive
Figure P12–1: P1—Gear <i>B</i> ; P14—Sheave <i>D</i>	P22
Figure P12–2: P2—Gear <i>C</i> ; P12—Sprocket <i>D</i> ; P13—Pulley <i>A</i>	P23
Figure P12–3: P3—Gear <i>B</i> ; P15—Sprocket <i>C</i> ; P16—Sheaves <i>D</i> , <i>E</i>	P24
Figure P12–4: P4—Gear <i>A</i> ; P19—Sprockets <i>C</i> , <i>D</i>	P25
Figure P12–5: P5—Gear <i>D</i> ; P20—Sheave <i>A</i> ; P21—Sprocket <i>E</i>	P26
Figure P12–6: P6—Gear <i>E</i> ; (No separate analysis of Sheave <i>A</i>)	P27 (Includes Sheave <i>A</i>)
Figure P12–7: P7—Gear <i>C</i> ; P8—Gear <i>A</i>	P28 (Includes Sheaves <i>D</i> , <i>E</i>)
Figure P12–9: P9—Gear <i>C</i> ; P10—Gear <i>D</i> ; P11—Gear <i>F</i>	P29 (Includes Sheave <i>B</i>)
Figure P12–17: P17—Sheave <i>C</i> ; P18—Pulley <i>D</i>	P30 (Includes Fan <i>A</i>)

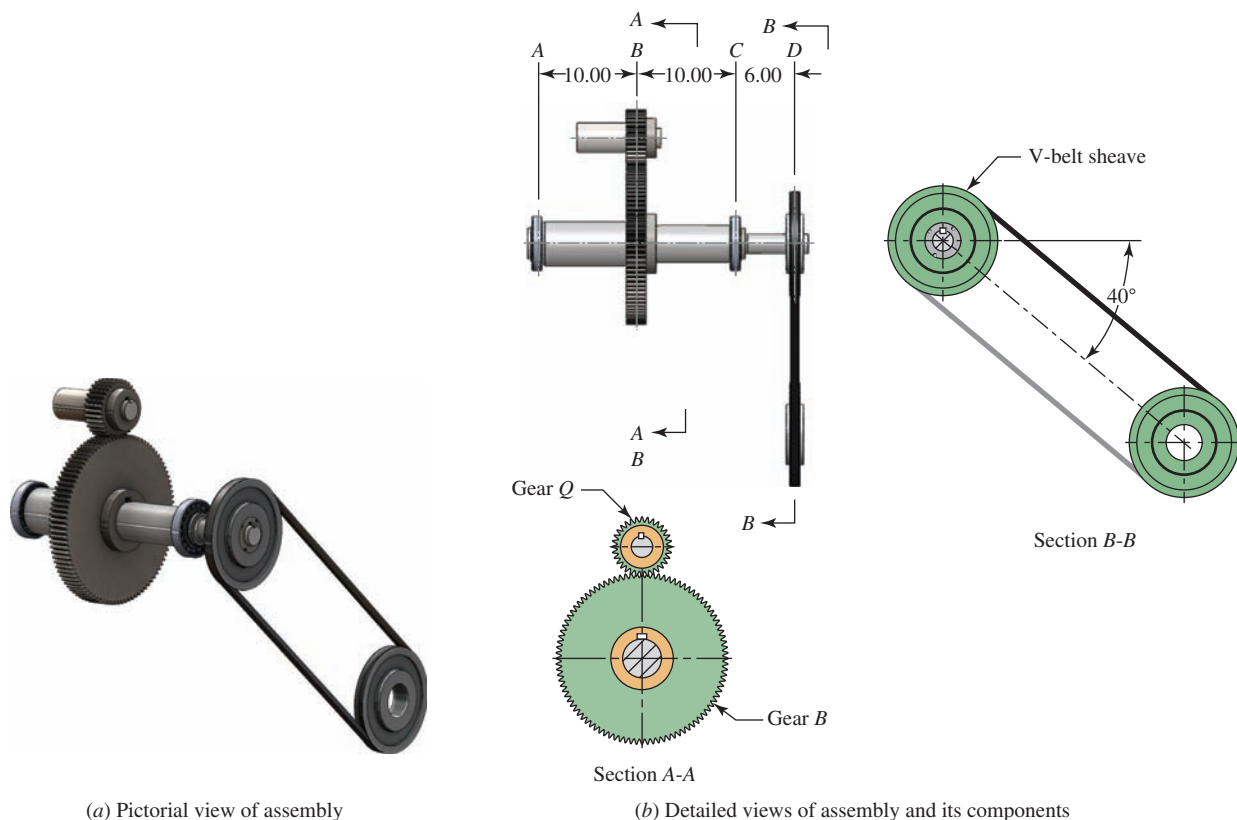


FIGURE P12–1 (Problems 1, 14, and 22) Shaft carrying a gear and a V-belt drive

Torques and Forces Acting Radial to Shaft

1. See Figure P12-1. The shaft rotating at 550 rpm carries a spur gear *B* having 96 teeth with a diametral pitch of 6. The teeth are of the 20°, full-depth, involute form. The gear receives 30 hp from a pinion directly above it. Compute the torque delivered to the shaft and the tangential and radial forces exerted on the shaft by the gear.
2. See Figure P12-2. The shaft rotating at 200 rpm carries a spur gear *C* having 80 teeth with a diametral pitch of 8. The teeth are of the 20°, full-depth, involute form. The gear delivers 6 hp to a pinion directly below it. Compute the torque delivered by the shaft to gear *C* and the tangential and radial forces exerted on the shaft by the gear.
3. See Figure P12-3. The shaft rotating at 480 rpm carries a spur pinion *B* having 24 teeth with a diametral pitch of 8. The teeth are of the 20°, full-depth, involute form. The pinion delivers 5 hp to a gear directly below it. Compute

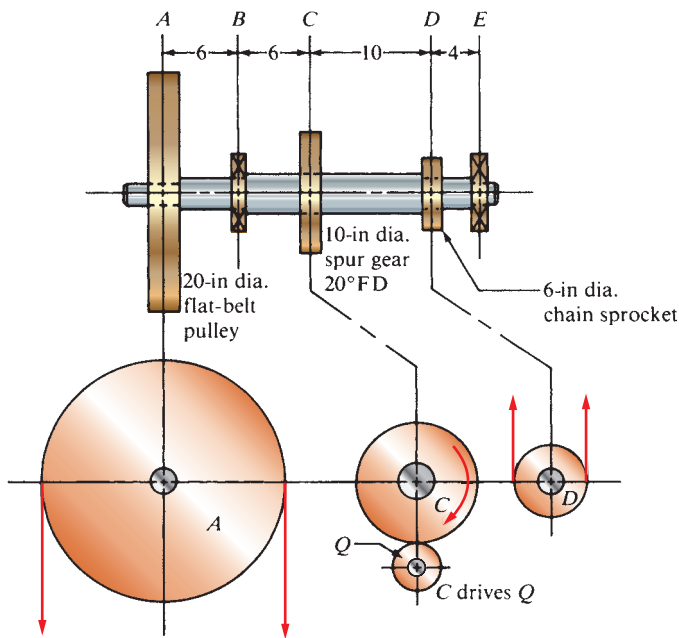


FIGURE P12-2 (Problems 2, 12, 13, and 23)

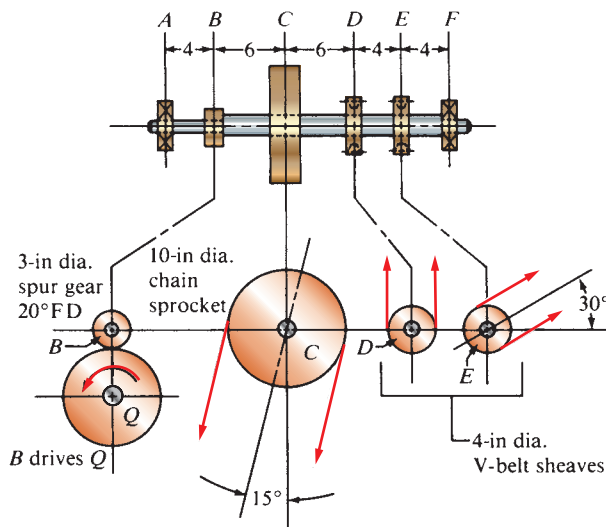


FIGURE P12-3 (Problems 3, 15, 16, and 24)

- the torque delivered by the shaft to pinion *B* and the tangential and radial forces exerted on the shaft by the pinion.
4. See Figure P12-4. The shaft rotating at 120 rpm carries a spur gear *A* having 80 teeth with a diametral pitch of 5. The teeth are of the 20°, full-depth, involute form. The gear receives 40 hp from a pinion to the right as shown. Compute the torque delivered to the shaft and the tangential and radial forces exerted on the shaft by the gear.
5. See Figure P12-5. The shaft rotating at 240 rpm carries a spur gear *D* having 48 teeth with a diametral pitch of 6. The teeth are of the 20°, full-depth, involute form. The gear receives 15 hp from pinion *Q* located as shown. Compute the torque delivered to the shaft and the tangential and radial forces exerted on the shaft by the gear. Resolve the tangential and radial forces into their horizontal and vertical components, and determine the net forces acting on the shaft at *D* in the horizontal and vertical directions.
6. See Figure P12-6. The shaft rotating at 310 rpm carries a spur pinion *E* having 36 teeth with a diametral pitch of 6. The teeth are of the 20°, full-depth, involute form. The pinion delivers 20 hp to a gear to the left as shown. Compute the torque delivered by the shaft to pinion *E* and the tangential and radial forces exerted on the shaft by the pinion. Include the weight of the pinion.
7. See Figure P12-7. The shaft rotating at 480 rpm carries a spur gear *C* having 50 teeth with a diametral pitch of 5. The teeth are of the 20°, full-depth, involute form. The gear receives 50 hp from a pinion directly below it. Compute the torque delivered to the shaft and the tangential and radial forces exerted on the shaft by the gear.
8. See Figure P12-7. The shaft rotating at 480 rpm carries a spur pinion *A* having 30 teeth with a diametral pitch of 6. The teeth are of the 20°, full-depth, involute form. The pinion delivers 30 hp to a gear to the left as shown. Compute the torque delivered by the shaft to pinion *A* and the tangential and radial forces exerted on the shaft by the pinion.
9. See Figure P12-9. The shaft rotating at 220 rpm carries a spur pinion *C* having 60 teeth with a diametral pitch of 10. The teeth are of the 20°, full-depth, involute form. The pinion delivers 5 hp to a gear directly above it. Compute the torque delivered by the shaft to pinion *C* and the tangential and radial forces exerted on the shaft by the pinion.
10. See Figure P12-9. The shaft rotating at 220 rpm carries a spur gear *D* having 96 teeth with a diametral pitch of 8. The teeth are of the 20°, full-depth, involute form. The gear receives 12.5 hp from a pinion directly below it. Compute the torque delivered to the shaft and the tangential and radial forces exerted on the shaft by the gear.
11. See Figure P12-9. The shaft rotating at 220 rpm carries a spur pinion *F* having 60 teeth with a diametral pitch of 10. The teeth are of the 20°, full-depth, involute form. The pinion delivers 5 hp to a gear located as shown. Compute the torque delivered by the shaft to pinion *F* and the tangential and radial forces exerted on the shaft by the pinion. Resolve the forces into their horizontal and vertical components, and determine the net forces acting on the shaft at *F* in the horizontal and vertical directions.
12. See Figure P12-2. The shaft rotating at 200 rpm carries a 6-in-diameter chain sprocket *D* that delivers 4 hp to a mating sprocket above. Compute the torque delivered by the shaft to sprocket *D* and the force exerted on the shaft by the sprocket.

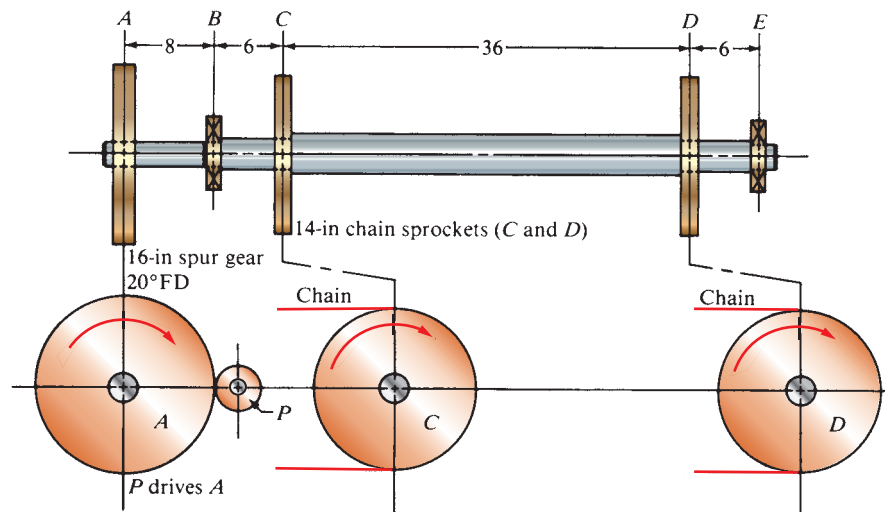


FIGURE P12-4 (Problems 4, 19, and 25)

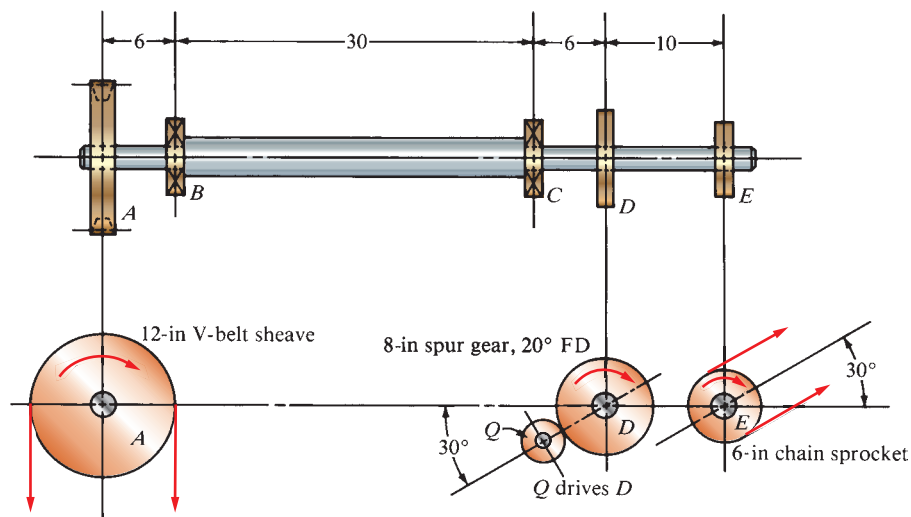


FIGURE P12-5 (Problems 5, 20, 21, and 26)

13. See Figure P12-2. The shaft rotating at 200 rpm carries a 20-in-diameter flat-belt pulley at A that receives 10 hp from below. Compute the torque delivered by the pulley to the shaft and the force exerted on the shaft by the pulley.
14. See Figure P12-1. The shaft rotating at 550 rpm carries a 10-in-diameter V-belt sheave at D that delivers 30 hp to a mating sheave as shown. Compute the torque delivered by the shaft to the sheave and the total force exerted on the shaft by the sheave. Resolve the force into its horizontal and vertical components, and show the net forces acting on the shaft at D in the horizontal and vertical directions.
15. See Figure P12-3. The shaft rotating at 480 rpm carries a 10-in-diameter chain sprocket at C that receives 11 hp from a mating sprocket below and to the left as shown. Compute the torque delivered to the shaft by the sprocket and the total force exerted on the shaft by the sprocket. Resolve the force into its horizontal and vertical components, and show the net forces acting on the shaft at C in the horizontal and vertical directions.
16. See Figure P12-3. The shaft rotating at 480 rpm carries two 4-in-diameter sheaves at D and E that each deliver 3 hp to mating sheaves as shown. Compute the torque delivered by the shaft to each sheave and the total force exerted on the shaft by each sheave. Resolve the force at E into its horizontal and vertical components, and show the net forces acting on the shaft at E in the horizontal and vertical directions.
17. See Figure P12-17. The shaft rotating at 475 rpm carries a 10-in V-belt sheave at C that receives 15.5 hp from a mating sheave to the left as shown. Compute the torque delivered to the shaft at C by the sheave and the total force exerted on the shaft at C by the sheave.
18. See Figure P12-17. The shaft rotating at 475 rpm carries a 6-in-diameter flat-belt pulley at D that delivers 3.5 hp to a mating pulley above and to the right as shown. Compute the torque delivered to the shaft by the pulley and the total force exerted on the shaft by the pulley. Resolve

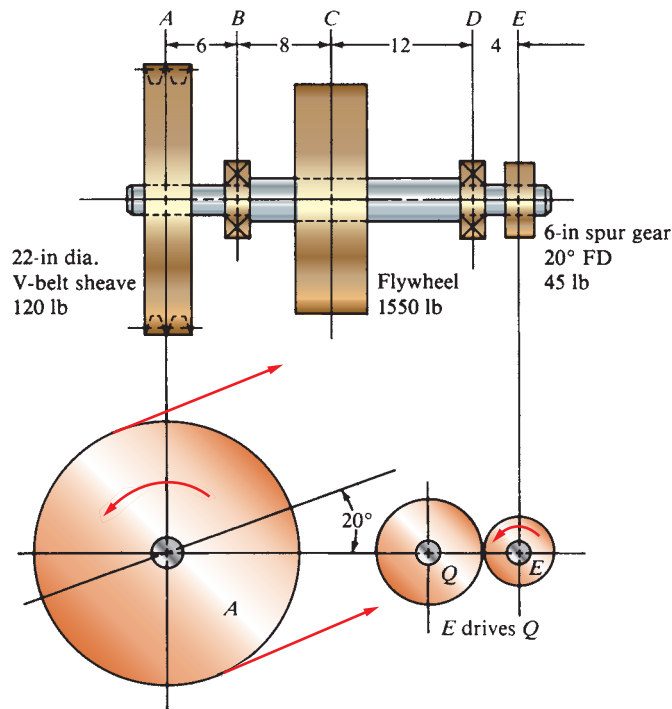


FIGURE P12-6 (Problems 6 and 27)

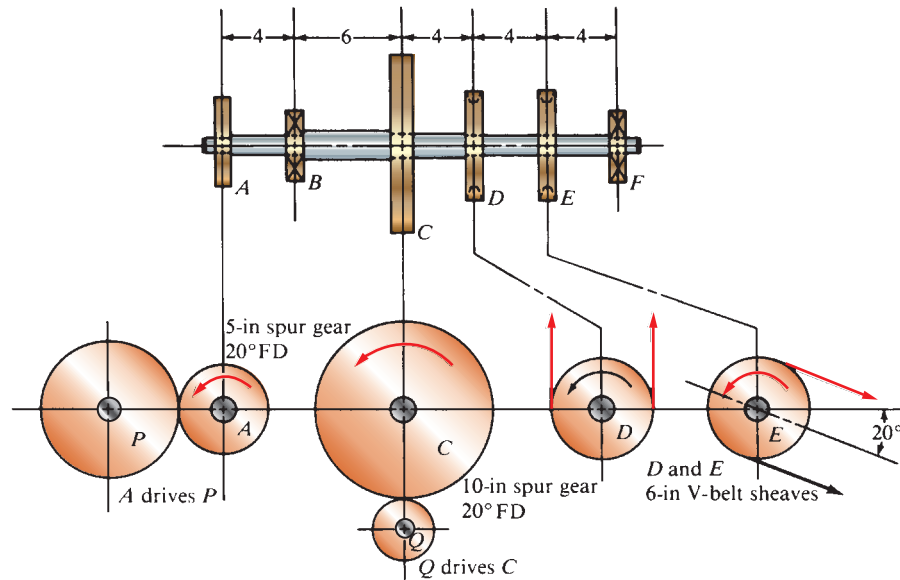


FIGURE P12-7 (Problems 7, 8, and 28)

the force into its horizontal and vertical components, and show the net forces acting on the shaft at D in the horizontal and vertical directions.

19. See Figure P12-4. The shaft rotating at 120 rpm carries two identical 14-in-diameter chain sprockets at C and D . Each sprocket delivers 20 hp to mating sprockets to the left as shown. Compute the torque delivered by the shaft to each sprocket and the total force exerted on the shaft by each sprocket.
20. See Figure P12-5. The shaft rotating at 240 rpm carries a 12-in-diameter V-belt sheave at A that delivers 10 hp to a

mating sheave directly below. Compute the torque delivered by the shaft to the sheave and the total force exerted on the shaft at A by the sheave.

21. See Figure P12-5. The shaft rotating at 240 rpm carries a 6-in-diameter chain sprocket at E that delivers 5.0 hp to a mating sprocket to the right and above as shown. Compute the torque delivered by the shaft to the sprocket and the total force exerted on the shaft by the sprocket. Resolve the force into its horizontal and vertical components, and show the net forces acting on the shaft at E in the horizontal and vertical directions.

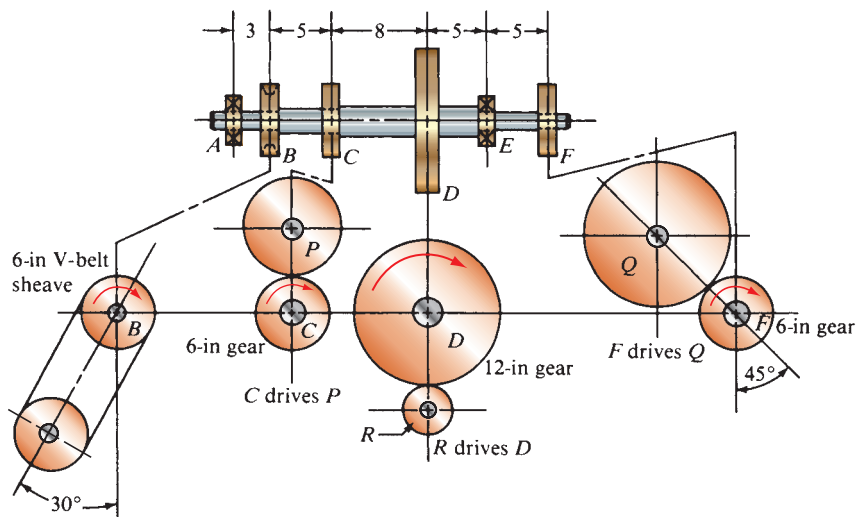


FIGURE P12-9 (Problems 9–11 and 29)

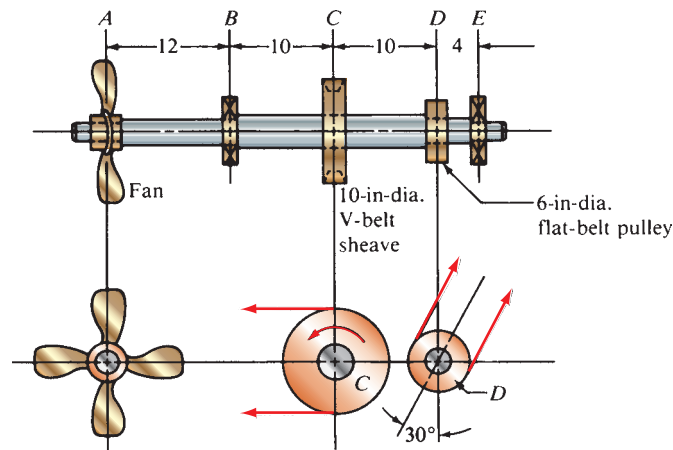


FIGURE P12-17 (Problems 17, 18, and 30)

Comprehensive Shaft Design Problems

For each of the following Problems 22 through 30, it will be necessary to do the following:

- Determine the magnitude of the torque in the shaft at all points.
- Compute the forces acting on the shaft at all power-transmitting elements.
- Compute the reactions at the bearings.
- Draw the complete load, shear, and bending moment diagrams.

Neglect the weight of the elements on the shafts, unless otherwise noted.

The objective of any problem, at the discretion of the instructor, could be any of the following:

- Design the complete shaft, including the specification of the overall geometry and the consideration of stress concentration factors. The analysis would show the minimum acceptable diameter at each point on the shaft to be safe from the standpoint of strength.

- Given a suggested geometry of one part of the shaft, specify the minimum acceptable diameter for the shaft at that point.
- Specify the required geometry at any selected element on the shaft: a gear, sheave, bearing, or other.
- Make a working drawing of the design for the shaft, following the appropriate stress analysis, and specify the final dimensions.
- Suggest how the given shaft can be redesigned by moving or reorienting the elements on the shaft to improve the design to produce lower stresses, smaller shaft size, more convenient assembly, and so on.
- Incorporate the given shaft into a more comprehensive machine, and complete the design of the entire machine. In most problems, the type of machine for which the shaft is being designed is suggested.

- The shaft in Figure P12-1 is part of a drive for an automated transfer system in a metal stamping plant. Gear *Q* delivers 30 hp to gear *B*. Sheave *D* delivers the power to its mating sheave as shown. The shaft carrying *B* and *D* rotates at 550 rpm. Use SAE 1040 cold-drawn steel.

23. The shaft in Figure P12-2 rotates at 200 rpm. Pulley *A* receives 10 hp from below. Gear *C* delivers 6 hp to the mating gear below it. Chain sprocket *D* delivers 4 hp to a shaft above. Use SAE 1117 cold-drawn steel.
24. The shaft in Figure P12-3 is part of a special machine designed to reclaim scrap aluminum cans. The gear at *B* delivers 5 hp to a chopper that cuts the cans into small pieces. The V-belt sheave at *D* delivers 3 hp to a blower that draws air through the chopper. V-belt sheave *E* delivers 3 hp to a conveyor that raises the shredded aluminum to an elevated hopper. The shaft rotates at 480 rpm. All power comes into the shaft through the chain sprocket at *C*. Use SAE 1137 OQT 1300 steel for the shaft. The elements at *B*, *C*, *D*, and *E* are held in position with retaining rings and keys in profile keyseats.
25. The shaft in Figure P12-4 is the drive shaft for a large, bulk material conveyor. The gear receives 40 hp and rotates at 120 rpm. Each chain sprocket delivers 20 hp to one side of the conveyor. Use SAE 1020 cold-drawn steel.
26. The shaft in Figure P12-5 is part of a conveyor drive system that feeds crushed rock into a railroad car. The shaft rotates at 240 rpm and is subjected to moderate shock during operation. All power is input to the gear at *D*. The V-belt sheave at *A* delivers 10.0 hp vertically downward. Chain sprocket *E* delivers 5.0 hp. Note the position of the gear *Q*, which drives gear *D*.
27. Figure P12-6 illustrates an intermediate shaft of a punch press that rotates at 310 rpm while transmitting 20 hp from the V-belt sheave to the gear. The flywheel is not absorbing or giving up any energy at this time. Consider the weight of all elements in the analysis.
28. The shaft in Figure P12-7 is part of a material-handling system aboard a ship. All power comes into the shaft through gear *C*, which rotates at 480 rpm. Gear *A* delivers 30 hp to a hoist. V-belt sheaves *D* and *E* each deliver 10 hp to hydraulic pumps. Use SAE 3140 OQT 1000 steel.
29. The shaft in Figure P12-9 is part of an automatic machining system. All power is input through gear *D*. Gears *C* and *F* drive two separate tool feed devices requiring 5.0 hp each. The V-belt sheave at *B* requires 2.5 hp to drive a coolant pump. The shaft rotates at 220 rpm. All gears are spur gears with 20°, full-depth teeth. Use SAE 1020 cold-drawn steel for the shaft.
30. The shaft in Figure P12-17 is part of a grain-drying system. At *A* is a propeller-type fan that requires 12 hp when rotating at 475 rpm. The fan weighs 34 lb, and its weight should be included in the analysis. The flat-belt pulley at *D* delivers 3.5 hp to a screw conveyor handling the grain. All power comes into the shaft through the V-belt sheave at *C*. Use SAE 1144 cold-drawn steel.

Shafts with both Radial and Axial Loads

Problems P31 through P34 deal with shafts carrying helical gears and wormgears that produce forces directed axially in addition to radial forces.

Torques and Forces Acting Radial and Axial to Shaft	Comprehensive
Figure P12-31: P31—Helical Gear <i>B</i>	P32
Figure P12-33: P33—Wormgear <i>C</i>	P34 (Includes Sheave <i>A</i>)

Forces and Torque on Shafts—Helical Gears

31. See Figure P12-31. The shaft is rotating at 650 rpm, and it receives 7.5 hp through a flexible coupling. The power is delivered to an adjacent shaft through a single helical gear *B* having a normal pressure angle of 20° and a helix angle of 15°. The pitch diameter for the gear is 4.141 in. Review the discussion of forces on helical gears in Chapter 10, and use Equations (12-1), (12-2), (12-4), and (12-5) in this chapter to verify the values for the forces given in the figure. Draw the complete free-body diagrams for the shaft in both the vertical plane and the horizontal plane. Then draw the complete shearing force and bending moment diagrams for the shaft in both planes.
32. Figure P12-31 shows a helical gear mounted on a shaft that rotates at 650 rpm while transmitting 7.5 hp. The gear is also analyzed in Example Problem 10-1, and the tangential, radial, and axial forces on it are shown in the figure. The pitch diameter of the gear is 4.141 in. The power is delivered from the shaft through a flexible coupling at its right end. A spacer is used to position the gear relative to bearing *C*. The thrust load is taken at bearing *A*.

Forces and Torque on Shafts—Wormgears

33. See Figure P12-33. The shaft is rotating at 1750 rpm, and it receives 7.5 hp through a 5.00-in V-belt drive from a mating belt sheave directly below. The power is delivered by a worm with a pitch diameter of 2.00 in. The forces on the worm were used in Design Example 12-3 and are shown in the figure. Review the description of these forces in Chapter 10. Draw the complete free-body diagrams for the shaft in both the vertical plane and the horizontal plane. Then draw the complete shearing force and bending moment diagrams for the shaft in both planes.
34. The shaft shown in Figure P12-33 is the input shaft of a wormgear drive. The V-belt sheave receives 7.5 hp from directly downward. The worm rotates at 1750 rpm and has a pitch diameter of 2.000 in. This is the driving worm

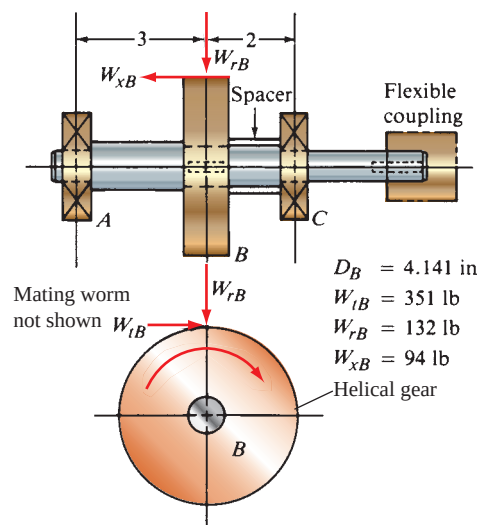


FIGURE P12-31 (Problems 31 and 32)

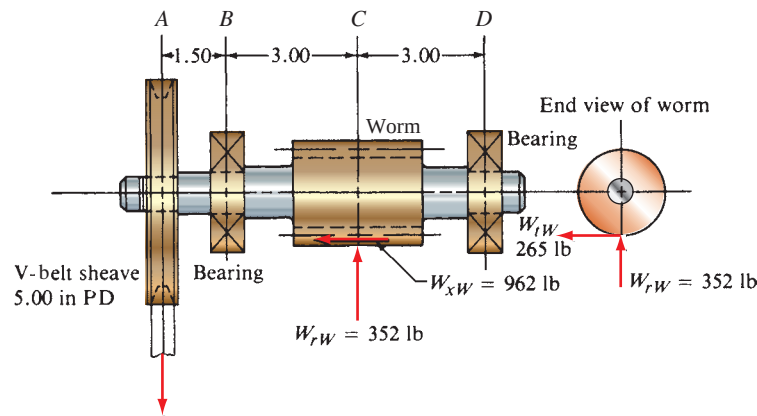


FIGURE P12-33 (Problems 33 and 34)

for the wormgear described in Design Example 12-3. The tangential, radial, and axial forces are shown in the figure. The worm is to be machined integrally with the shaft, and it has a root diameter of 1.614 in. Assume that the geometry of the root area presents a stress concentration factor of 1.5 for bending. Analyze the stress in the area of the worm thread root, and specify a suitable material for the shaft.

Other Comprehensive Design Problems

Problems 35 through 41 contain a variety of loading situations for which the general solution procedure must be adapted. Some of the problems involve more than one shaft, considering shafts for mating gears and multiple reductions.

Figure P12-35: P35—Double-reduction helical drive

Figure P10-8 in Chapter 10: P36—Bevel gear drive

Figure P12-37: P37—Bevel gear drive with two chain sprockets

Figure P12-38: P38—Double-reduction spur gear drive; design three shafts

Figure P12-39: P39—Drive system consisting of an electric motor, a V-belt drive, a double-reduction spur gear-type reducer, and a chain drive

Figure P12-40: P40—Shaft with three spur gears

Figure P12-41: P41—Shaft for windshield wiper mechanism with two levers

35. The double-reduction, helical gear reducer shown in Figure P12-35 transmits 5.0 hp. Shaft 1 is the input, rotating at 1800 rpm and receiving power directly from an electric motor through a flexible coupling. Shaft 2 rotates at 900 rpm. Shaft 3 is the output, rotating at 300 rpm. A chain sprocket is mounted on the output shaft as shown and delivers the power upward. The data for the gears are given in Table 12-5. Each gear has a $14\frac{1}{2}^\circ$ normal pressure angle and a 45° helix angle. The combinations of left- and right-hand helixes are arranged so that the axial forces oppose each other on shaft 2 as shown. Use SAE 4140 OQT 1200 for the shafts.

36. Complete the design of the shafts carrying the bevel gear and pinion shown in Figure 10-8. The forces on the gears, the bearing reactions, and the bending moment diagrams are developed in Example Problems 10-4 and 10-5 and are shown in Figures 10-8 through 10-12. Assume that

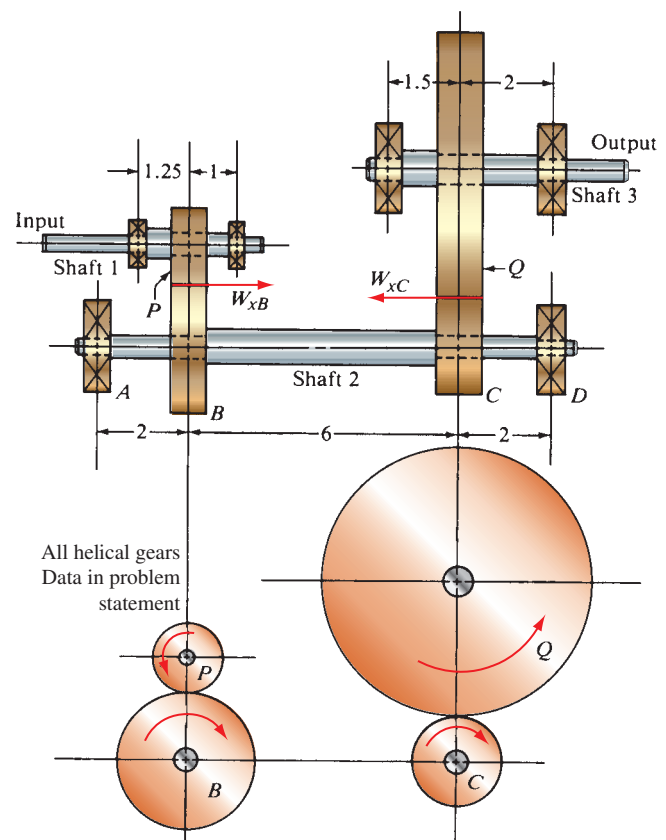


FIGURE P12-35 (Problem 35)

TABLE 12-5

Gear	Diametral pitch	Pitch diameter	Number of teeth	Face width
P	8	1.500 in	12	0.75 in
B	8	3.000 in	24	0.75 in
C	6	2.000 in	12	1.00 in
Q	6	6.000 in	36	1.00 in

the 2.50 hp enters the pinion shaft from the right through a flexible coupling. The power is delivered by way of the lower extension of the gear shaft through another flexible coupling. Use SAE 1040 OQT 1200 for the shafts.

37. The vertical shaft shown in Figure P12–37 is driven at a speed of 600 rpm with 4.0 hp entering through the bevel gear. Each of the two chain sprockets delivers 2.0 hp to the side to drive mixer blades in a chemical reactor vessel. The bevel gear has a diametral pitch of 5, a pitch diameter of 9.000 in, a face width of 1.31 in, and a pressure angle of 20° . Use SAE 4140 OQT 1000 steel for the shaft. See Chapter 10 for the methods for computing the forces on the bevel gear.
38. Figure P12–38 shows a sketch for a double-reduction spur gear train. Shaft 1 rotates at 1725 rpm, driven directly by an electric motor that transmits 15.0 hp to the reducer.

The gears in the train all have 20° , full-depth teeth and the following numbers of teeth and pitch diameters:

Gear A	Gear B
18 teeth	54 teeth
1.80-in dia.	5.40-in dia.
Gear C	Gear D
24 teeth	48 teeth
4.00-in dia.	8.00-in dia.

Note that the speed reduction for each gear pair is proportional to the ratio of the number of teeth. Therefore, shaft 2 rotates at 575 rpm and shaft 3 rotates at 287.5 rpm.

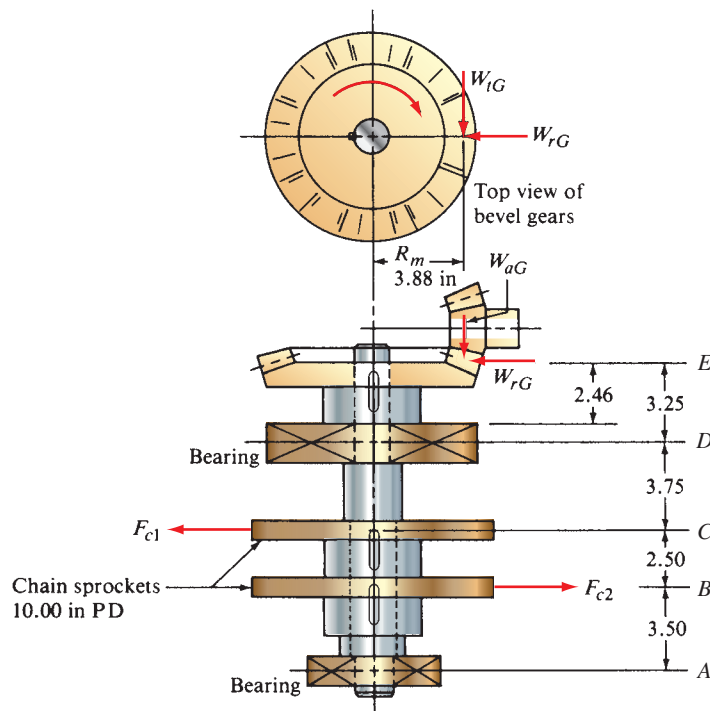


FIGURE P12–37 Bevel gears shown in section for Problem 37. See Chapter 10

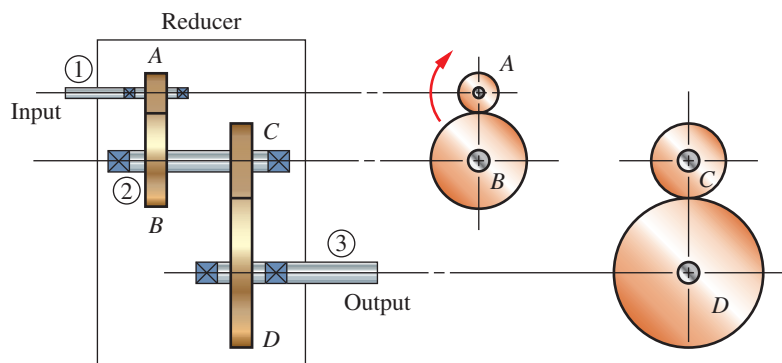


FIGURE P12–38 Gear reducer for Problems 38 and 39

Assume that all shafts carry 15.0 hp. The distance from the middle of each bearing to the middle of the face of the nearest gear is 3.00 in. Shaft 2 has a length of 5.00 in between the two gears, making the total distance between the middle of the two bearings equal to 11.00 in. The extensions of the input and the output shafts carry torque, but no bending loads are exerted on them. Complete the design of all three shafts. Provide for profile keyseats for each gear and sled-runner keyseats at the outboard ends of the input and output shafts. Provide for the location of each gear and bearing on the shaft.

39. Figure P12–39 shows a speed reducer with a V-belt drive delivering power to the input shaft and a chain drive

taking power from the output shaft and delivering it to a conveyor. The drive motor supplies 12.0 hp and rotates 1150 rpm. The speed reductions of the V-belt drive and the chain drive are proportional to the ratio of the diameters of the driving and driven sheaves or sprockets. The arrangement of the gears in the reducer is the same as that described in Figure P12–38 and Problem 38. Determine the forces applied to the motor shaft, to each of the three shafts of the reducer, and to the conveyor driveshaft. Then complete the design of the three shafts of the reducer, assuming that all shafts transmit 12.0 hp.

40. Figure P12–40 shows a drive for a system to crush coal and deliver it by conveyor to a railroad car. Gear A delivers

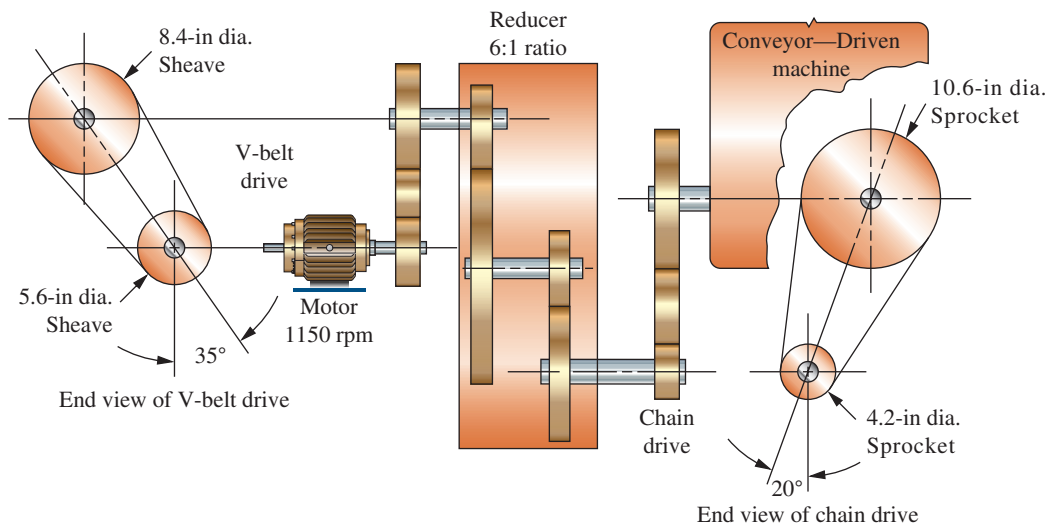


FIGURE P12–39 Drive system for Problem 39

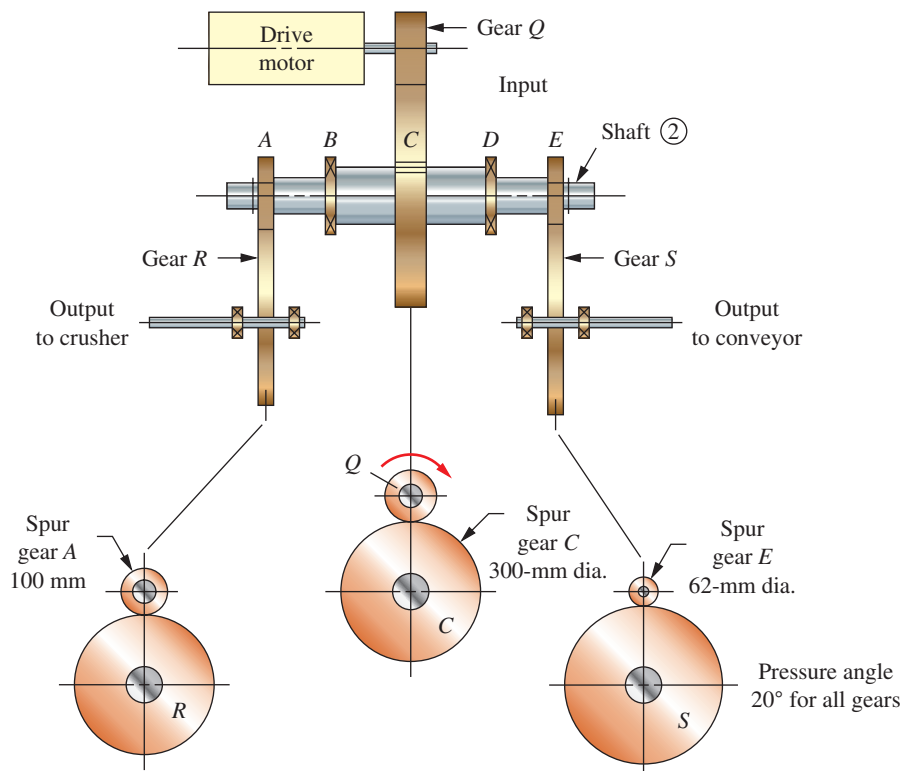


FIGURE P12–40 Drive system for Problem 40

15 kW to the crusher and gear E delivers 7.5 kW to the conveyor. All power enters the shaft through gear C . The shaft carrying gears A , C , and E rotates at 480 rpm. Design that shaft. The distance from the middle of each bearing to the middle of the face of the nearest gear is 100 mm.

41. A shaft is to be designed for a linkage of a windshield wiper mechanism for a truck (see Figure P12–41). A 20-N

force is applied to lever 1 by an adjacent link. The reaction force, F_2 , on lever 2 is transmitted to another link. The distance, d , between elements is 20.0 mm. Bearings A and C are straight, cylindrical, bronze bushings, 10.0 mm long. Design the shaft and levers 1 and 2.

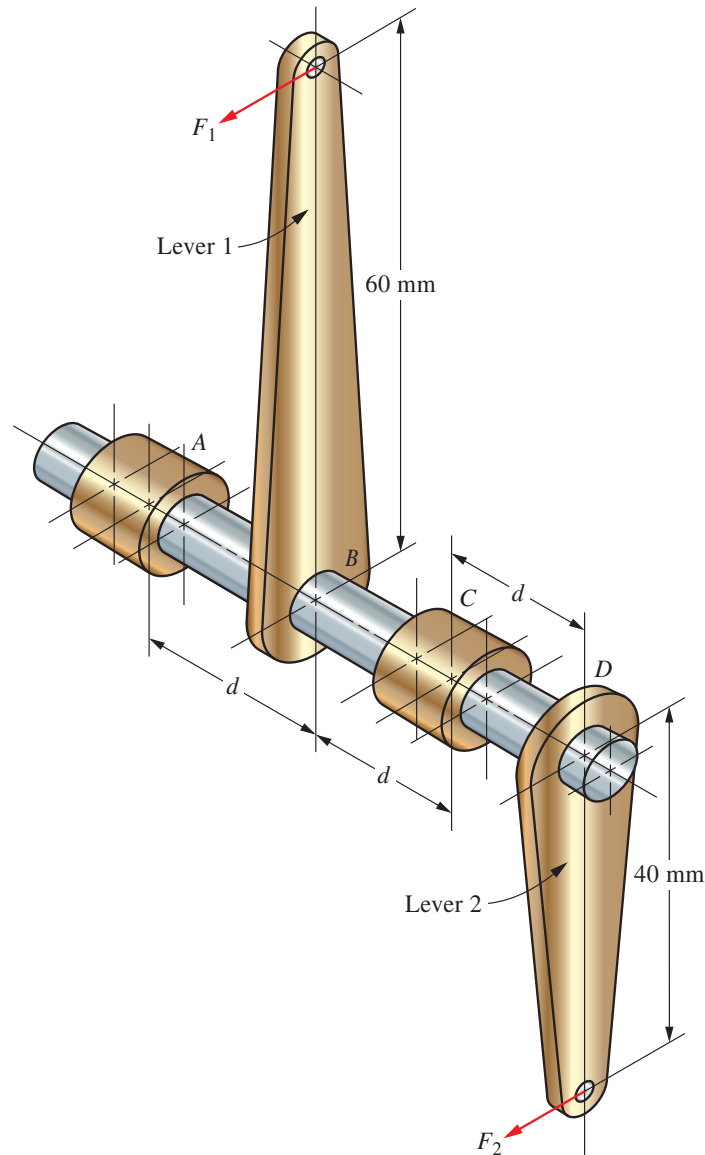


FIGURE P12–41 Shaft and levers for windshield wiper system for Problem 41

TOLERANCES AND FITS

The Big Picture

You Are the Designer

- 13-1 Objectives of This Chapter
- 13-2 Factors Affecting Tolerances and Fits
- 13-3 Tolerances, Production Processes, and Cost
- 13-4 Preferred Basic Sizes
- 13-5 Clearance Fits
- 13-6 Interference Fits
- 13-7 Transition Fits
- 13-8 Stresses for Force Fits
- 13-9 General Tolerancing Methods
- 13-10 Robust Product Design

THE BIG PICTURE

Tolerances and Fits

Discussion Map

- A *tolerance* is the permissible variation on key dimensions of mechanical parts. You, as a designer, must specify the tolerance for each dimension, considering how it will perform and how it will be manufactured.
- Two or more parts may be assembled together either with a *clearance fit*, allowing free relative motion, or with an *interference fit*, where the two parts must be pressed together and therefore will not move during operation of the device.

This chapter will help you acquire the competency to specify suitable fits for mating parts and the dimensional tolerances that will produce the desired fit.

The objective of most of the analysis procedures discussed in this book is to determine the minimum acceptable geometric size at which a component is safe and performs properly under specified conditions. As a designer, you must also then specify the final dimensions for the components, including the tolerances on those dimensions.

The term *tolerance* refers to the permissible deviation of a dimension from the specified basic size. The proper performance of a machine can depend on the tolerances specified for its parts, particularly those that must fit together for location or for suitable relative motion.

The term *fit* usually refers to the *clearances* that are permissible between mating parts in a mechanical device that must assemble easily and that must often move relative to each other during normal operation of the device. Such fits are usually called *running* or *sliding fits*. *Fit* can also refer to the amount of *interference*

Discover

Identify examples of products for which precise dimensional tolerances would be appropriate and some for which looser tolerances are permitted.

Look for examples of mechanical parts that have clearance fits and some that have interference fits.

Describe the reasons why the designer may have made the design in this way.

that exists when the inside part should be larger than the outside part. Interference fits are used to ensure that mating parts do not move relative to each other.

Look for examples of parts for mechanical devices that have clearance fits. Any machine having shafts that rotate in plain surface bearings should have such parts. This type of bearing is called a *journal bearing*, and there must be a small but reliable clearance between the shaft and the bearing to permit smooth rotation of the shaft. But the clearance cannot be too large, or the operation of the machine will appear to be too crude and rough.

Consider also any assembly having hinges that allow one part to rotate relative to the other, such as an access door or panel or the cover of a container. The hinge parts will have a clearance fit. Many types of measuring equipment, such as calipers, dial indicators, and electronic probes, that have moving parts

must be designed carefully to maintain the expected precision of the measurements while allowing reliable motion. On the other end of the spectrum is the rather loose fit that is typical of toys and some other types of recreation equipment. The fit of the wheel of a child's wagon onto its axle is typically quite large to allow free rotation and to permit easy assembly. We will show in this chapter that the wide range of fits that are encountered in practice are designed by specifying a *class of fit* and then determining the allowable range of key dimensions of mating parts.

Look also for examples where two parts are assembled so that they cannot move relative to one another. They are held tightly together because the inside part is larger than the outside part. Perhaps some parts of your car are like that.

This chapter will help you design parts that must have clearance or interference fits. When an interference fit occurs, it is often desirable to predict the level of stress to which the mating parts are subjected. That topic will also be covered in this chapter.

YOU ARE THE DESIGNER

Assume that you are responsible for the design of the gear-type speed reducer discussed in Chapter 1 and sketched in Figure 1–12. The sketch is repeated here as Figure 13–1. The input and output shafts each carry one gear while being supported on two bearings mounted in the housing. The gears are keyed to the shafts to permit the transmission of torque from the shaft to the gear, or vice versa. The material in Chapter 12 will help you to design the shafts themselves. Chapter 14 discusses bearing selection and application. The results of these design decisions include the calculation of the minimum acceptable diameter for the shaft at any section and the specification of suitable bearings to carry the loads applied with a reasonable life. The bearing manufacturer will specify the tolerances on the various dimensions of the bearing. Your job is to specify the final dimensions for the shaft at all points, including the tolerances on those dimensions.

Consider the part of the input shaft where the first gear in the train is mounted. What is a convenient nominal size to specify for the shaft diameter at the location of the gear? Do you want to be able to slide the gear easily over the left end of the shaft and up against the shoulder that locates it? If so, how much clearance should be allowed between the shaft and the bore of the gear to ensure ease of assembly while still providing accurate location of the gear and smooth operation? When the shaft is machined, what range of dimensions will you allow the manufacturing staff to produce? What surface finish should be specified for the shaft, and what manufacturing process is required to produce that finish? What is the relative cost of the manufacturing operation? Similar questions must be answered for the bore of the gear.

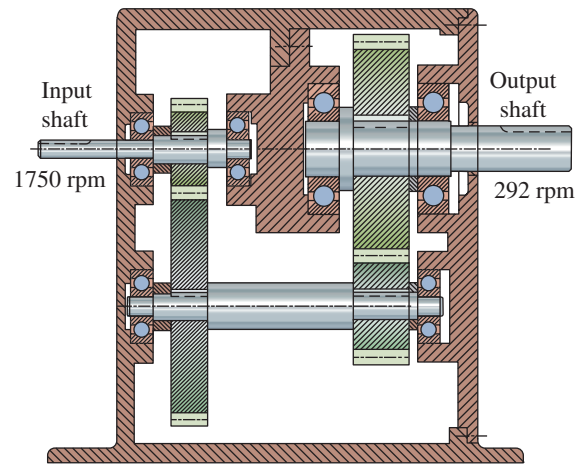


FIGURE 13–1 Conceptual design for a speed reducer

Rolling contact bearings, such as the familiar ball bearing, are designed to be installed on the shaft with an *interference fit*. That is, the inside diameter of the bearing is smaller than the outside diameter of the shaft where the bearing is to be seated. A significant force is required to press the bearing onto the shaft. What dimensions do you specify for the shaft at the bearing seat? How much interference should be specified? How much stress is produced in the shaft because of the interference fit? As the designer, you must answer these questions.

13-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Define the terms *tolerance*, *allowance*, *unilateral tolerance*, and *bilateral tolerance*.
2. Describe the relationships among tolerances, production processes, and cost.
3. Specify basic sizes for dimensions according to a set of preferred sizes.
4. Use the ANSI Standard B4.1, *Preferred Limits and Fits for Cylindrical Parts*, and ANSI B4.2 *Preferred Metric Limits and Fits* to specify tolerances, fits, and clearances.
5. Specify transitional, interference, and force fits.

6. Compute the pressure created between parts subjected to interference fits and the resulting stresses in the mating members.
7. Use a spreadsheet to assist in the calculation of stresses for interference fits.
8. Specify appropriate geometric dimensions and tolerancing controls for mating parts.

13-2 FACTORS AFFECTING TOLERANCES AND FITS

Consider the plain surface bearings designed in Chapter 16. A critical part of the design is the specification of the diametral clearance between the journal and the bearing, as illustrated in Figure 16–1. The typical value is just a few

thousandths of an inch. But some variation must be allowed on both the journal outside diameter and the bearing inside diameter for reasons of economy of manufacture. Thus, there will be a variation of the actual clearance in production devices, depending on where the individual mating components fall within their own tolerance bands. Such variations must be accounted for in the analysis of the bearing performance. Too small a clearance could cause seizing. Conversely, too large a clearance would reduce the precision of the machine and adversely affect the lubrication.

The mounting of power-transmission elements onto shafts is another situation in which tolerances and fits must be considered. A chain sprocket for general-purpose mechanical drives is usually produced with a bore that will easily slide into position on the shaft during assembly. But once in place, it will transmit power smoothly and quietly only if it is not excessively loose. A high-speed turbine rotor must be installed on its shaft with an interference fit, eliminating any looseness that would cause vibrations at the high rotational speeds.

When relative motion between two parts is required, a clearance fit is necessary. But here, too, there are differences. Some measuring instruments have parts that must move with no perceptible looseness (sometimes called *play*) between mating parts that would adversely affect the accuracy of measurement. An idler sheave in a belt drive system must rotate on its shaft reliably, without the tendency to seize, but with only a small amount of play. The requirement for the mounting of a wheel of a child's wagon on its axle is much different. A loose clearance fit is satisfactory for the expected use of the wheel, and it permits wide tolerances on the wheel bore and on the axle diameter for economy.

13-3 TOLERANCES, PRODUCTION PROCESSES, AND COST

A *unilateral tolerance* deviates in only one direction from the basic size. A *bilateral tolerance* deviates both above and below the basic size. The *total tolerance* is the difference between the maximum and minimum permissible dimensions.

The term *allowance* refers to an intentional difference between the maximum material limits of mating parts. For example, a positive allowance for a hole/shaft pair would define the minimum *clearance* between the mating parts from the largest shaft mating with the smallest hole. A negative allowance would result in the shaft being larger than the hole (*interference*).

The term *fit* refers to the relative looseness (clearance fit) or tightness (interference fit) of mating parts, especially as it affects the motion of the parts or the force between them after assembly. Specifying the degree of clearance or interference is one of the tasks of the designer.

It is costly to produce components with very small tolerances on dimensions. It is the designer's responsibility to set the tolerances at the highest possible level that results in satisfactory operation of the machine. Certainly judgment and experience must be exercised in

such a process. In quantity production situations, it may be cost-effective to test prototypes with a range of tolerances to observe the limits of acceptable performance.

In general, the production of parts with small tolerances on their dimensions requires multiple processing steps. A shaft may first have to be turned on a lathe and then ground to produce the final dimensions and surface finish. In extreme cases, lapping may be required. Each subsequent step in the manufacture adds cost. Even if different operations are not required, the holding of small tolerances on a single machine, such as a lathe, may require several passes, ending with a fine-finish cut. Cutting tool changes must be more frequent, also, as wear of the tool takes the part out of tolerance.

The production of part features with small tolerances usually involves finer surface finishes as well. Figure 13-2 shows the general relationship between the surface finish and the relative cost of producing a part. The typical total tolerance produced by the processes described is included in the figure. The increase in cost is dramatic for the small tolerances and fine finishes.

Figure 13-3 presents the relationship between the surface finish and the machining operations available to produce it.

The basic reference for tolerances and fits in the United States is the ANSI Standard B4.1-1967, *Preferred Limits and Fits for Cylindrical Parts*. Metric dimensions should conform to ANSI B4.2-1978, *Preferred Metric Limits and Fits*. The most recent version should be consulted. (See References 1 and 2.)

The International Organization for Standardization (ISO) gives metric data for limits and fits in ISO 286-1:1988, which is used in Europe and many other countries.

The term *tolerance grade* refers to a set of tolerances that can be produced with an approximately equal production capability. The actual total tolerance allowed within each grade depends on the nominal size of the dimension. Smaller tolerances can be achieved for smaller dimensions, and vice versa. Standards ISO R286 and ANSI B4.1 include complete data for tolerance grades from 01 through 16, as shown in Table 13-1. Tolerances are smaller for the smaller grade number.

A sampling of tolerance data for machined parts for a few grades and ranges of size is given in Tables 13-2 and 13-2M. Figure 13-2 shows the capability of selected manufacturing processes to produce work within the given tolerance grades. Reference 8 includes numerous tables of data for tolerance grades, limits, and fits.

Some projects may benefit from the ability to compute the amount of tolerance for a given basic dimension to achieve a particular IT grade. The following equation applies to metric dimensions.

✦ Tolerance for IT Grade

$$T = [0.045 \times (D)^{1/3} + 0.001 \times (D)][10^{0.2(ITG-1)}] \quad (13-1)$$

where

T = tolerance in mm

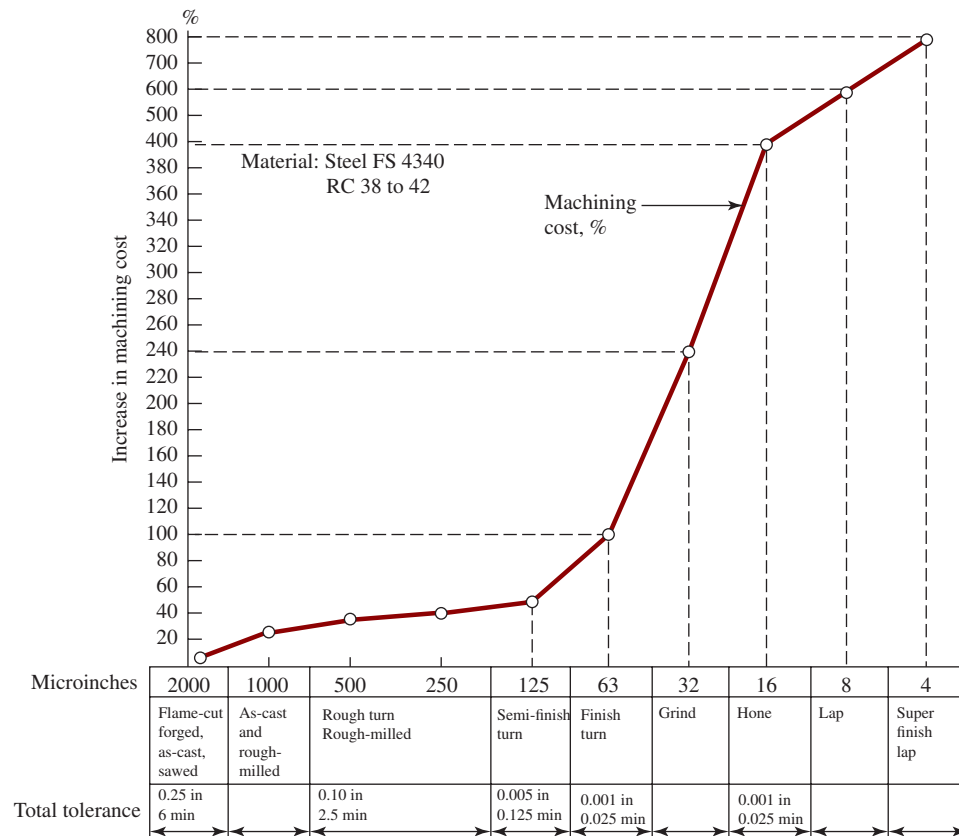


FIGURE 13-2 Machining costs versus surface finish specified

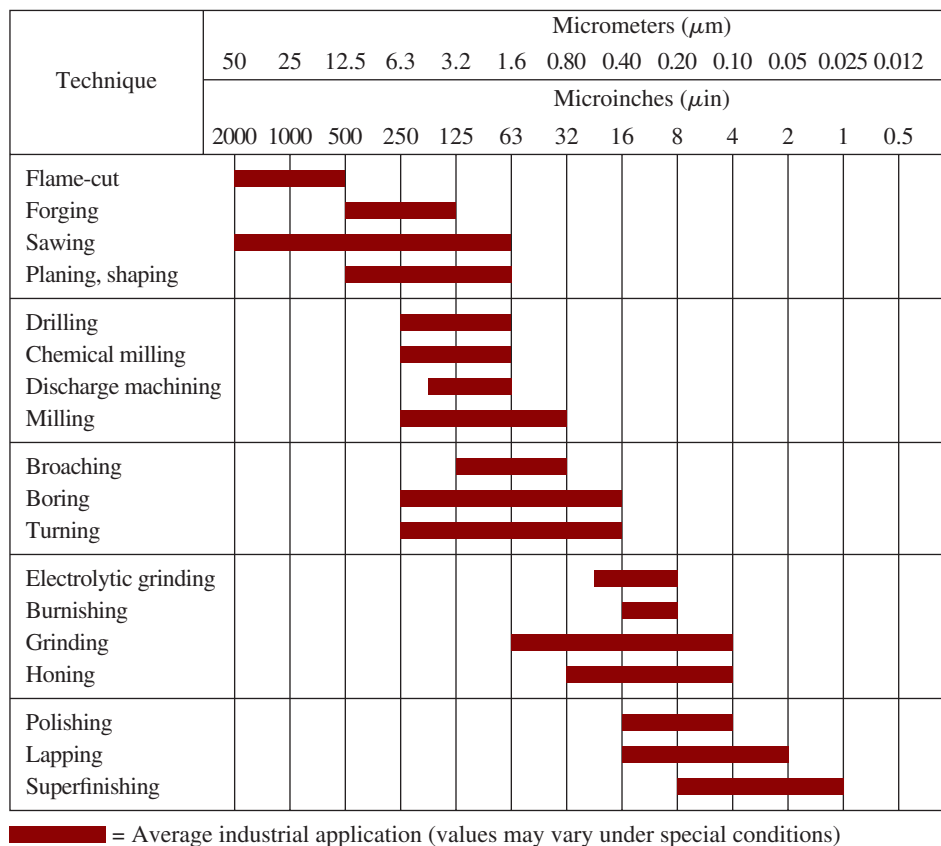
FIGURE 13-3 Finishes produced by various techniques (roughness average, R_a)

TABLE 13-1 Tolerance Grades

Application	Tolerance grades								
Measuring tools	01	0	1	2	3	4	5	6	7
Fits of machined parts	4	5	6	7	8	9	10	11	
Material, as supplied	8	9	10	11	12	13	14		
Rough forms (casting, sawing, forging, etc.)	12	13	14	15	16				

TABLE 13-2 Tolerances for Some Tolerance Grades

Nominal size (in)	Tolerance grade							
	4	5	6	7	8	9	10	11
	Tolerances in thousandths of an inch							
0.24–0.40	0.15	0.25	0.4	0.6	0.9	1.4	2.2	3.5
0.40–0.71	0.2	0.3	0.4	0.7	1.0	1.6	2.8	4.0
0.71–1.19	0.25	0.4	0.5	0.8	1.2	2.0	3.5	5.0
1.19–1.97	0.3	0.4	0.6	1.0	1.6	2.5	4.0	6.0
1.97–3.15	0.3	0.5	0.7	1.2	1.8	3.0	4.5	7.0
3.15–4.73	0.4	0.6	0.9	1.4	2.2	3.5	5.0	9.0
4.73–7.09	0.5	0.7	1.0	1.6	2.5	4.0	6.0	10.0

TABLE 13-2M Metric Tolerances for Some Tolerance Grades

Range of nominal size (mm)	Tolerance grade								Mean dimension (Reference) (mm)
	4	5	6	7	8	9	10	11	
	Tolerances in millimeters (mm)								
6–10	0.0036	0.0057	0.009	0.014	0.023	0.036	0.057	0.091	8
10–18	0.0044	0.0069	0.011	0.017	0.028	0.044	0.069	0.110	14
18–30	0.0053	0.0083	0.013	0.021	0.033	0.053	0.083	0.132	24
30–50	0.0063	0.010	0.016	0.025	0.040	0.063	0.100	0.158	40
50–80	0.0075	0.012	0.019	0.030	0.047	0.075	0.118	0.187	65
80–120	0.0087	0.014	0.022	0.035	0.055	0.087	0.138	0.219	100
120–180	0.0101	0.016	0.025	0.040	0.064	0.101	0.160	0.254	150

D = mean dimension in mm of the range of nominal sizes in Table 13-2M

ITG = International Tolerance Grade number (an integer from 1 to 14)

13-4 PREFERRED BASIC SIZES

The first step in specifying a dimension for a part is to decide on the basic size—that dimension to which the tolerances are applied. The analysis for strength, deflection, or performance of the part determines the nominal

or minimum size required. Unless special conditions exist, the basic size should be chosen from the lists of preferred basic sizes in Table A2-1 for fractional-inch sizes, decimal-inch sizes, and metric sizes from the SI. If possible, select from the first-choice column. If a size between two first-choice sizes is required, the second-choice column should be used. This will limit the number of sizes typically encountered in the manufacture of products and will lead to cost-effective standardization. The choice of system depends on the policies of the company and the market for the product.

13-5 CLEARANCE FITS

When there must always be a clearance between mating parts, a clearance fit is specified. The designation for standard clearance fits from ANSI/ASME Standard B4.1 for members that must move together is the *running* or *sliding clearance fit* (RC). Within this standard, there are nine classes, RC1 through RC9, with RC1 providing the smallest clearance and RC9 the largest. The following descriptions for the individual members of this class should help you to decide which is most appropriate for a given application.

RC1 (close sliding fit): Accurate location of parts that must assemble without perceptible play.

RC2 (sliding fit): Parts that will move and turn easily but are not intended to run freely. Parts may seize with small temperature changes, especially in the larger sizes.

RC3 (precision running fit): Precision parts operating at slow speeds and light loads that must run freely. Changes in temperature may cause difficulties.

RC4 (close running fit): Accurate location with minimum play for use under moderate

loads and speeds. A good choice for accurate machinery.

RC5 (medium running fit): Accurate machine parts for higher speeds and/or loads than for RC4.

RC6 (medium running fit): Similar to RC5 for applications in which larger clearance is desired.

RC7 (free running fit): Reliable relative motion under wide temperature variations in applications where accuracy is not critical.

RC8 (loose running fit): Permits large clearances, allowing the use of parts with commercial, “as-received” tolerances.

RC9 (loose running fit): Similar to RC8, with approximately 50% larger clearances.

The complete standard ANSI/ASME B4.1 lists the tolerances on the mating parts and the resulting limits of clearances for all nine classes and for sizes from 0 to 200 in. (See also References 4, 6, and 8.) Table 13–3 is abstracted from the standard. Let RC2 represent the precision fits (RC1, RC2, and RC3); let RC5 represent

TABLE 13–3 Clearance Fits (RC)

Nominal size range (in)	Class RC2			Class RC5			Class RC8			Nominal size range (in)
	Limits of clearance	Standard limits		Limits of clearance	Standard limits		Limits of clearance	Standard limits		
		Hole	Shaft		Hole	Shaft		Hole	Shaft	
Over To										Over To
0–0.12	0.1	+0.25	−0.1	0.6	+0.6	−0.6	2.5	+1.6	−2.5	0–0.12
	0.55	0	−0.3	1.6	−0	−1.0	5.1	0	−3.5	
0.12–0.24	0.15	+0.3	−0.15	0.8	+0.7	−0.8	2.8	+1.8	−2.8	0.12–0.24
	0.65	0	−0.35	2.0	−0	−1.3	5.8	0	−4.0	
0.24–0.40	0.2	+0.4	−0.2	1.0	+0.9	−1.0	3.0	+2.2	−3.0	0.24–0.40
	0.85	0	−0.45	2.5	−0	−1.6	6.6	0	−4.4	
0.40–0.71	0.25	+0.4	−0.25	1.2	+1.0	−1.2	3.5	+2.8	−3.5	0.40–0.71
	0.95	0	−0.55	2.9	−0	−1.9	7.9	0	−5.1	
0.71–1.19	0.3	+0.5	−0.3	1.6	+1.2	−1.6	4.5	+3.5	−4.5	0.71–1.19
	1.2	0	−0.7	3.6	−0	−2.4	10.0	0	−6.5	
1.19–1.97	0.4	+0.6	−0.4	2.0	+1.6	−2.0	5.0	+4.0	−5.0	1.19–1.97
	1.4	0	−0.8	4.6	−0	−3.0	11.5	0	−7.5	
1.97–3.15	0.4	+0.7	−0.4	2.5	+1.8	−2.5	6.0	+4.5	−6.0	1.97–3.15
	1.6	0	−0.9	5.5	−0	−3.7	13.5	0	−9.0	
3.15–4.73	0.5	+0.9	−0.5	3.0	+2.2	−3.0	7.0	+5.0	−7.0	3.15–4.73
	2.0	0	−1.1	6.6	−0	−4.4	15.5	0	−10.5	
4.73–7.09	0.6	+1.0	−0.6	3.5	+2.5	−3.5	8.0	+6.0	−8.0	4.73–7.09
	2.3	0	−1.3	7.6	−0	−5.1	18.0	0	−12.0	
7.09–9.85	0.6	+1.2	−0.6	4.0	+2.8	−4.0	10.0	+7.0	−10.0	7.09–9.85
	2.6	0	−1.4	8.6	−0	−5.8	21.5	0	−14.5	
9.85–12.41	0.7	+1.2	−0.7	5.0	+3.0	−5.0	12.0	+8.0	−12.0	9.85–12.41
	2.8	0	−1.6	10.0	−0	−7.0	25.0	0	−17.0	

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Note: Limits are in thousandths of an inch.

the accurate, reliable running fits (RC4 to RC7); and let RC8 represent the loose fits (RC8 and RC9).

The numbers in Table 13–3 are in thousandths of an inch. Thus, a clearance of 2.8 from the table means a difference in size between the inside and outside parts of 0.0028 in. The tolerances on the hole and the shaft are to be applied to the basic size to determine the limits of size for that dimension.

In applying tolerances and the allowance to the dimensions of mating parts, designers use the basic hole system and the basic shaft system. In the *basic hole system*, the design size of the hole is the basic size, and the allowance is applied to the shaft; the basic size is the minimum size of the hole. In the *basic shaft system*, the design

size of the shaft is the basic size, and the allowance is applied to the hole; the basic size is the maximum size of the shaft. The basic hole system is preferred.

Figure 13–4 shows a graphical display of the tolerances and fits for all nine RC classes when applied to a shaft/hole combination in which the basic size is 2.000 in and the basic hole system is used. Note that such a diagram shows the total tolerance on both the shaft and the hole. The tolerance for the hole always starts at the basic size, while the shaft tolerance is offset below the basic size to provide for the minimum clearance (the smallest hole combined with the largest shaft). The maximum clearance combines the largest hole with the smallest shaft. This figure also shows the dramatic

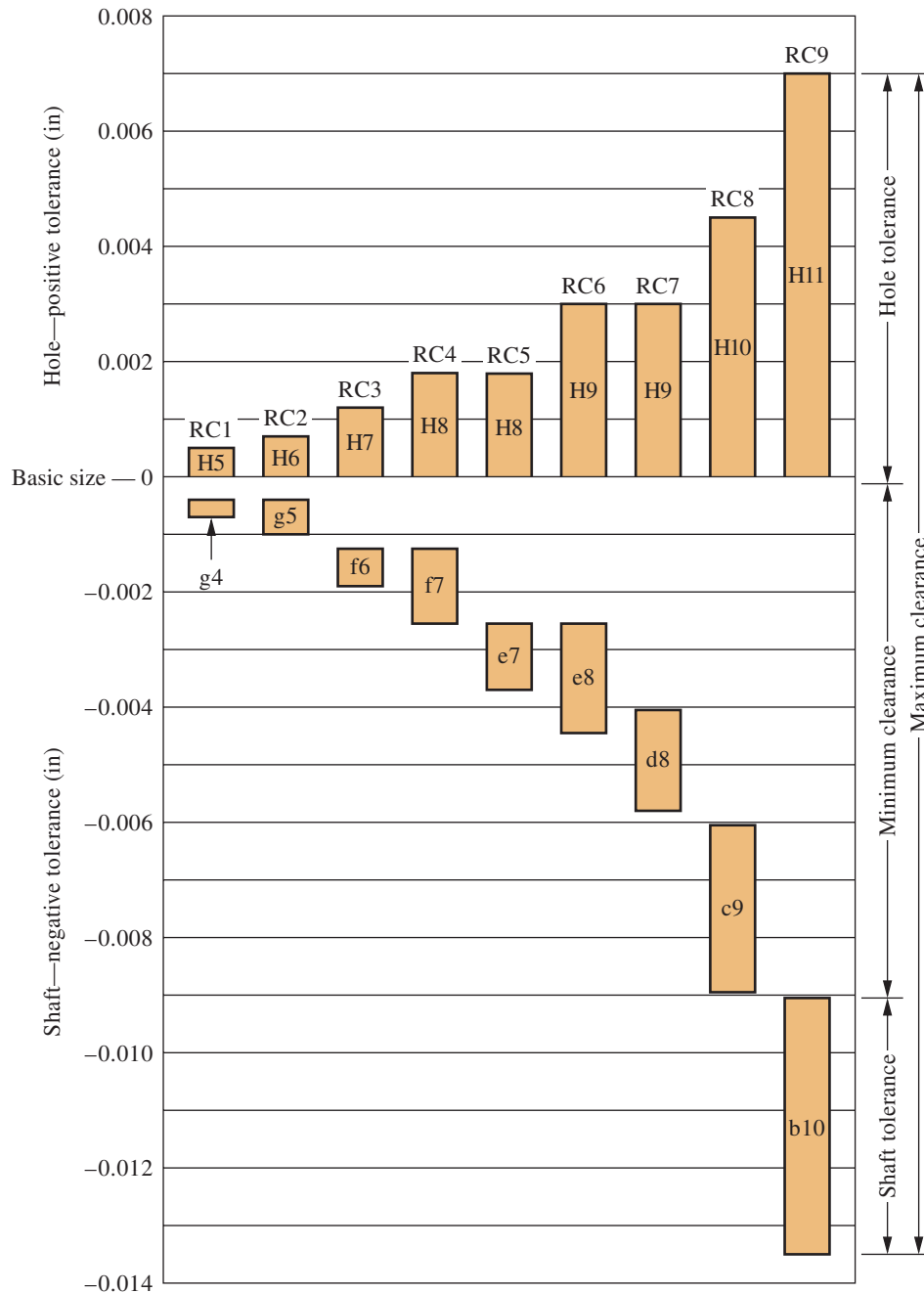


FIGURE 13–4 RC fits for basic hole size of 2.00 in showing tolerances for hole and shaft, minimum clearance, and maximum clearance

range of clearances provided by the nine classes within the RC system.

The codes within the tolerance bars in Figure 13–4 refer to the tolerance grades mentioned earlier. The capital H combined with a tolerance grade number is used for the hole in the basic hole system for which there is no fundamental deviation from the basic size. The lowercase letters in the shaft tolerance bars indicate some fundamental deviation of the shaft size from the basic size. Then the tolerance is added to the fundamental

deviation. The size of the tolerance is indicated by the number.

ISO standard R286 for limits and fits also uses the letter/number codes. For example, a specification for a shaft/hole combination with a basic size of 50 mm that is to provide a free running fit (similar to RC7) could be defined on a drawing as

Hole: $\varnothing 50$ H9 Shaft: $\varnothing 50$ d8

No tolerance values need to be given on the drawing.

Example Problem 13–1

A shaft carrying an idler sheave for a belt drive system is to have a nominal size of 2.00 in. The sheave must rotate reliably on the shaft, but with the smoothness characteristic of accurate machinery. Specify the limits of size for the shaft and the sheave bore, and list the limits of clearance that will result. Use the basic hole system.

Solution

An RC5 fit should be satisfactory in this application. From Table 13–3, the hole tolerance limits are +1.8 and –0. The sheave hole then should be within the following limits:

Sheave Hole

$$2.0000 + 0.0018 = 2.0018 \text{ in (largest)}$$

$$2.0000 - 0.0000 = 2.0000 \text{ in (smallest)}$$

Notice that the smallest hole is the basic size.

The shaft tolerance limits are –2.5 and –3.7. The resulting size limits are as follows:

Shaft Diameter

$$2.0000 - 0.0025 = 1.9975 \text{ in (largest)}$$

$$2.0000 - 0.0037 = 1.9963 \text{ in (smallest)}$$

Figure 13–5 illustrates these results.

Combining the smallest shaft with the largest hole gives the largest clearance. Conversely, combining the largest shaft with the smallest hole gives the smallest clearance. Therefore, the limits of clearance are

$$2.0018 - 1.9963 = 0.0055 \text{ in (largest)}$$

$$2.0000 - 1.9975 = 0.0025 \text{ in (smallest)}$$

These values check with the limits of clearance in Table 13–3. Notice that the total tolerance for the shaft is 0.0012 in, and for the hole 0.0018 in, both relatively small values.

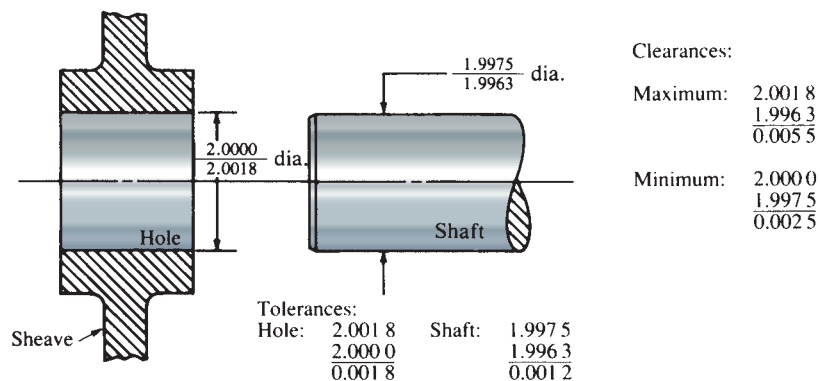


FIGURE 13–5 An RC5 fit using the basic hole system

Locational Clearance Fits

Another clearance fit system is available for parts for which control of the location is desired although the parts will normally not move in relation to each other in operation. Called *locational clearance (LC)* fits, they include 11 classes. The first four, LC1 to LC4, have a zero clearance

(size to size) as the lower limit of the fit, regardless of the size or class. The upper limit of fit increases with both the size of the parts and the class number. Classes LC5 through LC11 provide some positive clearance for all sizes, increasing with the size of the parts and with the class. Numerical values for the tolerances and fits for these classes have been published. (See References 1, 2, 4, 6, and 8.)

13-6 INTERFERENCE FITS

Interference fits are those in which the inside member is larger than the outside member, requiring the application of force during assembly. There is some deformation of the parts after assembly, and a pressure exists at the mating surfaces.

Force fits are designed to provide a controlled pressure between mating parts throughout the range of sizes for a given class. They are used where forces or torques are transmitted through the joint. Instead of being assembled through the application of force, similar fits are obtained by *shrink fitting*, in which one member is

heated to expand it while the other remains cool. The parts are then assembled with little or no force. After cooling, the same dimensional interference exists as for the force fit. *Locational interference fits* are used for location only. There is no movement between parts after assembly, but there is no special requirement for the resulting pressure between mating parts.

Force Fits (FN)

As shown in Table 13-4, five classes of force fits are defined in ANSI/ASME Standard B4.1. (See References 1, 2, 4, 6, and 8.)

TABLE 13-4 Force and Shrink Fits (FN)

Nominal size range (in)	Class FN1			Class FN2			Class FN3			Class FN4			Class FN5		
	Limits of interference	Standard limits		Limits of interference	Standard limits		Limits of interference	Standard limits		Limits of interference	Standard limits		Limits of interference	Standard limits	
		Hole	Shaft		Hole	Shaft		Hole	Shaft		Hole	Shaft		Hole	Shaft
0-0.12	0.05	+0.25	+0.5	0.2	+0.4	+0.85				0.3	+0.4	+0.95	0.3	+0.6	+1.3
	0.5	-0	+0.3	0.85	-0	+0.6				0.95	-0	+0.7	1.3	-0	+0.9
0.12-0.24	0.1	+0.3	+0.6	0.2	+0.5	+1.0				0.4	+0.5	+1.2	0.5	+0.7	+1.7
	0.6	-0	+0.4	1.0	-0	+0.7				1.2	-0	+0.9	1.7	-0	+1.2
0.24-0.40	0.1	+0.4	+0.75	0.4	+0.6	+1.4				0.6	+0.6	+1.6	0.5	+0.9	+2.0
	0.75	-0	+0.5	1.4	-0	+1.0				1.6	-0	+1.2	2.0	-0	+1.4
0.40-0.56	0.1	+0.4	+0.8	0.5	+0.7	+1.6				0.7	+0.7	+1.8	0.6	+1.0	+2.3
	0.8	-0	+0.5	1.6	-0	+1.2				1.8	-0	+1.4	2.3	-0	+1.6
0.56-0.71	0.2	+0.4	+0.9	0.5	+0.7	+1.6				0.7	+0.7	+1.8	0.8	+1.0	+2.5
	0.9	-0	+0.6	1.6	-0	+1.2				1.8	-0	+1.4	2.5	-0	+1.8
0.71-0.95	0.2	+0.5	+1.1	0.6	+0.8	+1.9				0.8	+0.8	+2.1	1.0	+1.2	+3.0
	1.1	-0	+0.7	1.9	-0	+1.4				2.1	-0	+1.6	3.0	-0	+2.2
0.95-1.19	0.3	+0.5	+1.2	0.6	+0.8	+1.9	0.8	+0.8	+2.1	1.0	+0.8	+2.3	1.3	+1.2	+3.3
	1.2	-0	+0.8	1.9	-0	+1.4	2.1	-0	+1.6	2.3	-0	+1.8	3.3	-0	+2.5
1.19-1.58	0.3	+0.6	+1.3	0.8	+1.0	+2.4	1.0	+1.0	+2.6	1.5	+1.0	+3.1	1.4	+1.6	+4.0
	1.3	-0	+0.9	2.4	-0	+1.8	2.6	-0	+2.0	3.1	-0	+2.5	4.0	-0	+3.0
1.58-1.97	0.4	+0.6	+1.4	0.8	+1.0	+2.4	1.2	+1.0	+2.8	1.8	+1.0	+3.4	2.4	+1.6	+5.0
	1.4	-0	+1.0	2.4	-0	+1.8	2.8	-0	+2.2	3.4	-0	+2.8	5.0	-0	+4.0
1.97-2.56	0.6	+0.7	+1.8	0.8	+1.2	+2.7	1.3	+1.2	+3.2	2.3	+1.2	+4.2	3.2	+1.8	+6.2
	1.8	-0	+1.3	2.7	-0	+2.0	3.2	-0	+2.5	4.2	-0	+3.5	6.2	-0	+5.0
2.56-3.15	0.7	+0.7	+1.9	1.0	+1.2	+2.9	1.8	+1.2	+3.7	2.8	+1.2	+4.7	4.2	+1.8	+7.2
	1.9	-0	+1.4	2.9	-0	+2.2	3.7	-0	+3.0	4.7	-0	+4.0	7.2	-0	+6.0
3.15-3.94	0.9	+0.9	+2.4	1.4	+1.4	+3.7	2.1	+1.4	+4.4	3.6	+1.4	+5.9	4.8	+2.2	+8.4
	2.4	-0	+1.8	3.7	-0	+2.8	4.4	-0	+3.5	5.9	-0	+5.0	8.4	-0	+7.0
3.94-4.73	1.1	+0.9	+2.6	1.6	+1.4	+3.9	2.6	+1.4	+4.9	4.6	+1.4	+6.9	5.8	+2.2	+9.4
	2.6	-0	+2.0	3.9	-0	+3.0	4.9	-0	+4.0	6.9	-0	+6.0	9.4	-0	+8.0
4.73-5.52	1.2	+1.0	+2.9	1.9	+1.6	+4.5	3.4	+1.6	+6.0	5.4	+1.6	+8.0	7.5	+2.5	+11.6
	2.9	-0	+2.2	4.5	-0	+3.5	6.0	-0	+5.0	8.0	-0	+7.0	11.6	-0	+10.0
5.52-6.30	1.5	+1.0	+3.2	2.4	+1.6	+5.0	3.4	+1.6	+6.0	5.4	+1.6	+8.0	9.5	+2.5	+13.6
	3.2	-0	+2.5	5.0	-0	+4.0	6.0	-0	+5.0	8.0	-0	+7.0	13.6	-0	+12.0

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Note: Limits are in thousandths of an inch.

TABLE 13-5 Coefficient of Thermal Expansion

Material	Coefficient of thermal expansion, α	
	in/in · °F	mm/mm · °C
Steel:		
SAE 1020	6.5×10^{-6}	11.7×10^{-6}
SAE 1050	6.1×10^{-6}	11.0×10^{-6}
SAE 4140	6.2×10^{-6}	11.2×10^{-6}
Stainless steel:		
SAE 301	9.4×10^{-6}	16.9×10^{-6}
SAE 430	5.8×10^{-6}	10.4×10^{-6}
Aluminum:		
2014	12.8×10^{-6}	23.0×10^{-6}
6061	13.0×10^{-6}	23.4×10^{-6}
Bronze:		
	10.0×10^{-6}	18.0×10^{-6}

FN1 (light drive fit): Only light pressure required to assemble mating parts. Used for fragile parts and where no large forces must be transmitted across the joint.

FN2 (medium drive fit): General-purpose class used frequently for steel parts of moderate cross section.

FN3 (heavy drive fit): Used for heavy steel parts.

FN4 (force fit): Used for high-strength assemblies where high resulting pressures are required.

FN5 (force fit): Similar to FN4 for higher pressures.

The use of shrink fit methods is desirable in most cases of interference fits and is virtually required in the heavier classes and larger size parts. The temperature increase required to produce a given expansion for assembly can be computed from the basic definition of the coefficient of thermal expansion:

$$\delta = \alpha L(\Delta t) \quad (13-1)$$

where δ = total deformation desired (in or mm)
 α = coefficient of thermal expansion (in/in · °F or mm/mm · °C)
 L = nominal length of member being heated (in or mm)
 Δt = temperature difference (°F or °C)

For cylindrical parts, L is the diameter and δ is the change in diameter required. Table 13-5 gives the values for α for several materials. (See Reference 8 for more data.)

13-7 TRANSITION FITS

The *location transition (LT)* fit is used where accuracy of location is important, but where a small amount of clearance or a small amount of interference is acceptable.

There are six classes, LT1 to LT6. In any class, there is an overlap in the tolerance limits for both the hole and the shaft so that possible combinations produce a small clearance, a small interference, or even a size-to-size fit. Complete tables of data for these fits are published in References 1, 2, 4, 6, and 8.

13-8 STRESSES FOR FORCE FITS

When force fits are used to secure mechanical parts, the interference creates a pressure acting at the mating surfaces. The pressure causes stresses in each part. Under heavy force fits or even with lighter fits in fragile parts, the stresses developed can be great enough to yield ductile materials. A permanent set results, which normally destroys the usefulness of the assembly. With brittle materials such as cast iron, actual fracture may result.

The stress analysis applicable to force fits is related to the analysis of thick-walled cylinders. The outer member expands under the influence of the pressure at the mating surface, with the tangential tensile stress developed being a maximum at the mating surface. There is a radial stress equal to the pressure itself. Also, the inner member contracts because of the pressure and is subjected to a tangential compressive stress along with the radial compressive stress equal to the pressure. (See Reference 5.)

The usual objective of the analysis is to determine the magnitude of the pressure due to a given interference fit that would be developed at the mating surfaces. Then the stresses due to this pressure in the mating members are computed. The following procedure can be used:

PROCEDURE FOR COMPUTING STRESSES FOR FORCE FITS ▼

1. Determine the amount of interference from the design of the parts. For standard force fits, Table 13-4 can be used. The maximum limit of interference would, of course, give the maximum stresses for the parts. Note that the interference values are based on the total interference on the diameter, which is the sum of the expansion of the outer ring plus the contraction of the inner member (see Figure 13-6).
2. Compute the pressure at the mating surface from Equation (13-2) if both members are of the same material:

➤ Pressure Created by Force Fit

$$p = \frac{E\delta}{2b} \left[\frac{(c^2 - b^2)(b^2 - a^2)}{2b^2(c^2 - a^2)} \right] \quad (13-2)$$

Use Equation (13-3) if they are of different materials:

➤ Pressure Created by Force Fit, Two Different Materials

$$p = \frac{\delta}{2b \left[\frac{1}{E_o} \left(\frac{c^2 + b^2}{c^2 - b^2} + \nu_o \right) + \frac{1}{E_i} \left(\frac{b^2 + a^2}{b^2 - a^2} - \nu_i \right) \right]} \quad (13-3)$$

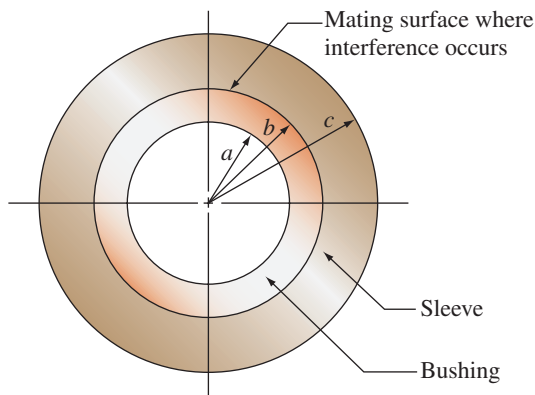


FIGURE 13-6 Terminology for interference fit

where p = pressure at the mating surface
 δ = total diametral interference
 E = modulus of elasticity of each member if they are the same
 E_o = modulus of elasticity of outer member
 E_i = modulus of elasticity of inner member
 ν_o = Poisson's ratio for outer member
 ν_i = Poisson's ratio for member

3. Compute the tensile stress in the outer member from

⇒ Tensile Stress in Outer Member

$$\sigma_o = p \left(\frac{c^2 + b^2}{c^2 - b^2} \right) \quad \text{(tensile at the inner surface in the tangential direction)} \quad (13-4)$$

4. Compute the compressive stress in the inner member from

⇒ Compressive Stress in Inner Member

$$\sigma_i = -p \left(\frac{b^2 + a^2}{b^2 - a^2} \right) \quad \text{(compressive at the outer surface in the tangential direction)} \quad (13-5)$$

5. If desired, the increase in the diameter of the outer member due to the tensile stress can be computed from

$$\delta_o = \frac{2bp}{E_o} \left[\frac{c^2 + b^2}{c^2 - b^2} + \nu_o \right] \quad (13-6)$$

6. If desired, the decrease in the diameter of the inner member due to the compressive stress can be computed from

$$\delta_i = \frac{2bp}{E_i} \left[\frac{b^2 + a^2}{b^2 - a^2} - \nu_i \right] \quad (13-7)$$

The stresses computed from Equations (13-4) and (13-5) were derived assuming that the two cylinders are of equal length. If the outer member is shorter than the inner member, the stresses are higher at its ends by as much as a factor of 2.0. Such a factor should be applied as a stress concentration factor.

In the absence of applied shear stresses, the tensile stress in the tangential direction in the outer member is the maximum principal stress and can be compared with the yield strength of the material to determine the resulting design factor.

Example Problem 13-2 shows the application of these relationships. Then a spreadsheet is listed to solve the same equations, with sample output showing the solution to Example Problem 13-2.

Example Problem 13-2

A bronze bushing is to be installed into a steel sleeve as indicated in Figure 13-6. The bushing has an inside diameter of 2.000 in and a nominal outside diameter of 2.500 in. The steel sleeve has a nominal inside diameter of 2.500 in and an outside diameter of 3.500 in.

1. Specify the limits of size for the outside diameter of the bushing and the inside diameter of the sleeve in order to obtain a heavy drive fit, FN3. Determine the limits of interference that would result.
2. For the maximum interference from 1, compute the pressure that would be developed between the bushing and the sleeve, the stress in the bushing and the sleeve, and the deformation of the bushing and the sleeve. Use $E = 30 \times 10^6$ psi for the steel and $E = 17 \times 10^6$ for the bronze. Use $\nu = 0.27$ for both materials.

Solution

For 1, from Table 13-4, for a part size of 2.50 in at the mating surface, the tolerance limits on the hole in the outer member are +1.2 and -0. Applying these limits to the basic size gives the dimension limits for the hole in the steel sleeve:

$$2.5012 \text{ in}$$

$$2.5000 \text{ in}$$

For the bronze insert, the tolerance limits are +3.2 and +2.5. Then the size limits for the outside diameter of the bushing are

$$2.5032 \text{ in}$$

$$2.5025 \text{ in}$$

The limits of interference would be 0.0013 to 0.0032 in.

For 2, the maximum pressure would be produced by the maximum interference, 0.0032 in. Then, using $a = 1.00$ in, $b = 1.25$ in, $c = 1.75$ in, $E_o = 30 \times 10^6$ psi, $E_i = 17 \times 10^6$ psi, and $\nu_o = \nu_i = 0.27$ from Equation (13-3),

$$p = \frac{\delta}{2b \left[\frac{1}{E_o} \left(\frac{c^2 + b^2}{c^2 - b^2} + \nu_o \right) + \frac{1}{E_i} \left(\frac{b^2 + a^2}{b^2 - a^2} - \nu_i \right) \right]}$$

$$p = \frac{0.0032}{(2)(1.25) \left[\frac{1}{30 \times 10^6} \left(\frac{1.75^2 + 1.25^2}{1.75^2 - 1.25^2} + 0.27 \right) + \frac{1}{17 \times 10^6} \left(\frac{1.25^2 + 1.00^2}{1.25^2 - 1.00^2} - 0.27 \right) \right]}$$

$$p = 3518 \text{ psi}$$

The tensile stress in the steel sleeve is

$$\sigma_o = p \left(\frac{c^2 + b^2}{c^2 - b^2} \right) = 3518 \left(\frac{1.75^2 + 1.25^2}{1.75^2 - 1.25^2} \right) = 10\,846 \text{ psi}$$

The compressive stress in the bronze bushing is

$$\sigma_i = -p \left(\frac{b^2 + a^2}{b^2 - a^2} \right) = -3518 \left(\frac{1.25^2 + 1.00^2}{1.25^2 - 1.00^2} \right) = -16\,025 \text{ psi}$$

The increase in the diameter of the sleeve is

$$\delta_o = \frac{2bp}{E_o} \left[\frac{c^2 + b^2}{c^2 - b^2} + \nu_o \right]$$

$$\delta_o = \frac{2(1.25)(3518)}{30 \times 10^6} \left[\frac{1.75^2 + 1.25^2}{1.75^2 - 1.25^2} + 0.27 \right] = 0.000\,98 \text{ in}$$

The decrease in the diameter of the bushing is

$$\delta_i = \frac{2bp}{E_i} \left[\frac{b^2 + a^2}{b^2 - a^2} - \nu_i \right]$$

$$\delta_i = \frac{2(1.25)(3518)}{17 \times 10^6} \left[\frac{1.25^2 + 1.00^2}{1.25^2 - 1.00^2} - 0.27 \right] = 0.002\,22 \text{ in}$$

Note that the sum of δ_o and δ_i equals 0.0032, the total interference, δ .

The spreadsheet to analyze force fits is shown in Figure 13-7. The data shown are from Example Problem 13-2.

13-9 GENERAL TOLERANCING METHODS

It is the designer's responsibility to establish tolerances on each dimension of each component in a mechanical device. The tolerances must ensure that the component fulfills its function. But it should also be as large as practical to permit economical manufacture. This pair of conflicting principles must be dealt with. See comprehensive texts on technical drawing and interpretation of engineering drawings for general principles (References 4, 6, and 9. Also see Internet site 8).

Special attention should be paid to the features of a component that mate with other components and with which they must operate reliably or with which they must be accurately located. The fit of the inner races of the bearings on the shafts is an example of such features. Other fits provide clearances between parts that must be assembled together easily but that must not have large relative motion during operation.

Where no other component mates with certain features of a given component, the tolerances should be as large as practical so that they can be produced with basic machining, molding, or casting processes without the need for subsequent finishing. It is often recommended that blanket tolerances be given for such dimensions and that the precision with which the basic size is stated on the drawing implies a certain tolerance. For decimal

STRESSES FOR FORCE FITS Refer to Figure 13–6 for geometry.		Data from: Example Problem 13–2	
Input Data:		Numerical values in <i>italics</i> must be inserted for each problem.	
Inside radius of inner member: a		=	1.0000 in
Outside radius of inner member: b		=	1.2500 in
Outside radius of outer member: c		=	1.7500 in
Total interference: d		=	0.0032 in
Modulus of outer member: E_o		=	3.00E+07 psi
Modulus of inner member: E_i		=	1.70E+07 psi
Poisson's ratio for outer member: ν_o		=	0.27
Poisson's ratio for inner member: ν_i		=	0.27
Computed Results			
Pressure at mating surface: p		=	3518 psi
		Using Equation (13–3)	
Tensile stress in outer member: σ_o		=	10 846 psi
		Using Equation (13–4)	
Compressive stress in inner member: σ_i		=	–16 025 psi
		Using Equation (13–5)	
Increase in diameter of outer member: δ_o		=	0.000 98 in
		Using Equation (13–6)	
Decrease in diameter of inner member: δ_i		=	0.002 22 in
		Using Equation (13–7)	

FIGURE 13–7 Spreadsheet solution for pressure, stresses, and deformations in cylindrical members mating with a force fit

dimensions in U.S. Customary units, a note similar to the following is often given:

DIMENSIONS IN in TOLERANCES ARE AS FOLLOWS UNLESS OTHERWISE STATED.

$$XX.X = \pm 0.050$$

$$XX.XX = \pm 0.010$$

$$XX.XXX = \pm 0.005$$

$$XX.XXXX = \pm 0.0005$$

$$\text{ANGLES: } \pm 0.50^\circ$$

where X represents a specified digit.

For example, if a given dimension has a basic size of 2.5 in, the dimension can be stated on the drawing in any of four ways with different interpretations:

2.5 means 2.5 ± 0.050 or limits of 2.550 to 2.450 in

2.50 means 2.50 ± 0.010 or limits of 2.510 to 2.490 in

2.500 means 2.500 ± 0.005 or limits of 2.505 to 2.495 in

2.5000 means 2.5000 ± 0.0005 or limits of 2.5005 to 2.4995 in

Any other desired tolerance must be specified on the dimension. Of course, you may select different standard tolerances according to the needs of the system being designed.

Similar data for metric drawings would appear as follows:

DIMENSIONS IN mm TOLERANCES ARE AS FOLLOWS UNLESS OTHERWISE STATED.

$$XX = \pm 1.0$$

$$XX.X = \pm 0.25$$

$$XX.XX = \pm 0.15$$

$$XX.XXX = \pm 0.012$$

$$\text{ANGLES: } \pm 0.50^\circ$$

Some tolerance notes also relate the degree of precision to the nominal size of the feature, with tighter tolerances on smaller dimensions and looser tolerances on larger dimensions. The international tolerance (IT) grades, discussed in Section 13–3, use this approach.

Geometric Dimensioning and Tolerancing (GD&T).

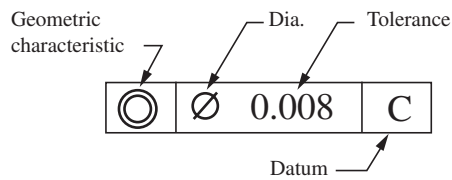
The geometric dimensioning and tolerancing (GD&T) method is used to control location, form, profile, orientation, and runout on a dimensioned feature. Its purpose is to ensure proper assembly and/or operation of parts, and it is especially useful in quantity production of interchangeable parts. The complete definition of the method is given in the ASME Standard Y14.5M-2009 (Reference 3). References 4 and 6–9 show some applications and demonstrate the interpretation of the numerous symbols. Also see Internet sites 1, 2, and 4.

Figure 13–8(a) shows some of the more common geometric symbols. Part (b) illustrates their use in a *feature control frame*, containing the symbol for the feature geometric characteristic being controlled, the tolerance on the dimension or form, and the *datum* to which the given feature is to be related. For example, in Part (c), the smaller diameter is to be concentric with the larger diameter (datum –A–) within a tolerance of 0.010 in.

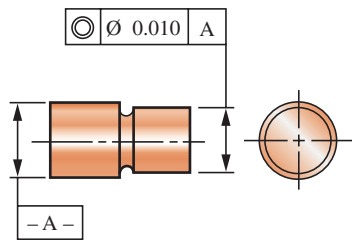
Surface Finish. The designer should also control the surface finish of all features critical to the performance of the device being designed. This includes the mating surfaces that have already been discussed. But also any surface that experiences relatively high stresses, particularly reversed bending, should have a smooth surface.

Tolerance	Characteristic	Symbol
Form	Straightness	—
	Flatness	▱
	Circularity	○
	Cylindricity	⊘
Profile	Profile of a line	⌒
	Profile of a surface	⌓
Orientation	Angularity	∠
	Perpendicularity	⊥
	Parallelism	//
Location	Position/symmetry	⊕
	Concentricity	⊗
Runout	Circular runout	↗
	Total runout	↘

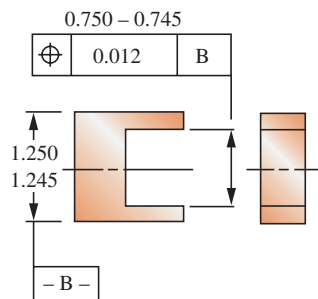
(a) Geometric symbols



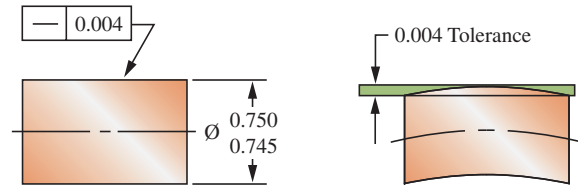
(b) Feature control frame



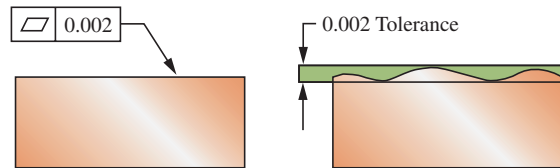
(c) Concentricity



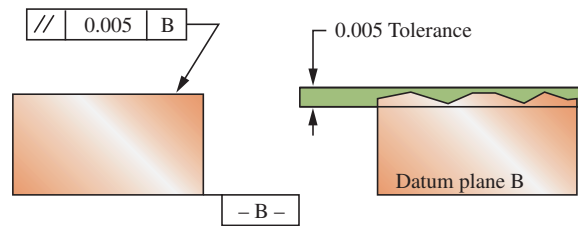
(d) Symmetry



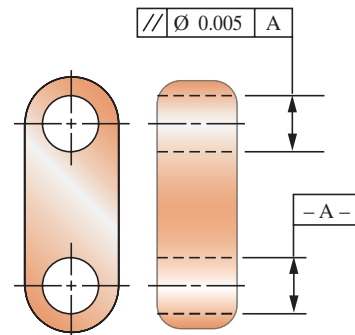
(e) Straightness



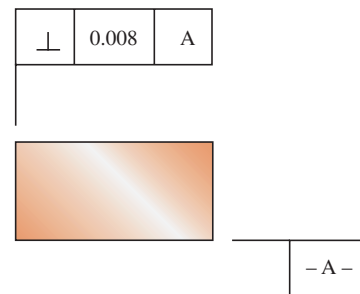
(f) Flatness



(g) Parallelism: planes



(h) Parallelism: cylinders



(i) Perpendicularity

FIGURE 13-8 Examples of geometric tolerancing

See Figure 5–11 for a rough indication of the effect of surface finishes between, for example, *machined* and *ground* surfaces on the basic endurance strength of steels. In general, *ground* surfaces have an average roughness, R_a , of $16\text{ }\mu\text{in}$ ($0.4\text{ }\mu\text{m}$). Figure 13–3 shows the expected range of surface finish for many kinds of machining processes. Note that turning is reported to be capable of producing that level of surface finish, but it is at the limit of its capability and will likely require very fine-finish cuts with sharp-edged tooling having a broad radius nose. The more nominal surface finish from turning, milling, broaching, and boring is $63\text{ }\mu\text{in}$ ($1.6\text{ }\mu\text{m}$). This would correspond to the *machined* category in Figure 5–11.

Bearing seats on shafts for accurate machinery are typically ground, particularly in the smaller sizes under 3.0 in (80 mm), with a maximum allowable average roughness of $16\text{ }\mu\text{in}$ ($0.4\text{ }\mu\text{m}$). Above that size and up to 20 in (500 mm), $32\text{ }\mu\text{in}$ ($0.8\text{ }\mu\text{m}$) is allowed. See manufacturers' catalogs.

13-10 ROBUST PRODUCT DESIGN

Careful control of the tolerances on the dimensions of machine parts is the responsibility of both the product designer and the manufacturing staff. The goal is to ensure the functionality of the product while permitting economical manufacture. Often these goals seem incompatible.

Robust product design is a tool that can help. (See Reference 10 and Internet sites 3, 5, and 6.) It is a technique in which a sequence of experiments is undertaken to determine the variables in a product's design that most affect its performance. Then the optimum limits on those variables are defined. The concepts can be used for dimensions, material properties, control settings, and many other factors.

The design of the experiments used to implement a robust product is critical to the success of the design. An initial screening experiment is used to select the most significant variables. Then additional experiments are run to determine the response of the system to combinations of these variables. Computer-generated plots of results establish limits for the significant variables.

Robust product design is based on the *Taguchi method*, an important element in the process of improving manufacturing quality developed by Dr. Genichi Taguchi. He showed how design of experiments can be used to produce products of consistently high quality that are insensitive to the operating environment. (See Reference 10.)

Controlling dimensions and tolerances to minimize variation can be accomplished using statistical analysis tools. The designer models the type and magnitude of the expected size variation for a set of mating components. Simulation software predicts the final assembly

configuration considering the “tolerance stack” of the individual components. The analysis can be done in one, two, or three dimensions. (See Internet site 5.)

The major goal of these methods is the allocation of tolerances on critical dimensions of mating parts that ensure satisfactory operation under all foreseeable conditions of component manufacture, assembly, environment, and use. Maintaining reasonable manufacturing costs is also an important consideration. Many manufacturers practice these methods, particularly in the automotive, aerospace, military, and high-production consumer products markets. Computer-assisted techniques are available to facilitate the process. Most comprehensive, solid modeling, computer-aided design systems include statistically based tolerance analysis as a standard feature. Internet site 7 describes the activities of the Association for the Development of Computer-Aided Tolerancing Systems (ADCATS), a consortium of several industries and Brigham Young University.

Robust design is an important part of the overall *product realization process* that was introduced in Chapter 1, Section 1–2. It is also important to the following concepts:

- *Design for manufacture and assembly* (see Reference 27 in Chapter 1)
- *Design for manufacturability* (see Reference 11)
- *Concurrent engineering* (see Internet site 9 and Reference 11)
- *ISO 9001–Part 7.1 Planning of Product Realization, and Part 7.5.1 Control of Production and Service Provision* (see Internet site 10)

Each of these concepts calls for careful specification of part dimensional characteristics and the rigorous measurement and control of the variations that occur in production systems. A related process of assessing the capability of production equipment to produce a part to required precision is called C_{pk} and it requires in-depth knowledge of tolerancing and accurate measurement of part features. The entire process of product design must be closely linked with manufacturing process planning and design to ensure production of a robust product.

REFERENCES

1. The American Society of Mechanical Engineers. ANSI/ASME. Standard B4.1-1967 (R2009). *Preferred Limits and Fits for Cylindrical Parts*. New York: American Society of Mechanical Engineers, 2009.
2. The American Society of Mechanical Engineers. ANSI/ASME. Standard B4.2-1978 (R2009). *Preferred Metric Limits and Fits*. New York: American Society of Mechanical Engineers, 2009.
3. The American Society of Mechanical Engineers. ASME. Standard Y14.5-2009. *Dimensioning and Tolerancing*. New York: American Society of Mechanical Engineers, 2009.

- Bertoline, G. R., E. N. Wiebe, N. W. Hartman, and W. A. Ross. *Fundamentals of Graphics Communication*. 6th ed. New York: McGraw-Hill, 2011.
- Budynas, R. G., and K. J. Nisbett. *Shigley's Mechanical Engineering Design*. 10th ed. New York: McGraw-Hill, 2015.
- Giesecke, F. E., I. L. Hill, A. E. Mitchell, J. E. Novak, H. C. Spencer, and J. T. Dygdon. *Technical Drawing with Engineering Graphics*. 14th ed. Upper Saddle River, NJ: Pearson/Prentice-Hall, 2012.
- Madsen, D. A., and D. P. Madsen. *Geometric Dimensioning and Tolerancing*. 9th ed. Tinley Park, IL: Goodheart-Willcox Co., 2012.
- Oberg, E. et al. *Machinery's Handbook*. 30th ed. New York: Industrial Press, 2015.
- Puncochar, D. E. *Interpretation of Geometric Dimensioning and Tolerancing*. 3rd ed. New York: Industrial Press, 2010.
- Chowdhury, Subir and Genichi Taguchi. *Robust Engineering*. New York: McGraw-Hill, 1999.
- Anderson, D. M. *Design for Manufacturability & Concurrent Engineering: How to Use Concurrent Engineering to Rapidly Develop Low-Cost, High-Quality Products for Lean Production*. Boca Raton, FL: Productivity Press (Part of CRC Press), 2014.
- Drafting Zone**. Information on mechanical drafting standards and practices. Genium Group, Inc.
- Sage Publishing**. Publisher of Concurrent Engineering Research and Applications that provides information about the field, including collaborative decision making, information modeling, teaming and sharing, and management of concurrent engineering.
- The 9000store.com**. Documents and standards related to ISO 9001 quality standards and processes. Part 7.1 deals with Planning for Product Realization and Part 7.5.1 deals with Control of Production and Service Provision.

PROBLEMS

Clearance Fits

- Specify the class of fit, the limits of size, and the limits of clearance for the bore of a slow-moving but heavily loaded conveyor roller that must rotate freely on a stationary shaft under heavy load. The nominal diameter of the shaft is 3.500 in. Use the basic hole system.
- A sine plate is a measuring device that pivots at its base, allowing the plate to assume different angles that are set with gage blocks for precision. The pin in the pivot has a nominal diameter of 0.5000 in. Specify the class of fit, the limits of size, and the limits of clearance for the pivot. Use the basic hole system.
- A child's toy wagon has an axle with a nominal diameter of 5/8 in. For the wheel bore and axle, specify the class of fit, the limits of size, and the limits of clearance. Use the basic shaft system.
- The planet gear of an epicyclic gear train must rotate reliably on its shaft while maintaining accurate position with respect to the mating gears. For the planet gear bore and its shaft, specify the class of fit, the limits of size, and the limits of clearance. Use the basic hole system. The nominal shaft diameter is 0.800 in.
- The base of a hydraulic cylinder is mounted to the frame of a machine by means of a clevis joint, which allows the cylinder to oscillate during operation. The clevis must provide reliable motion, but some play is acceptable. For the clevis holes and the pin, which have a nominal diameter of 1.25 in, specify the class of fit, the limits of size, and the limits of clearance. Use the basic hole system.
- A heavy door on a furnace swings upward to permit access to the interior of the furnace. During various modes of operation, the door and its hinge assembly see temperatures of 50°F to 500°F. The nominal diameter of each hinge pin is 4.00 in. For the hinge and its pin, specify the class of fit, the limits of size, and the limits of clearance. Use the basic shaft system.
- The stage of an industrial microscope pivots to permit the mounting of a variety of shapes. The stage must move with precision and reliability under widely varying temperatures. For the pin mount for the stage, having a nominal diameter of 0.750 in, specify the class of fit, the limits of size, and the limits of clearance. Use the basic hole system.
- An advertising sign is suspended from a horizontal rod and is allowed to swing under wind forces. The rod is to be commercial bar stock, nominally 1.50 in in diameter. For the mating hinges on the sign, specify the class of fit,

INTERNET SITES RELATED TO TOLERANCES AND FITS

- Engineering Fundamentals**. A comprehensive site offering design information on numerous topics, including geometric dimensioning and tolerancing.
- Engineers Edge**. A comprehensive site offering design information on numerous topics, including geometric dimensioning and tolerancing.
- iSixSigma**. A free information resource about quality management, including robust design, Taguchi methods, and six sigma quality management.
- Engineering Bookstore**. A site offering technical books in many fields, including geometric dimensioning and tolerancing.
- Dimensional Control Systems, Inc.** A developer and vendor of software for managing and controlling the dimensions and tolerances of components and assemblies in three dimensions. Variations can be modeled and simulated to ensure that quality objectives for fit, finish, and functionality are met.
- DRM Associates**. An extensive site on the subject of new product development (NPD), including robust design, design of experiments, Taguchi methods, geometric dimensioning and tolerancing, variability reduction, and tolerance design and analysis for components and assemblies. Select the navigation button for *NPD Body of Knowledge* for a comprehensive index of topics.
- ADCATS**. The Association for the Development of Computer-Aided Tolerancing Systems (ADCATS) at Brigham Young University.

the limits of size, and the limits of clearance. Use the basic shaft system.

9. For any of Problems 1–8, make a diagram of the tolerances, fits, and clearances for your specifications, using a method similar to that in Figure 13–4.

Force Fits

10. A spacer made of SAE 1020 hot-rolled steel is in the form of a hollow cylinder with a nominal inside diameter of 3.25 in and an outside diameter of 4.000 in. It is to be mounted on a solid steel shaft with a heavy force fit. Specify the dimensions for the shaft and sleeve, and compute the stress in the sleeve after installation. Use the basic hole system.
11. A bronze bushing, with an inside diameter of 3.50 in and a nominal outside diameter of 4.00 in, is pressed into a steel sleeve having an outside diameter of 4.50 in. For an FN3 class of fit, specify the dimensions of the parts, the limits of interference, and the stresses created in the bushing and the sleeve. Use the basic hole system.
12. It is proposed to install a steel rod with a nominal diameter of 3.00 in in a hole of an aluminum cylinder with an outside diameter of 5.00 in, with an FN5 force fit. Would this be satisfactory?
13. The allowable compressive stress in the wall of an aluminum tube is 8500 psi. Its outside diameter is 2.000 in, and the wall thickness is 0.065 in. What is the maximum amount of interference between this tube and a steel sleeve that can be tolerated? The sleeve has an outside diameter of 3.00 in.
14. To what temperature would the spacer of Problem 10 have to be heated so that it would slide over the shaft with a clearance of 0.002 in? The ambient temperature is 75°F.
15. For the bronze bushing and the steel sleeve of Problem 11, the ambient temperature is 75°F. How much would the bronze bushing shrink if placed in a freezer at -220°F ? Then to what temperature would the steel sleeve have to be heated to provide a clearance of 0.004 in for assembly on the cold bushing?
16. For the bronze bushing of Problem 11, what would be the final inside diameter after assembly into the sleeve if it started with an inside diameter of 3.5000 in?

ROLLING CONTACT BEARINGS

The Big Picture

You Are the Designer

- 14–1 Objectives of This Chapter
- 14–2 Types of Rolling Contact Bearings
- 14–3 Thrust Bearings
- 14–4 Mounted Bearings
- 14–5 Bearing Materials
- 14–6 Load/Life Relationship
- 14–7 Bearing Manufacturers' Data
- 14–8 Design Life
- 14–9 Bearing Selection: Radial Loads Only
- 14–10 Bearing Selection: Radial and Thrust Loads Combined
- 14–11 Bearing Selection from Manufacturers' Catalogs
- 14–12 Mounting of Bearings
- 14–13 Tapered Roller Bearings
- 14–14 Practical Considerations in the Application of Bearings
- 14–15 Importance of Oil Film Thickness in Bearings
- 14–16 Life Prediction under Varying Loads
- 14–17 Bearing Designation Series

THE BIG PICTURE

Rolling Contact Bearings

Discussion Map

- Bearings are used to support a load while permitting relative motion between two elements of a machine.
- Some bearings use rolling elements, such as spherical balls or cylindrical or tapered rollers. This results in a very low coefficient of friction.

This chapter presents information about such bearings and gives methods for analyzing bearings and for selecting commercially available bearings.

Discover

Look for examples of bearings on machines, cars, trucks, bicycles, and consumer products.

Describe the bearings, including how they are installed and what kinds of forces are exerted on them.

The purpose of a bearing is to support a load while permitting relative motion between two elements of a machine. The term *rolling contact bearings* refers to the wide variety of bearings that use spherical balls cylindrical rollers, or some other type of roller between the stationary and the moving elements. The most common use for bearings is to support a rotating shaft, resisting purely radial loads, or a combination of radial and axial (thrust) loads. Some bearings are

designed to carry only thrust loads. Most bearings are used in applications involving rotation, but some are used in linear motion applications.

The components of a typical rolling contact bearing are the inner race, the outer race, and the rolling elements. Figure 14–1 shows the common single-row, deep-groove ball bearing. Usually the outer race is stationary and is held by the housing of the machine. The inner race is pressed onto the rotating shaft and thus

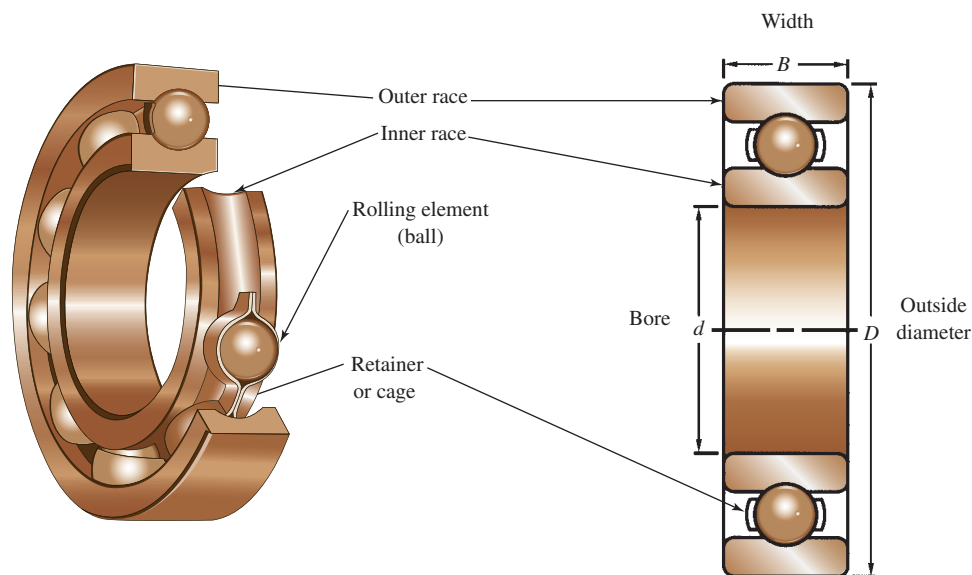


FIGURE 14-1 Single-row, deep-groove ball bearing

rotates with it. Then the balls roll between the outer and inner races. The load path is from the shaft, to the inner race, to the balls, to the outer race, and finally to the housing. The presence of the balls allows a very smooth, low-friction rotation of the shaft. The typical coefficient of friction for a rolling contact bearing is approximately 0.001 to 0.005. These values reflect only the rolling elements themselves and the means of retaining them in the bearing. The presence of seals, excessive lubricant, or unusual loading increases these values.

Look for consumer products, industrial machinery, or transportation equipment (cars, trucks, bicycles, and so on), and identify any uses of rolling contact bearings. Be sure that the machine is turned off, and then try to gain access to the mechanical drive shafts used to transmit power from a motor or an engine to some moving parts of the machine. Are these shafts supported in ball or roller bearings? Or are they supported in plain surface bearings in which the shaft passes through cylindrical members called *bushings* or *bearings*, typically with lubricants present between the rotating shaft and the stationary bearing? Plain surface bearings are discussed in Chapter 16.

For the shafts that are supported in ball or roller bearings, describe the bearing. How is it mounted on the shaft? How is it mounted in the housing of the machine? Can you identify what kinds of forces are acting on the bearing and in what directions they act? Are the forces directed radially toward the centerline of the shaft? Is there any force that acts parallel to the axis of the shaft? Compare the bearings that you find with the photographs that are in this chapter. Which varieties of bearings have you found? Measure or estimate the physical size of the bearings, particularly the diameter of the bore that is in contact with the shaft, the outside diameter, and the width. Can you see the rolling elements—balls or rollers? If so, sketch them and estimate their diameters and/or lengths. Are any of the rolling elements tapered rollers like those shown later in this chapter in Figure 14-7?

When you complete this chapter, you will be able to identify several kinds of rolling contact bearings and specify suitable bearings to carry specified loads. You will also be able to apply such bearings properly, planning for their installation on shafts and in housings.

YOU ARE THE DESIGNER

In Chapter 12, you were the designer of a shaft that was rotating at 600 rpm carrying two gears as a part of a power transmission system. Figures 12-1 and 12-2 showed the basic layout that was proposed. The shaft was designed to be supported on two bearings at points *B* and *D*. Then, in Example Problem 12-1, we completed the force analysis by computing the forces applied to the shaft by the gears and then computing the reactions at the bearings. Figure 12-12 showed the results, summarized here:

$$\begin{aligned} R_{Bx} &= 458 \text{ lb} & R_{By} &= 4620 \text{ lb} \\ R_{Dx} &= 1223 \text{ lb} & R_{Dy} &= 1680 \text{ lb} \end{aligned}$$

where *x* refers to the horizontal direction and *y* refers to the vertical direction. All forces on the bearings are in the radial direction. Your task now is to specify suitable rolling contact bearings for the shaft to withstand those forces and transfer them from the shaft to the housing of the speed reducer.

What kind of bearing should be selected? How do the forces just identified affect the choice? What life expectancy is reasonable for the bearings, and how does that affect the selection of the bearings? What size should be specified? How are the bearings to be installed on the shaft, and how does that affect the detailed design of the shaft? What limit dimensions should be defined for the

bearing seats on the shaft? How is the bearing to be located axially on the shaft? How is it to be installed in the housing and located there? How is lubrication to be provided for the bearings? Is there a need for shields and seals to keep contaminants out of the bearings? The information in this chapter will help you make these and other design decisions.

14-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Identify the types of rolling contact bearings that are commercially available, and select the appropriate type for a given application, considering the manner of loading and installation conditions.
2. Use the relationship between forces on bearings and the life expectancy for the bearings to determine critical bearing selection factors.
3. Use manufacturers' data for the performance of ball bearings to specify suitable bearings for a given application.
4. Recommend appropriate values for the design life of bearings.
5. Compute the *equivalent load* on a bearing corresponding to combinations of radial and thrust loads applied to it.
6. Specify mounting details for bearings that affect the design of the shaft onto which the bearing is to be seated and the housing into which it is to be installed.
7. Compute the equivalent loads on tapered roller bearings.
8. Describe the special design of thrust bearings.
9. Describe several types of commercially available mounted bearings and their application to machine design.
10. Understand certain practical considerations involved in the application of bearings, including lubrication, sealing, limiting speeds, bearing tolerance classes, and standards related to the manufacture and application of bearings.
11. Consider the effects of varying loads on the life expectancy and specification of bearings.

14-2 TYPES OF ROLLING CONTACT BEARINGS

Here we will discuss seven different types of rolling contact bearings and the applications in which each is typically used. Many variations on the designs shown are available. As each is discussed, refer to Table 14-1 for a comparison of the performance relative to the others.

Radial loads act toward the center of the bearing along a radius. Such loads are typical of those created by power transmission elements on shafts such as spur gears, V-belt drives, and chain drives. *Thrust loads* are those that act parallel to the axis of the shaft. The axial components of the forces on helical gears, worms and wormgears, and bevel gears are thrust loads. Also, bearings supporting shafts with vertical axes are subjected to thrust loads due to the weight of the shaft and the elements on the shaft as well as from axial operating forces. *Misalignment* refers to the angular deviation of the axis of the shaft at the bearing from the true axis of the bearing itself. An excellent rating for misalignment in Table 14-1 indicates that the bearing can accommodate up to 4.0° of angular deviation. A bearing with a fair rating can withstand up to 0.15° , while a poor rating indicates that rigid shafts with less than 0.05° of misalignment are required. Manufacturers' catalogs should be consulted for specific data. Check Internet sites 1-8.

Single-Row, Deep-Groove Ball Bearing

Sometimes called *Conrad bearings*, the single-row, deep-groove ball bearing (Figure 14-1) is what most people think of when the term *ball bearing* is used. The inner race is typically pressed on the shaft at the bearing seat with a slight interference fit to ensure that it rotates with the shaft. The spherical rolling elements, or balls, roll in a deep groove in both the inner and the outer races. The spacing of the balls is maintained by retainers or "cages." Although designed primarily for radial load-carrying

TABLE 14-1 Comparison of Bearing Types

Bearing type	Radial load capacity	Thrust load capacity	Misalignment capability
Single-row, deep-groove ball	Good	Fair	Fair
Double-row, deep-groove ball	Excellent	Good	Fair
Angular contact	Good	Excellent	Poor
Cylindrical roller	Excellent	Poor	Fair
Needle	Excellent	Poor	Poor
Spherical roller	Excellent	Fair/good	Excellent
Tapered roller	Excellent	Excellent	Poor

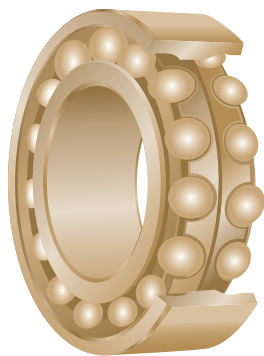


FIGURE 14-2 Double-row, deep-groove ball bearing

capacity, the deep groove allows it to carry a fairly sizable thrust load. The thrust load would be applied to one side of the inner race by a shoulder on the shaft. The load would pass across the side of the groove, through the ball, to the opposite side of the outer race, and then to the housing. The radius of the ball is slightly smaller than the radius of the groove to allow free rolling of the balls. The contact between a ball and the race is theoretically at a point, but it is actually a small circular area because of the deformation of the elements. Because the load is carried on a small area, very high local contact stresses occur. To increase the capacity of a single-row bearing, a bearing with a greater number of balls, or larger balls operating in larger diameter races, should be used.

Double-Row, Deep-Groove Ball Bearing

Adding a second row of balls (Figure 14-2) increases the radial load-carrying capacity of the deep-groove type of bearing compared with the single-row design because more balls share the load. Thus, a greater load can be carried in the same space, or a given load can be carried in a smaller space. The greater width of double-row bearings often adversely affects the misalignment capability.

Angular Contact Ball Bearing

One side of each race in an angular contact bearing is higher to allow the accommodation of greater thrust loads compared with the standard single-row, deep-groove bearing. The sketch in Figure 14-3 shows the preferred angle of the resultant force (radial and thrust loads combined), with commercially available bearings having angles of 15° to 40° .

Cylindrical Roller Bearing

Replacing the spherical balls with cylindrical rollers (Figure 14-4), with corresponding changes in the design of the races, gives a greater radial load capacity. The pattern of contact between a roller and its race is theoretically a line, and it becomes a rectangular shape as the members deform under load. The resulting contact stress levels are lower than for equivalent-size ball bearings, allowing smaller bearings to carry a given load or a given size bearing to carry a higher load. Thrust load capacity is poor because any thrust load would be applied to the side

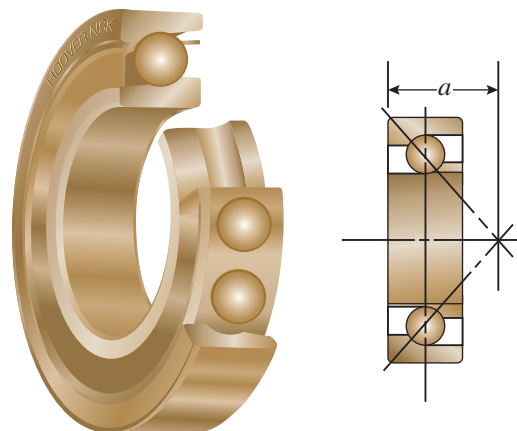


FIGURE 14-3 Angular contact ball bearing

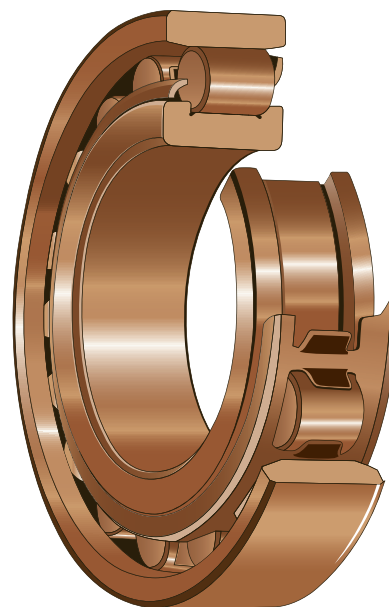
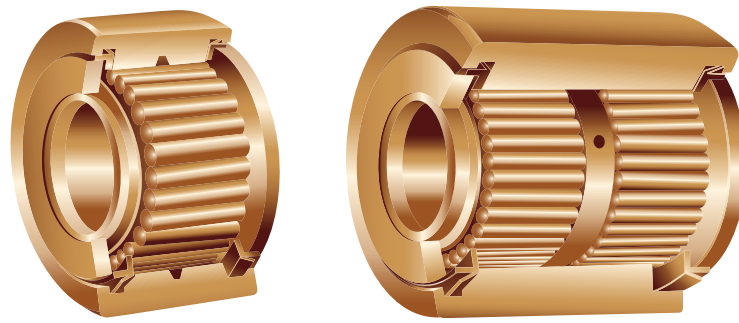


FIGURE 14-4 Cylindrical roller bearing

of the rollers, causing rubbing, not true rolling motion. It is recommended that *no* thrust load be applied. Roller bearings are often fairly wide, giving them only fair ability to accommodate angular misalignment.

Needle Bearing

Needle bearings (Figure 14-5) are actually roller bearings, but they have much smaller diameter rollers, as you can see by comparing Figures 14-4 and 14-5. A smaller radial space is typically required for needle bearings to carry a given load than for any other type of rolling contact bearing. This makes it easier to design them into many types of equipment and components such as pumps, universal joints, precision instruments, and household appliances. The cam follower shown in Figure 14-5(b) is another example in which the antifriction operation of needle bearings can be built in with little radial space required. As with other roller bearings, thrust and misalignment capabilities are poor.



(a) Single- and double-row needle bearings

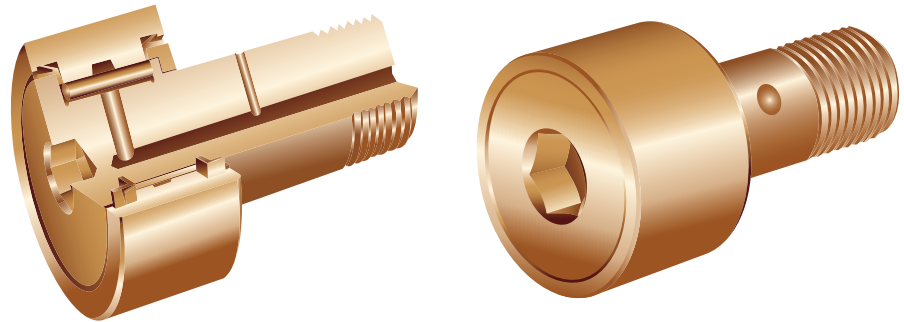


FIGURE 14-5 Needle bearings

(b) Needle bearings adapted to cam followers

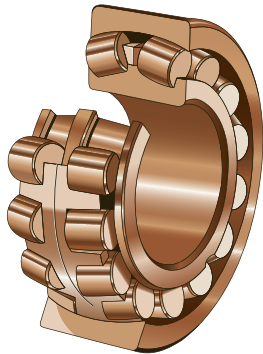


FIGURE 14-6 Spherical roller bearing

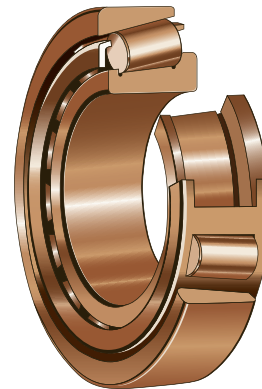


FIGURE 14-7 Tapered roller bearing

Spherical Roller Bearing

The spherical roller bearing (Figure 14-6) is one form of *self-aligning bearing*, so called because there is actual relative rotation of the outer race relative to the rollers and the inner race when angular misalignments occur. This gives the excellent rating for misalignment capability while retaining virtually the same ratings on radial load capacity.

Tapered Roller Bearing

Tapered roller bearings (Figure 14-7) are designed to take substantial thrust loads along with high radial loads, resulting in excellent ratings on both. They are often used in wheel bearings for vehicles and mobile equipment and in heavy-duty machinery having inherently high thrust loads. Section 14-13 gives additional information about their application. Figures 8-3, 9-31, 10-1, and 10-2 show tapered roller bearings applied in gear-type speed reducers.

14-3 THRUST BEARINGS

The bearings discussed so far in this chapter have been designed to carry radial loads or a combination of radial and thrust loads. Many machine design projects demand a bearing that resists only thrust loads, and several types of standard thrust bearings are commercially available. The same types of rolling elements are used: spherical balls, cylindrical rollers, and tapered rollers (see Figure 14-8).

Most thrust bearings can take little or no radial load. Then the design and the selection of such bearings are dependent only on the magnitude of the thrust load and the design life. The data for basic dynamic load rating and basic static load rating are reported in manufacturers' catalogs in the same way as they are for radial bearings.

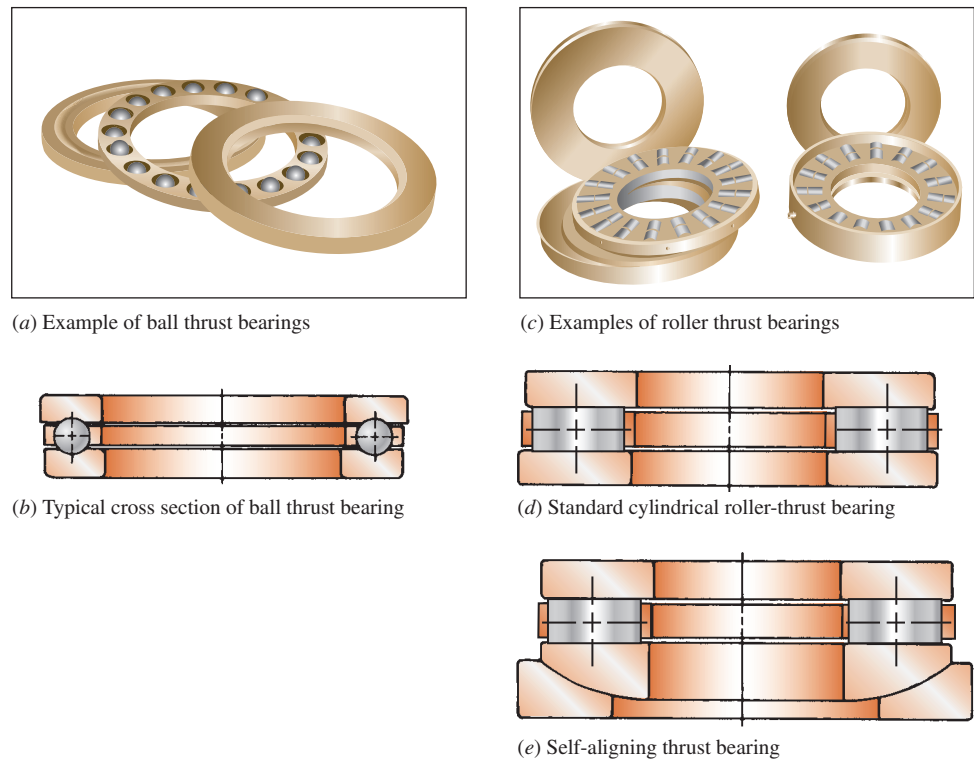


FIGURE 14-8 Thrust bearings

14-4 MOUNTED BEARINGS

In many types of heavy machines and special machines produced in small quantities, mounted bearings rather than unmounted bearings are selected. The mounted bearings, sometimes called housed bearings, provide a means of attaching the bearing unit directly to the frame of the machine with bolts rather than inserting it into a machined recess in a housing as is required in unmounted bearings.

Figure 14-9 shows the most common configuration for a mounted bearing: the *pillow block*. The housing is made from formed steel, cast iron, or cast steel, with

holes or slots provided for attachment during assembly of the machine, at which time alignment of the bearing unit is adjusted. The bearings themselves can be of virtually any of the types discussed in the preceding sections; ball, roller, tapered roller, or spherical roller. Misalignment capability is an important application consideration because of the conditions of use of such bearings. This capability is provided either in the bearing construction itself or in the housing as shown in Figure 14-9.

Because the bearing itself is similar to those already discussed, the selection process is also similar. Most

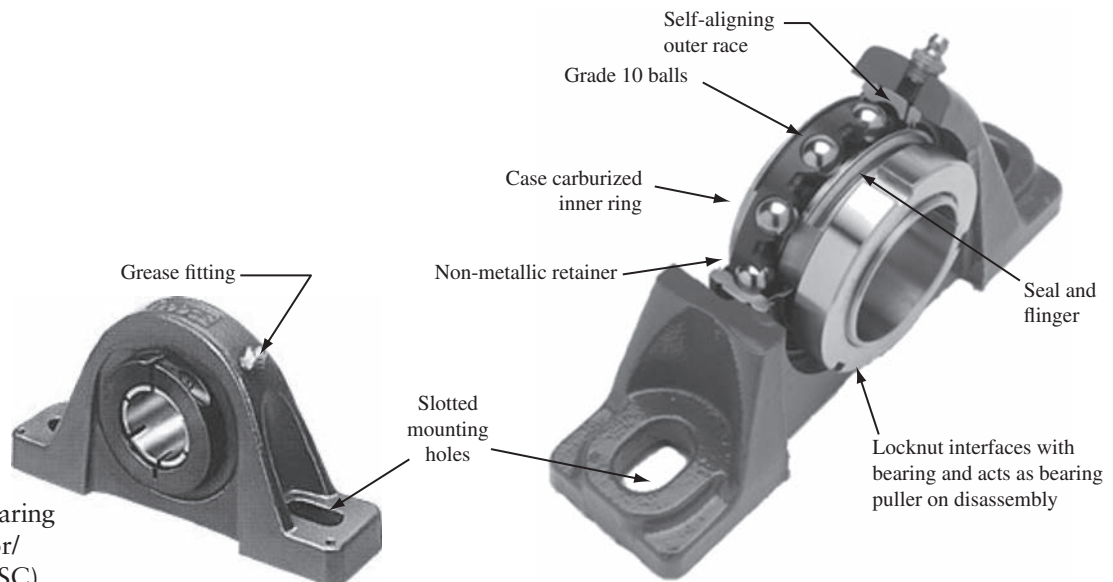
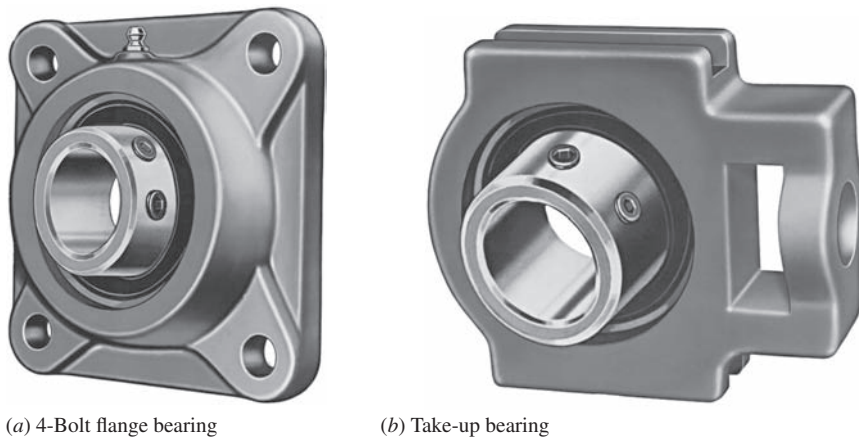
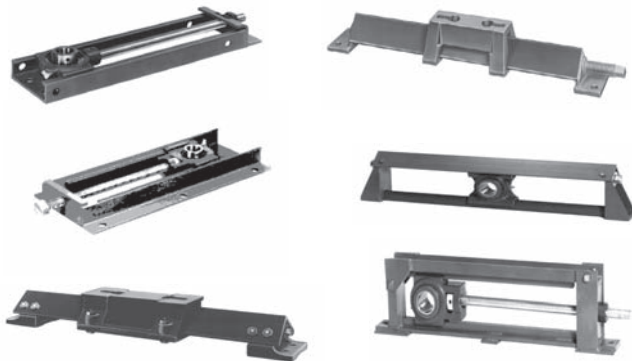


FIGURE 14-9 Ball bearing pillow block (Baldor/Dodge, Greenville, SC)



(a) 4-Bolt flange bearing

(b) Take-up bearing



(c) Selected styles of take-up frames

FIGURE 14-10 Forms of mounted bearings (Baldor/Dodge, Greenville, SC)

catalogs provide extensive data listing the load-carrying capacity at specified rated life values. Internet site 7 includes an example.

Other forms of mounted bearings are shown in Figure 14-10. The *flange units* are designed to be mounted on the vertical side frames of machines, holding horizontal shafts. Again, several bearing types and sizes are available. The term *take-up unit* refers to a bearing mounted in a housing, which in turn is mounted in a frame that allows movement of the bearing with the shaft in place. Used on conveyors, chain drives, belt drives, and similar applications, the take-up unit permits adjustment of the center distance of the drive components at the time of installation and during operation to accommodate wear or stretch of parts of the assembly.

14-5 BEARING MATERIALS

The load applied to a rolling contact bearing passes through the outer race to the rolling elements (balls or rollers) and then to the inner race, resulting in forces exerted on very small areas of contact. Contact stresses of approximately 300 000 psi (2070 MPa) are not uncommon. To withstand such high stresses, the balls, rollers, and races are typically made from very hard, high-strength steel or ceramic. Other metallic materials used include some titanium/nickel alloys and Monel metal (alloys of primarily nickel, copper, and cobalt). Where lighter loads are encountered or when resistance to corrosion by

specific materials is needed, plastic bearings can be used. See Table 14-2 for properties of selected materials.

The most widely used bearing material is SAE 52100 steel having a very high carbon content of 0.95% to 1.10% along with 1.3% to 1.6% chromium, 0.25% to 0.45% manganese, 0.2% to 0.35% silicon, and other alloying elements in low, but controlled, amounts. Impurities are carefully minimized. The steel is through hardened to the range of 58 to 65 HRC to give it the ability to resist high contact stress. Some tool steels, particularly M1 and M50, are also used. Steel alloys such as SAE 3310, 4620, and 8620, case hardened by carburizing to achieve high surface hardness and a tough core, are sometimes used. Careful control of case depth is required to withstand critical subsurface stresses. Some more lightly loaded bearings and those exposed to corrosive environments are made from SAE 440C stainless steel.

Rolling elements, races, and other components can also be made from ceramic materials such as silicon nitride (Si_3N_4), zirconia oxide (ZrO_2), alumina oxide (Al_2O_3), and silicon carbide (SiC). Compared with typical bearing steels, these materials have much lower density, higher hardness, and higher modulus of elasticity values leading to lighter weight, lower wear, and higher stiffness. They typically have better corrosion resistance and can operate at much higher temperatures as well. Their applications include machine tools, railroad equipment, semiconductor manufacturing, aerospace devices, and any equipment operating at high temperatures.

TABLE 14-2 Comparison of Bearing Materials

Property	Unit	Steels		Titanium/ Nickel	Cermics			Monel	Plastic	
		Beariing Steel	Stainless Steel	Silicon Nitride	Zirconia Oxide	Alumina Oxide	Silicon Carbide	K-500 Metal	Polyamide (Nylon 66)	
		52 100	440C	60NiTi	Si ₃ N ₄	ZrO ₂	Al ₂ O ₃	SiC		
Density	kg/m ³	7680	7750	6700	3230	6050	3920	3120	8434	1360
	lbm/ft ³	480	484	418	202	378	245	195	527	85
Modulus of elasticity	GPa	207	200	114	300	210	340	440	179	4.2
	ksi	30 000	29 000	16 500	43 500	30 500	49 300	63 800	25 950	610
Hardness	Vickers	700	700	650	1500	1200	1650	2800	263	—
Flexural strength	MPa	2240	1930	900	450	210	230	380	965	82
	ksi	325	280	131	65	31	33	55	140	11.9
Maximum use temperature	°C	300	350	400	1050	750	1500	1700	315	130
	°F	570	660	750	1920	1380	2730	3100	600	270

Note: Data taken from a variety of sources and are representative only. Actual properties highly dependent on specific composition, processing, thermal treatment, and form.

Ceramics can be used for all bearing elements (balls, rollers, races, cages) or hybrid designs consisting of ceramic rolling elements in steel races with steel or



(a) Full ceramic—radial ball bearings



(b) Full ceramic—spherical ball bearings

FIGURE 14-11 Ceramic bearings in radial ball and spherical ball forms (Boca Bearings Company, Delray Beach, FL)

polymeric cages. Figure 14-11 shows examples of full ceramic bearings in radial ball type and in spherical ball type. Note that silicon nitride is shiny black, zirconia oxide is shiny white, alumina oxide is dull white, and silicon carbide is dull black. Internet site 1 shows ceramic bearings with bores as small as 3.0 mm (0.1181 in) and up to 50 mm (1.9685 in). Inch bore sizes are as small as 0.0937 in (2.38 mm) up to 1.25 in (31.75 mm).

Alloys of titanium and nickel, sometimes called Nitinol, are applied where their anti-corrosive properties are desirable. The lower density compared with steel is also advantageous. One alloy is listed in Table 14-2. Development continues for other titanium alloys.

Monel metal is an alloy of primarily nickel, copper, and cobalt, with smaller amounts of iron, silicon, manganese, carbon, aluminum, titanium, and sulfur. Its resistance to fresh water, sea water, most acids, calcium chloride, and petroleum oils make it attractive to marine applications, chemical plants, oil and gas industries, and water and wastewater treatment facilities. One alloy is listed in Table 14-2. (See Internet site 12.)

Bearings made from plastics are typically used for lightly loaded, moderate precision applications requiring good corrosion resistance and light weight. Several types of plastic formulations can be used such as acetal, polyamide (PA, nylon), and thermoplastic polyester (PBT). Hybrid designs using plastic housing elements with stainless steel rolling balls or rollers are often used. One glass-reinforced polyamide material is listed in Table 14-2. (See Internet sites 13 and 14 for some suppliers.)

14-6 LOAD/LIFE RELATIONSHIP

Despite using steels with very high strength, all bearings have a finite life and will eventually fail due to fatigue because of the high contact stresses. But, obviously, the lighter the load, the longer the life, and vice versa. The

relationship between load, P , and life, L , for rolling contact bearings can be stated as

➤ **Relationship between Bearing Load and Life**

$$\frac{L_2}{L_1} = \left(\frac{P_1}{P_2} \right)^k \quad (14-1)$$

where $k = 3.00$ for ball bearings
 $k = 3.33$ for roller bearings

14-7 BEARING MANUFACTURERS' DATA

The selection of a rolling contact bearing from a manufacturer's catalog involves considerations of load-carrying capacity and the geometry of the bearing. Table 14-3 shows data for three classes of single-row, deep-groove ball bearings that are similar to those in manufacturers' catalogs. *It is essential that published data from specific manufacturers be used in any real application.*

Standard bearings are available in several classes, typically extremely light, extra-light, light, medium, and heavy classes. The designs differ in the size and number of load-carrying elements (balls or rollers) in the bearing. The bearing number usually indicates the class and the size of the bore of the bearing. Most bearings are manufactured with nominal dimensions in metric units, and the last two digits of the bearing number indicate the nominal bore size. The bore-size convention can be seen from the data in Table 14-3. Note that for bore sizes 04 and above, the nominal bore dimension in millimeters is five times the last two digits in the bearing number.

The number preceding the last two digits indicates the class. Typically, the 100 series is for extra-light, 200 for light, 300 for medium, and 400 for heavy-duty. Figure 14-12 shows the relative size of the different classes of bearings. The three digits may be preceded by others to indicate a design code for the type of bearing. In Table 14-3, the leading number 6 indicates *single-row, deep-groove ball bearings*.

Note also in Figure 14-12 that the designation number for each of the bearings shown has the last two digits of 00. Then note that each bearing shown has the same inside diameter, or bore, of 10 mm. The overall sizes and outside diameters of the heavier series are progressively larger allowing them to carry higher loads.

Inch-type bearings are available with bores ranging from 0.125 through 15.000 in.

Considering load-carrying capacity first, the data reported for each bearing design will include a basic dynamic load rating, C , and a basic static load rating, C_o .

The *basic static load rating*, C_o , is the load that the bearing can withstand without permanent deformation of any component. If this load is exceeded, the most probable result would be the indentation of one of the bearing races by the rolling elements. The deformation would be similar to that produced in the Brinell hardness test, and the failure is sometimes referred to as *brinelling*. Operation of the bearing after brinelling would be very noisy, and the impact loads on the indented area would produce rapid wear and progressive failure of the bearing.

To understand the *basic dynamic load rating*, C , it is necessary first to discuss the concept of the rated life of a bearing. Fatigue occurs over a large number of cycles of loading; for a bearing, that would be a large number of revolutions. Also, fatigue is a statistical phenomenon with considerable spread of the actual life of a group of bearings of a given design. The rated life is the standard means of reporting the results of many tests of bearings of a given design. It represents the life that 90% of the bearings would achieve successfully at a rated load. Note that it also represents the life that 10% of the bearings would not achieve. The rated life is thus typically referred to as the L_{10} life at the rated load.

Now the *basic dynamic load rating* C , can be defined as that load to which the bearings can be subjected while achieving a rated life (L_{10}) of 1 million revolutions (rev). Thus, the manufacturer supplies you with one set of data relating load and life. Equation (14-1) can be used to compute the expected life at any other load.

You should be aware that different manufacturers use other bases for the rated life. For example, some use 90 million cycles as the rated life and determine the rated load for that life. Also, some will report *average life*, for which 50% of the bearings will not survive. Thus, average life can be called the L_{50} life, not the L_{10} . Note that the average life is approximately five times as long as the L_{10} life. (See Internet site 2.) Be sure that you understand the basis for the ratings in any given catalog. (See References 2, 5, 6, 7, 10, and 12 for additional analysis of roller bearing performance.)

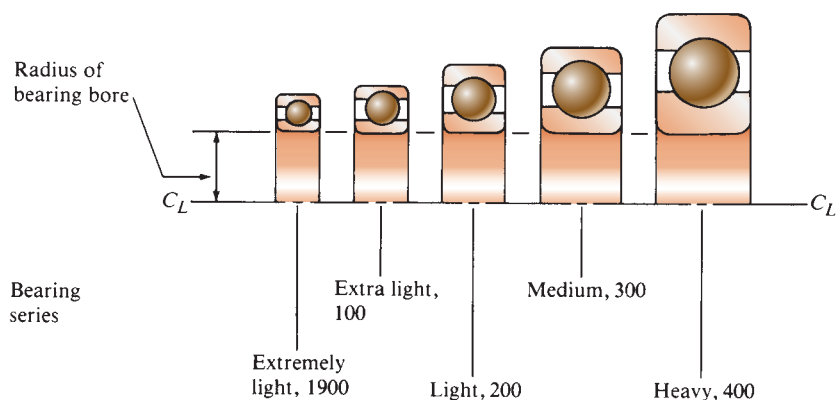


FIGURE 14-12 Relative sizes of bearing series

TABLE 14-3 Dimensions for Single Row, Deep-groove Ball Bearings

Bearing number	Nominal bearing dimensions						Basic load ratings				Maximum fillet		Minimum shaft		Maximum housing		Bearing mass	
	Bore, <i>d</i>		Outside dia., <i>D</i>		Width, <i>B</i>		Static, <i>C₀</i>		Dynamic, <i>C</i>		radius <i>r_{max}¹</i>		shoulder diameter, <i>S</i>		shoulder diameter, <i>H</i>			
mm	in	mm	in	mm	in	kN	lb _f	kN	lb _f	mm	in	mm	in	mm	in	kg	lb _m	
6000	10	0.3937	26	1.0236	8	0.3150	1.96	441	4.62	1039	0.3	0.012	12	0.472	24	0.945	0.019	0.042
6200	10	0.3937	30	1.1811	9	0.3543	2.36	531	5.07	1140	0.6	0.024	14	0.551	26	1.024	0.032	0.071
6300	10	0.3937	35	1.3780	11	0.4331	8.06	1812	3.40	764	0.6	0.024	14	0.551	31	1.220	0.053	0.117
6001	12	0.4724	28	1.1024	8	0.3150	2.36	531	5.07	1140	0.3	0.012	14	0.551	26	1.024	0.022	0.049
6201	12	0.4724	32	1.2598	10	0.3937	3.10	697	6.89	1549	0.6	0.024	16	0.630	28	1.102	0.037	0.082
6301	12	0.4724	37	1.4567	12	0.4724	4.15	933	9.75	2192	1.0	0.039	17	0.669	32	1.260	0.060	0.132
6002	15	0.5906	32	1.2598	9	0.3543	2.85	641	5.59	1257	0.3	0.012	17	0.669	30	1.181	0.030	0.066
6202	15	0.5906	35	1.3780	11	0.4331	3.75	843	7.80	1754	0.6	0.024	19	0.748	31	1.220	0.045	0.099
6302	15	0.5906	42	1.6535	13	0.5118	5.40	1214	11.40	2563	1.0	0.039	20	0.787	37	1.457	0.082	0.181
6003	17	0.6693	35	1.3780	10	0.3937	3.25	731	6.05	1360	0.3	0.012	19	0.748	33	1.299	0.039	0.086
6203	17	0.6693	40	1.5748	12	0.4724	4.75	1068	9.56	2149	0.6	0.024	21	0.827	36	1.417	0.065	0.143
6303	17	0.6693	47	1.8504	14	0.5512	6.55	1473	13.50	3035	1.0	0.039	22	0.866	42	1.654	0.120	0.265
6004	20	0.7874	42	1.6535	12	0.4724	5.00	1124	9.36	2104	0.6	0.024	24	0.945	38	1.496	0.069	0.152
6204	20	0.7874	47	1.8504	14	0.5512	6.55	1473	12.70	2855	1.0	0.039	25	0.984	42	1.654	0.110	0.243
6304	20	0.7874	52	2.0472	15	0.5906	7.80	1754	15.90	3575	1.0	0.039	27	1.063	45	1.772	0.140	0.309
6005	25	0.9843	47	1.8504	12	0.4724	6.55	1473	11.20	2518	0.6	0.024	29	1.142	43	1.693	0.080	0.176
6205	25	0.9843	52	2.0472	15	0.5906	7.80	1754	14.00	3147	1.0	0.039	30	1.181	47	1.850	0.130	0.287
6305	25	0.9843	62	2.4409	17	0.6693	11.60	2608	22.50	5058	1.0	0.039	32	1.260	55	2.165	0.230	0.507
6006	30	1.1811	55	2.1654	13	0.5118	8.30	1866	13.30	2990	1.0	0.039	35	1.378	50	1.969	0.160	0.353
6206	30	1.1811	62	2.4409	16	0.6299	11.2	2518	19.5	4384	1.0	0.039	35	1.378	57	2.244	0.200	0.441
6306	30	1.1811	72	2.8346	19	0.7480	16.0	3597	28.1	6317	1.0	0.039	37	1.457	65	2.559	0.350	0.772
6007	35	1.3780	62	2.4409	14	0.5512	10.2	2293	15.9	3575	1.0	0.039	40	1.575	57	2.244	0.160	0.353
6207	35	1.3780	72	2.8346	17	0.6693	15.3	3440	25.5	5733	1.0	0.039	42	1.654	65	2.559	0.290	0.639
6307	35	1.3780	80	3.1496	21	0.8268	19.0	4272	33.2	7464	1.5	0.059	43	1.693	72	2.835	0.460	1.014
6008	40	1.5748	68	2.6772	15	0.5906	11.6	2608	16.8	3777	1.0	0.039	45	1.772	63	2.480	0.190	0.419
6208	40	1.5748	80	3.1496	18	0.7087	19.0	4272	30.7	6902	1.0	0.039	47	1.850	73	2.874	0.370	0.816
6308	40	1.5748	90	3.5433	23	0.9055	24.0	5396	41.0	9218	1.5	0.059	48	1.890	82	3.228	0.630	1.389

TABLE 14-3 Dimensions for Single Row, Deep-groove Ball Bearings (continued)

Bearing number	Nominal bearing dimensions				Basic load ratings				Maximum fillet radius r_{max}^1		Minimum shaft shoulder diameter, S		Maximum housing shoulder diameter, H		Bearing mass	
	Bore, d		Outside dia., D		Width, B		Static, C_0		Dynamic, C		mm		in		mm	
	mm	in	mm	in	mm	in	kN	lb _f	kN	lb _f	mm	in	mm	in	mm	in
6009	45	1.7717	75	2.9528	16	0.6299	14.6	3282	20.8	4676	1.0	0.039	50	1.969	70	2.756
6209	45	1.7717	85	3.3465	19	0.7480	21.6	4856	33.2	7464	1.0	0.039	52	2.047	78	3.071
6309	45	1.7717	100	3.9370	25	0.9843	31.5	7082	52.7	11 848	1.5	0.059	53	2.087	92	3.622
6010	50	1.9685	80	3.1496	16	0.6299	16.0	3597	21.6	4856	1.0	0.039	55	2.165	75	2.953
6210	50	1.9685	90	3.5433	20	0.7874	23.2	5216	35.1	7891	1.0	0.039	57	2.244	83	3.268
6310	50	1.9685	110	4.3307	27	1.0630	38.0	8543	61.8	13 894	2.0	0.079	59	2.323	101	3.976
6011	55	2.1654	90	3.5433	18	0.7087	21.2	4766	28.1	6317	1.0	0.039	62	2.441	83	3.268
6211	55	2.1654	100	3.9370	21	0.8268	29.0	6520	43.6	9802	1.5	0.059	63	2.480	92	3.622
6311	55	2.1654	120	4.7244	29	1.1417	45.0	10 117	71.5	16 075	2.0	0.079	64	2.520	111	4.370
6012	60	2.3622	95	3.7402	18	0.7087	23.2	5216	29.6	6655	1.0	0.039	67	2.638	88	3.465
6212	60	2.3622	110	4.3307	22	0.8661	32.5	7307	47.5	10 679	1.5	0.059	68	2.677	102	4.016
6312	60	2.3622	130	5.1181	31	1.2205	52.0	11 691	81.9	18 413	2.0	0.079	71	2.795	119	4.685
6013	65	2.5591	100	3.9370	18	0.7087	25.0	5621	30.7	6902	1.0	0.039	72	2.835	93	3.661
6213	65	2.5591	120	4.7244	23	0.9055	40.5	9105	55.9	12 567	1.5	0.059	73	2.874	112	4.409
6313	65	2.5591	140	5.5118	33	1.2992	60.0	13 489	92.3	20 751	2.0	0.079	76	2.992	129	5.079
6014	70	2.7559	110	4.3307	20	0.7874	31.0	6969	37.7	8476	1.0	0.039	77	3.031	103	4.055
6214	70	2.7559	125	4.9213	24	0.9449	45.0	10 117	60.5	13 602	1.5	0.059	78	3.071	117	4.606
6314	70	2.7559	150	5.9055	35	1.3780	68.0	15 288	104.0	23 381	2.0	0.079	81	3.189	139	5.472
6015	75	2.9528	115	4.5276	20	0.7874	33.5	7531	39.7	8925	1.0	0.039	82	3.228	108	4.252
6215	75	2.9528	130	5.1181	25	0.9843	49.0	11 016	66.3	14 906	1.5	0.059	83	3.268	122	4.803
6315	75	2.9528	160	6.2992	37	1.4567	76.5	17 199	114.0	25 629	2.0	0.079	86	3.386	149	5.866
6016	80	3.1496	125	4.9213	22	0.8661	40.0	8993	47.5	10 679	1.0	0.039	87	3.425	118	4.646
6216	80	3.1496	140	5.5118	26	1.0236	55.0	12 365	70.2	15 782	2.0	0.079	89	3.504	131	5.157
6316	80	3.1496	170	6.6929	39	1.5354	86.5	19 447	124.0	27 878	2.0	0.079	91	3.583	159	6.260

(continued)

TABLE 14-3 Dimensions for Single Row, Deep-groove Ball Bearings (continued)

Bearing number	Nominal bearing dimensions						Basic load ratings				Maximum fillet radius r_{max}^1		Minimum shaft shoulder diameter, S		Maximum housing shoulder diameter, H		Bearing mass		
	Bore, d		Outside dia., D		Width, B		Static, C_0		Dynamic, C		mm		in		mm			in	
							kN	lb _f	kN	lb _f									
6017	mm	in	mm	in	mm	in	kN	lb _f	kN	lb _f	mm	in	mm	in	mm	in	kg	lb _m	
	85	3.3465	130	5.1181	22	0.8661	43.0	9667	49.4	11 106	1.0	0.039	92	3.622	123	4.843	0.890	1.962	
6217	85	3.3465	150	5.9055	28	1.1024	64.0	14 388	83.2	18 705	2.0	0.079	94	3.701	141	5.551	1.800	3.969	
6317	85	3.3465	180	7.0866	41	1.6142	96.5	21 695	133.0	29 901	2.5	0.098	98	3.858	167	6.575	4.250	9.371	
6018	90	3.5433	140	5.5118	24	0.9449	50.0	11 241	58.5	13 152	1.5	0.059	98	3.858	132	5.197	1.150	2.536	
6218	90	3.5433	160	6.2992	30	1.1811	73.5	16 524	95.6	21 493	2.0	0.079	99	3.898	151	5.945	2.150	4.741	
6318	90	3.5433	190	7.4803	43	1.6929	108.0	24 281	143.0	32 149	2.5	0.098	103	4.055	177	6.969	4.900	10.805	
6019	95	3.7402	145	5.7087	24	0.9449	54.0	12 140	60.5	13 602	1.5	0.059	103	4.055	137	5.394	1.200	2.646	
6219	95	3.7402	170	6.6929	32	1.2598	81.5	18 323	108.0	24 281	2.0	0.079	106	4.173	159	6.260	2.600	5.733	
6319	95	3.7402	200	7.8740	45	1.7717	118.0	26 529	153.0	34 397	2.5	0.098	108	4.252	187	7.362	5.650	12.458	
6020	100	3.9370	150	5.9055	24	0.9449	54.0	12 140	60.5	13 602	1.5	0.059	108	4.252	142	5.591	1.250	2.756	
6220	100	3.9370	180	7.0866	34	1.3386	93.0	20 908	124.0	27 878	2.0	0.079	111	4.370	169	6.654	3.150	6.946	
6320	100	3.9370	215	8.4646	47	1.8504	140.0	31 475	174.0	39 119	2.5	0.098	113	4.449	202	7.953	7.000	15.435	
6021	105	4.1339	160	6.2992	26	1.0236	65.5	14 726	72.8	16 367	2.0	0.079	114	4.488	151	5.945	1.600	3.528	
6221	105	4.1339	190	7.4803	36	1.4173	104.0	23 381	133.0	29 901	2.0	0.079	116	4.567	179	7.047	3.700	8.159	
6321	105	4.1339	225	8.8583	49	1.9291	153.0	34 397	182.0	40 917	2.5	0.098	118	4.646	212	8.346	8.250	18.191	
6022	110	4.3307	170	6.6929	28	1.1024	73.5	16 524	81.9	18 413	2.0	0.079	119	4.685	161	6.339	1.950	4.300	
6222	110	4.3307	200	7.8740	38	1.4961	118.0	26 529	143.0	32 149	2.0	0.079	121	4.764	189	7.441	4.350	9.592	
6322	110	4.3307	240	9.4488	50	1.9685	180.0	40 468	203.0	45 638	2.5	0.098	123	4.843	227	8.937	9.550	21.058	
6024	120	4.7244	180	7.0866	28	1.1024	80.0	17 986	85.2	19 155	2.0	0.079	129	5.079	171	6.732	2.050	4.520	
6224	120	4.7244	215	8.4646	40	1.5748	118.0	26 529	146.0	32 824	2.0	0.079	131	5.157	204	8.031	5.150	11.356	
6324	120	4.7244	260	10.2362	55	2.1654	186.0	41 817	208.0	46 763	2.5	0.098	133	5.236	247	9.724	14.500	31.973	
6026	130	5.1181	200	7.8740	33	1.2992	100.0	22 482	106.0	23 831	2.0	0.079	139	5.472	191	7.520	3.150	6.946	
6226	130	5.1181	230	9.0551	40	1.5748	132.0	29 676	156.0	35 072	2.5	0.098	143	5.630	217	8.543	5.800	12.789	
6326	130	5.1181	280	11.0236	58	2.2835	216.0	48 561	229.0	51 484	3.0	0.118	146	5.748	264	10.394	18.000	39.690	

¹Maximum fillet on shaft shoulder that will clear radius on bearing race

Example Problem 14-1 A catalog lists the basic dynamic load rating for a ball bearing to be 7050 lb for a rated life of 1 million rev. What would be the expected L_{10} life of the bearing if it were subjected to a load of 3500 lb?

Solution In Equation (14-1),

$$\begin{aligned} P_1 &= C = 7050 \text{ lb} && \text{(basic dynamic load rating)} \\ P_2 &= P_d = 3500 \text{ lb} && \text{(design load)} \\ L_1 &= 10^6 \text{ rev} && (L_{10} \text{ life at load } C) \\ k &= 3 && \text{(ball bearing)} \end{aligned}$$

Then letting the life, L_2 , be called the *design life*, L_d , at the design load,

$$L_2 = L_d = L_1 \left(\frac{P_1}{P_2} \right)^k = 10^6 \left(\frac{7050}{3500} \right)^{3.00} = 8.17 \times 10^6 \text{ rev}$$

This must be interpreted as the L_{10} life at a load of 3500 lb.

14-8 DESIGN LIFE

We will use the method developed in Example Problem 14-1 to refine the procedure for computing the required basic dynamic load rating C for a given design load P_d and a given design life L_d . If the reported load data in the manufacturer's literature is for 10^6 revolutions, Equation (14-1) can be written as

✦ **Design Life**

$$L_d = (C/P_d)^k (10^6) \quad (14-2)$$

The required C for a given design load and life would be

✦ **Basic Dynamic Load Rating**

$$C = P_d (L_d / 10^6)^{1/k} \quad (14-3)$$

Most people do not think in terms of the number of revolutions that a shaft makes. Rather, they consider the speed of rotation of the shaft, usually in rpm, and the design life of the machine, usually in hours of operation. The design life is specified by the designer, considering the application. As a guide, Table 14-4 can be used. Now, for a specified design life in hours, h , and a known speed of rotation in rpm, the number of design revolutions for the bearing would be

$$L_d = (h)(\text{rpm})(60 \text{ min/h})$$

Example Problem 14-2 Compute the required basic dynamic load rating, C , for a ball bearing to carry a radial load of 650 lb from a shaft rotating at 600 rpm that is part of an assembly conveyor in a manufacturing plant.

Solution From Table 14-4, let's select a design life of 30 000 hours. Then L_d is

$$L_d = (30\,000 \text{ h})(600 \text{ rpm})(60 \text{ min/h}) = 1.08 \times 10^9 \text{ rev}$$

From Equation (14-3),

$$C = (650 \text{ lb})(1.08 \times 10^9 / 10^6)^{1/3} = 6670 \text{ lb}$$

TABLE 14-4 Recommended Design Life for Bearings

Application	Design life L_{10} , h
Domestic appliances, instruments, medical apparatus	1000–2000
Aircraft engines	1000–4000
Automotive	1500–5000
Agricultural equipment, hoists, construction machines	3000–6000
Elevators, industrial fans, multipurpose gearing, rotary crushers, cranes	8000–15 000
Electric motors, industrial blowers, general industrial machines, conveyors	20 000–30 000
Pumps and compressors, textile machinery, rolling mill drives	40 000–60 000
Critical equipment in continuous, 24-h operation; power plants, ship drives	100 000–200 000

Source: Eugene A. Avallone and Theodore Baumeister III, eds., *Marks' Standard Handbook for Mechanical Engineers*, 9th ed. New York: McGraw-Hill, 1986.

14-9 BEARING SELECTION: RADIAL LOADS ONLY

The selection of a bearing takes into consideration the load capacity, as discussed, and the geometry of the bearing ensuring that it can be installed conveniently in the machine. We will first consider unmounted bearings carrying radial loads only. Then we will consider unmounted bearings carrying a combination of radial and thrust loads. The term *unmounted* refers to the case in which the designer must provide for the proper support and installation of the bearing onto the shaft and into a housing.

PROCEDURE FOR SELECTING A BEARING—RADIAL LOAD ONLY ▼

1. The bearing is normally selected after the shaft design has progressed to the point where the required minimum diameter of the shaft has been determined, using the techniques presented in Chapter 12. The radial loads are also known, along with the orientation of the bearings with respect to other elements in the machine.

2. Specify the design load on the bearing, usually called *equivalent load*. The method of determining the equivalent load when only a radial load, R , is applied takes into account whether the inner or the outer race rotates.

➤ Equivalent Load, Radial Load Only

$$P = VR \quad (14-4)$$

The factor V is called a *rotation factor* and takes the value of 1.0 if the inner race of the bearing rotates, which is usually the case. Use $V = 1.2$ if the outer race rotates.

3. Determine the minimum acceptable diameter of the shaft that will limit the bore size of the bearing.
4. Select the type of bearing, using Table 14-1 as a guide.
5. Specify the design life of the bearing, using Table 14-4.
6. Compute the required basic dynamic load rating, C , from Equation (14-3).
7. Identify a set of candidate bearings that have the required basic dynamic load rating.
8. Select the bearing having the most convenient geometry, also considering its cost and availability.
9. Determine mounting conditions, such as shaft seat diameter and tolerance, housing bore diameter and tolerance, means of locating the bearing axially, and special needs such as seals or shields.

Example Problem 14-3

Select a single-row, deep-groove ball bearing to carry 650 lb of pure radial load from a shaft that rotates at 600 rpm. The design life is to be 30 000 h. The bearing is to be mounted on a shaft with a minimum acceptable diameter of 1.48 in.

Solution

Note that this is a pure radial load and the inner race is to be pressed onto the shaft and rotate with it. Therefore, the factor $V = 1.0$ in Equation (14-4), and the design load is equal to the radial load. These are the same data used in Example Problem 14-2, where we found the required basic dynamic load rating, C , to be 6670 lb. From Table 14-3, giving design data for three classes of bearings, we find that we could use a bearing 6208 having a bore of 40 mm (1.5748 in).

Summary of Data for the Selected Bearing

Bearing number: 6208, single-row, deep-groove ball bearing

Bore: $d = 40$ mm (1.5748 in)

Outside diameter: $D = 80$ mm (3.1496 in)

Width: $B = 18$ mm (0.7087 in)

Maximum fillet radius: $r = 1.0$ mm (0.039 in)

Basic dynamic load rating: $C = 6902$ lb.

14-10 BEARING SELECTION: RADIAL AND THRUST LOADS COMBINED

When both radial and thrust loads are exerted on a bearing, the equivalent load is the constant radial load that would produce the same rated life for the bearing as the combined loading. The method of computing the equivalent load, P , for such cases is presented in the manufacturer's catalog and takes the form

➤ Equivalent Load with Radial and Thrust Loads

$$P = VXR + YT \quad (14-5)$$

where P = equivalent load

V = rotation factor (as defined in Section 14-9)

R = applied radial load

T = applied thrust load

X = radial factor

Y = thrust factor

TABLE 14-5 Radial and Thrust Factors for Single-row, Deep-groove Ball Bearings

e	T/C_o	Y	e	T/C_o	Y
0.19	0.014	2.30	0.34	0.170	1.31
0.22	0.028	1.99	0.38	0.280	1.15
0.26	0.056	1.71	0.42	0.420	1.04
0.28	0.084	1.55	0.44	0.560	1.00
0.30	0.110	1.45			

Note: $X = 0.56$ for all values of Y .

The values of X and Y vary with the specific design of the bearing and with the magnitude of the thrust load relative to the radial load. For relatively small thrust loads, $X = 1$ and $Y = 0$, so the equivalent load equation reverts to the form of Equation (14-4) for pure radial loads. To indicate the limiting thrust load for which this is the case, manufacturers list a factor called e . If the ratio $T/R > e$, Equation (14-5) must be used to compute P . If $T/R < e$, Equation (14-4) is used. Table 14-5 shows one set of data for a single-row, deep-groove ball bearing, for which the value of $X = 0.56$ for all values of Y . Note that both e and Y depend on the ratio T/C_o , where C_o is the static load rating of a particular bearing. This presents a difficulty in bearing selection because the value of C_o is not known until the bearing has been selected. Therefore, a simple trial-and-error method is applied. If a significant thrust load is applied to a bearing along with a radial load, perform the following steps:

PROCEDURE FOR SELECTING A BEARING— RADIAL AND THRUST LOAD ▼

1. Assume a value of Y from Table 14-5. The value $Y = 1.50$ is reasonable, being at about the middle of the range of possible values.
2. Compute $P = VXR + YT$.
3. Compute the required basic dynamic load rating C from Equation (14-3).
4. Select a candidate bearing having a value of C at least equal to the required value.
5. For the selected bearing, determine C_o .
6. Compute T/C_o .
7. From Table 14-5, determine e .
8. If $T/R > e$, then determine Y from Table 14-5.
9. If the new value of Y is different from that assumed in Step 1, repeat the process.
10. If $T/R < e$, use Equation (14-4) to compute P , and proceed as for a pure radial load.

Example Problem 14-4

Select a single-row, deep-groove ball bearing from Table 14-3 to carry a radial load of 1850 lb and a thrust load of 675 lb. The shaft is to rotate at 1150 rpm, and a design life of 20 000 h is desired. The minimum acceptable diameter for the shaft is 3.10 in.

Solution Use the procedure outlined above.

Step 1. Use $X = 0.56$ and assume $Y = 1.50$.

Step 2. $P = VXR + YT = (1.0)(0.56)(1850 \text{ lb}) + (1.50)(675 \text{ lb}) = 2049 \text{ lb}$.

Step 3. The required basic dynamic load rating C is found from Equation (14-3).

$$L_d = (h)(\text{rpm})(60 \text{ min/h}) = (20\,000 \text{ h})(1150 \text{ rev/min})(60 \text{ min/h}) = 1.38 \times 10^9 \text{ rev}$$

$$C = P_d(L_d/10^6)^{1/k} = (2049 \text{ lb})(1.38 \times 10^9/10^6)^{1/3} = 22\,812 \text{ lb}$$

Step 4. From Table 14-3, we can use bearing number 6316 having a bore of 80 mm (3.1496 in), just larger than the stated $D_{\min} = 3.10$ in.

Step 5. For bearing number 6316, $C = 27\,878 \text{ lb}$ and $C_o = 19\,447 \text{ lb}$.

Step 6. $T/C_o = (675 \text{ lb})/(19\,447 \text{ lb}) = 0.0347$.

Step 7. From Table 14-5, $e = 0.25$ approximately.

Step 8. $T/R = (675 \text{ lb})/(1850 \text{ lb}) = 0.365$. Because $T/R > e$, we can find $Y = 1.78$ from Table 14–5 by interpolation based on $T/C_o = 0.0347$.

Step 9. Recompute $P = (1.0)(0.56)(1850 \text{ lb}) + (1.78)(675 \text{ lb}) = 2238 \text{ lb}$

$$C = P_d(L_d/10^6)^{1/k} = 2238 \text{ lb}(1.38 \times 10^9/10^6)^{1/3} = 24\,910 \text{ lb}$$

The value of $C = 27\,878 \text{ lb}$ for bearing 6316 is still satisfactory. Therefore, we specify that bearing for this application.

Adjustment of Life Rating for Reliability

Thus far we have used the basic L_{10} life for selecting rolling contact bearings. This is the general industrial practice and the basis for data published by most bearing manufacturers. Recall that L_{10} life indicates a 90% probability that the selected bearing would carry its rated dynamic load for the specified number of design hours. That leaves a 10% probability that any given bearing would have a lower life.

Certain applications call for greater reliability. Examples can be found in the aerospace, military, instrumentation, and medical fields. It is then desirable to be able to adjust the expected life of a bearing for higher reliability. The following equation provides a method.

$$L_{aR} = C_R L_{10} \quad (14-6)$$

where

L_{10} = Life in millions of revolutions for 90% reliability

L_{aR} = Life adjusted for reliability

C_R = Adjustment factor for reliability

Table 14–6 gives values of C_R for reliabilities between 90% and 99%.

It should be noted that one result of designing for higher reliability is that the bearings would be larger and more expensive.

14-11 BEARING SELECTION FROM MANUFACTURERS' CATALOGS

Manual methods of selection of single-row, deep-groove ball bearings in previous sections demonstrate the basic fundamentals and use the example rating tables in Table 14–3. The required basic dynamic load rating C was found from Equation (14–3) after determining the values of the design load P_d and the design life L_d . Adjustment of the life rating for reliability other than the typical L_{10} life (90% reliability) was added in Equation (14–6). All of these data were determined using standard methods of rating bearings developed by industry standards such as that described in Reference 13.

TABLE 14-6 Life Adjustment Factors for Reliability, C_R

Reliability (%)	C_R	Life designation
90	1.0	L_{10}
95	0.62	L_5
96	0.53	L_4
97	0.44	L_3
98	0.33	L_2
99	0.21	L_1

Most bearing manufacturers produce large numbers of styles and sizes of bearings and report performance data in comprehensive catalogs. Furthermore, other factors beyond those listed in this book are included. Bearing standards permit manufacturers to define additional adjustment factors for special bearing properties such as:

- The quality of steel or other materials used in the bearing.
- Special manufacturing processes.
- Unique design features of the bearings.
- Lubrication effectiveness represented by the nature of the oil film between the rolling elements and the races.
- Temperature of operation and the resulting viscosity of the oil.

It is impractical to report all these factors in a textbook and some are likely different for each particular manufacturer. Therefore, it is recommended that online bearing selection software provided by particular manufacturers be used in selecting bearings for problems in this book and for design projects. See Internet sites 2–8 for examples.

14-12 MOUNTING OF BEARINGS

Up to this point, we have considered the load-carrying capacity of the bearings and the bore size in selecting a bearing for a given application. Although these are the most critical parameters, the successful application

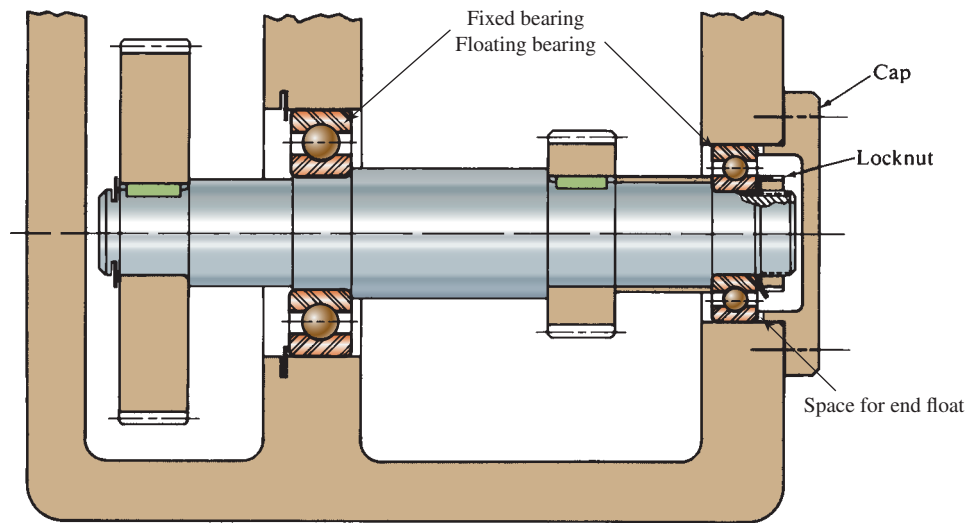


FIGURE 14-13 Bearing mounting illustration

of a bearing must consider its proper mounting. Bearings are precision machine elements. Great care must be exercised in their handling, mounting, installation, and lubrication.

The primary considerations in the mounting of a bearing are as follows:

- The shaft seat diameter and tolerances.
- The housing internal bore and tolerances.
- The shaft shoulder diameter against which the inner race of the bearing will be located.
- The housing shoulder diameter provided for locating the outer race.
- The radius of the fillets at the base of the shaft and housing shoulders.
- The means of retaining the bearing in position.

In a typical installation, the bore of the bearing makes a light interference fit on the shaft, and the outside diameter of the outer race makes a close clearance fit in the housing bore. To ensure proper operation and life, the mounting dimensions must be controlled to a total tolerance of only *a few ten-thousandths of an inch*. Most catalogs specify the limit dimensions for both the shaft seat diameter and the housing bore diameter.

Likewise, the catalog will specify the desirable shoulder diameters for the shaft and the housing that will provide a secure surface against which to locate the bearing while ensuring that the shaft shoulder contacts only the inner race and the housing shoulder contacts only the outer race. Table 14-3 includes these values.

The fillet radius specified in the catalog (see r in Table 14-3) is the maximum permissible radius *on the shaft and in the housing* that will clear the external radius

on the bearing races. Using too large a radius would not permit the bearing to seat tightly against the shoulder. Of course, the actual fillet radius should be made as large as possible up to the maximum to minimize the stress concentration at the shoulder.

Bearings can be retained in the axial direction by many of the means described in Chapter 11. Three popular methods are retaining rings, retaining caps, end caps, and locknuts. Figure 14-13 shows one possible arrangement. Note that for the left bearing, the shaft diameter is slightly smaller to the left of the bearing seat. This allows the bearing to be slid easily over the shaft up to the place where it must be pressed on. The retaining ring for the outer race could be supplied as a part of the outer race instead of as a separate piece.

The right bearing is held on the shaft with a locknut threaded on the end of the shaft. See Figure 14-14 for the design of standard locknuts. The internal tab on the lockwasher engages a groove in the shaft, and one of the external tabs is bent into a groove on the nut after it is seated to keep the nut from backing off. The external cap protects the bearing.

Care must be exercised to ensure that the bearings are not overly constrained. If both bearings are held tightly, any changes in dimensions due to thermal expansion or unfavorable tolerance stackup would cause binding and could lead to dangerous unexpected loads on the bearings. It is desirable to give one bearing complete location while allowing the other bearing to float axially. In Figure 14-3, the left bearing is held against the shaft shoulder on the inner race. The outer race employs a retaining ring to maintain axial location within the housing. However, for the right bearing, the inner race is seated against the shaft shoulder while the outer race may float in the axial

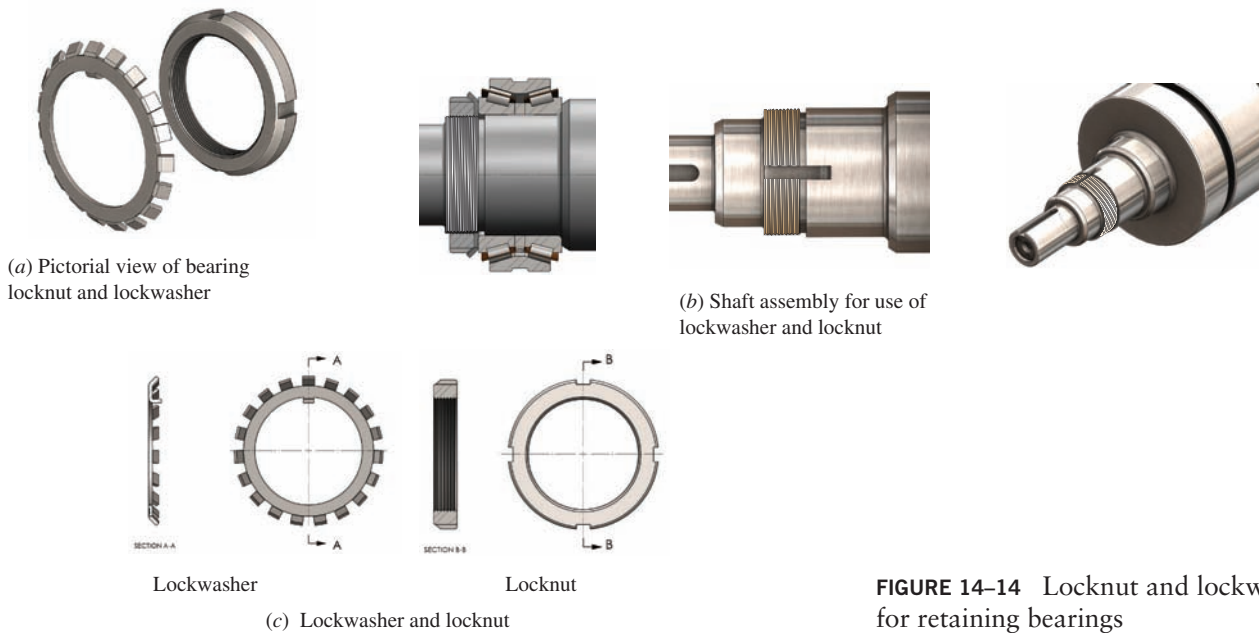


FIGURE 14-14 Locknut and lockwasher for retaining bearings

direction. Recall that the inner race is press-fit on its seat, whereas the outer race has a small clearance, permitting axial movement often called float.

14-13 TAPERED ROLLER BEARINGS

The taper on the rollers of tapered roller bearings, evident in Figure 14-7, results in a load path different from that for the bearings discussed thus far. Figure 14-15 shows two tapered roller bearings supporting a shaft with a combination of a radial load and a thrust load. The design of the shaft is such that the thrust load is resisted by the left bearing. But a peculiar feature of this

type of bearing is that a radial load on one of the bearings creates a thrust on the opposing bearing, also; this feature must be considered in analysis of the bearing.

The location of the radial reaction must also be determined with care. Part (b) of Figure 14-15 shows a dimension a that is found by the intersection of a line perpendicular to the axis of the roller and the centerline of the shaft. The radial reaction at the bearing acts through this point. The distance a is reported in the tables of data for the bearings.

The American Bearings Manufacturers' Association (ABMA) recommends the following approach in computing the equivalent loads on a tapered roller bearing:

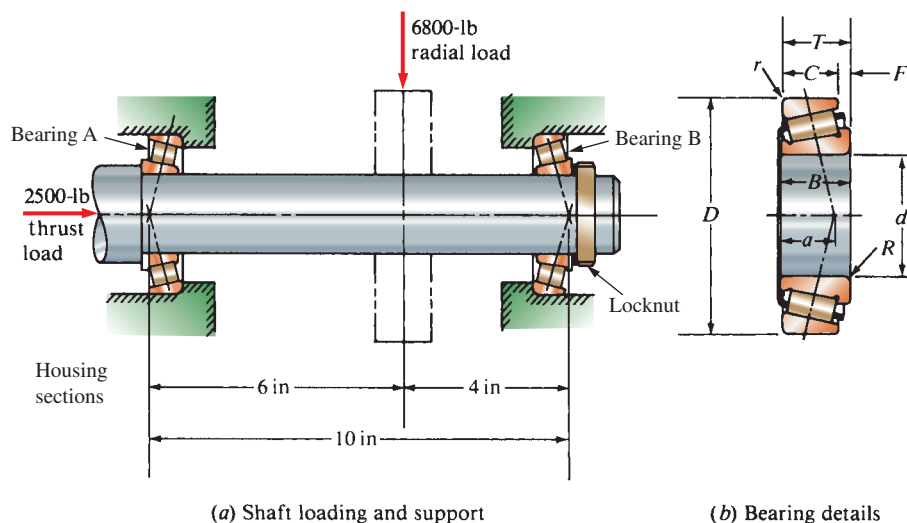


FIGURE 14-15 Example of tapered roller bearing installation

TABLE 14-7 Tapered Roller Bearing Data

Bore, d	Outside diameter, D	Width, T	a	Thrust factor, Y	Basic dynamic load rating, C
1.0000	2.5000	0.8125	0.583	1.71	8370
1.5000	3.0000	0.9375	0.690	1.98	12 800
1.7500	4.0000	1.2500	0.970	1.50	21 400
2.0000	4.3750	1.5000	0.975	2.02	26 200
2.5000	5.0000	1.4375	1.100	1.65	29 300
3.0000	6.0000	1.6250	1.320	1.47	39 700
3.5000	6.3750	1.8750	1.430	1.76	47 700

Note: Dimensions are in inches. Load C is in pounds for an L_{10} life of 1 million rev.

⇨ Equivalent Load for Tapered Roller Bearing

$$P_A = 0.4F_{rA} + 0.5 \frac{Y_A}{Y_B} F_{rB} + Y_A T_A \quad (14-7)$$

$$P_B = F_{rB} \quad (14-8)$$

where

P_A = equivalent radial load on bearing A

P_B = equivalent radial load on bearing B

F_{rA} = applied radial load on bearing A

F_{rB} = applied radial load on bearing B

T_A = thrust load on bearing A

Y_A = thrust factor for bearing A from tables

Y_B = thrust factor for bearing B from tables

Table 14-7 shows an abbreviated set of example data to illustrate the method of computing equivalent loads.

For the several hundred designs of standard tapered roller bearings available commercially, the value of the thrust factor varies from as small as 1.07 to as high as 2.26. In design problems, a trial-and-error procedure is usually necessary. Example Problem 14-5 illustrates one approach.

Example Problem 14-5

The shaft shown in Figure 14-15 carries a transverse load of 6800 lb and a thrust load of 2500 lb. The thrust is resisted by bearing A . The shaft rotates at 350 rpm and is to be used in a piece of agricultural equipment. Specify suitable tapered roller bearings for the shaft.

Solution The radial loads on the bearings are

$$F_{rA} = (6800 \text{ lb})(4 \text{ in}/10 \text{ in}) = 2720 \text{ lb}$$

$$F_{rB} = (6800 \text{ lb})(6 \text{ in}/10 \text{ in}) = 4080 \text{ lb}$$

$$T_A = 2500 \text{ lb}$$

To use Equation (14-7), we must assume values of Y_A and Y_B . Let's use $Y_A = Y_B = 1.75$. Then

$$P_A = 0.40(2720 \text{ lb}) + 0.5 \frac{1.75}{1.75} (4080 \text{ lb}) + 1.75(2500 \text{ lb}) = 7503 \text{ lb}$$

$$P_B = F_{rB} = 4080 \text{ lb}$$

Using Table 14-4 as a guide, let's select 4000 h as a design life. Then the number of revolutions would be

$$L_d = (4000 \text{ h})(350 \text{ rpm})(60 \text{ min/h}) = 8.4 \times 10^7 \text{ rev}$$

The required basic dynamic load rating can now be calculated from Equation (14-3), using $k = 3.33$. Note that the exponent $1/k$ is $1/3.33$, resulting in a value of 0.30.

$$C_A = P_A(L_d/10^6)^{1/k}$$

$$C_A = (7503 \text{ lb})(8.4 \times 10^7/10^6)^{0.30} = 28\,400 \text{ lb}$$

Similarly,

$$C_B = (4080 \text{ lb})(8.4 \times 10^7/10^6)^{0.30} = 15\,400 \text{ lb}$$

From Table 14–7, we can choose the following bearings.

Bearing A

$$\begin{aligned} d &= 2.5000 \text{ in} & D &= 5.0000 \text{ in} \\ C &= 29\,300 \text{ lb} & Y_A &= 1.65 \end{aligned}$$

Bearing B

$$\begin{aligned} d &= 1.7500 \text{ in} & D &= 4.0000 \text{ in} \\ C &= 21\,400 \text{ lb} & Y_B &= 1.50 \end{aligned}$$

We can now recompute the equivalent loads:

$$\begin{aligned} P_A &= 0.40(2720 \text{ lb}) + 0.5 \frac{1.65}{1.50}(4080 \text{ lb}) + 1.65(2500 \text{ lb}) = 7457 \text{ lb} \\ P_B &= F_{rB} = 4080 \text{ lb} \end{aligned}$$

From these, the new values of $C_A = 28\,200 \text{ lb}$ and $C_B = 15\,400 \text{ lb}$ are still satisfactory for the selected bearings.

One caution must be observed in using the equations for equivalent loads for tapered roller bearings. If, from Equation (14–7), the equivalent load on bearing *A* is less than the applied radial load, the following equation should be used.

If $P_A < F_{rA}$, then let $P_A = F_{rA}$ and compute P_B .

$$P_B = 0.4F_{rB} + 0.5 \frac{Y_B}{Y_A} F_{rA} - Y_B T_A \quad (14-9)$$

A similar analysis is used for angular contact ball bearings in which the design of the races results in a load path similar to that for tapered roller bearings. Figure 14–3 shows an angular contact bearing and the angle through the pressure center. This is equivalent to the line perpendicular to the axis of the tapered roller bearing. The radial reaction on the bearing acts through the intersection of this line and the axis of the shaft. Also, a radial load on one bearing induces a thrust load on the opposing bearing, requiring the application of the equivalent load formulas of the type used in Equations (14–7) and (14–9). The angle of the load line in commercially available angular contact bearings ranges from 15° to 40° .

14-14 PRACTICAL CONSIDERATIONS IN THE APPLICATION OF BEARINGS

This section discusses lubrication of bearings, installation, preloading, stiffness, operation under varying loads, sealing, limiting speeds, and standards and bearing tolerance classes related to the manufacture and application of bearings.

Lubrication

The functions of lubrication in a bearing unit are as follows:

1. To provide a low-friction film between the rolling elements and the races of the bearing and at points of contact with cages, guiding surfaces, retainers, and so on.

2. To protect the bearing components from corrosion.
3. To absorb heat from the bearing unit.
4. To carry heat away from the bearing unit.
5. To help disperse contaminants and moisture from the bearing.

Rolling contact bearings are usually lubricated with either grease or oil. Under normal ambient temperatures (approximately 70°F) and relatively slow speeds (under 500 rpm), grease is satisfactory. Note that grease typically needs to be replenished and grease fittings should be designed into the system. At higher speeds or higher ambient temperatures, oil lubrication applied in a continuous flow is required, possibly with external cooling of the oil.

Oils used in bearing lubrication are usually clean, stable mineral oils. Under lighter loads and lower speeds, light oil is used. Heavier loads and/or higher speeds require heavier oils up to SAE 30. A recommended upper limit for lubricant temperature is 160°F . The choice of the correct oil or grease depends on many factors, so each application should be discussed with the bearing manufacturer. In general, a kinematic viscosity of 13 to 21 centistokes should be maintained at the operating temperature of the lubricant in the bearing. Manufacturer's recommendations should be sought.

In some critical applications such as bearings in jet engines and high-speed devices, lubricating oil is pumped under pressure to an enclosed housing for the bearing where the oil is directed at the rolling elements themselves. A controlled return path is also provided. The temperature of the oil in the sump is monitored and controlled with heat exchangers or refrigeration to maintain oil viscosity within acceptable limits. Such systems provide reliable lubrication and ensure the removal of heat from the bearing.

See Section 14–15 for additional discussion about the importance of oil film thickness in bearings. References 1, 3, 5, 6, and 11 contain an extensive amount of information on this subject.

Greases used in bearings are mixtures of lubricating oils and thickening agents, usually soaps such as lithium or barium. The soaps act as carriers for the oil which is drawn out at the point of need within the bearing. Additives to resist corrosion or oxidation of the oil itself are sometimes added. Classifications of greases specify the operating temperatures to which the greases will be exposed, as defined by the American Bearing Manufacturers' Association (ABMA), and outlined in Table 14–8.

Internet site 1 lists many styles of greases, oils, and solid lubricants for ceramic bearings, metal bearings, hybrids, and other precision equipment. Grease thickeners include barium complex soap, polyurea, PTFE, and lithium. Also listed is an Ultra-Dry Lube DL-5 made from a modified tungsten disulfide (WS_2) in lamellar form that is chemically inert, non-toxic, and corrosion resistant. It can operate over a temperature range of $-188^{\circ}C$ to $+538^{\circ}C$ ($-350^{\circ}F$ to $+1000^{\circ}F$). The nominal coefficient of friction is 0.030. Some full ceramic bearings can operate well without lubrication, an important feature for applications in food processing, semiconductor production, aerospace devices, and high vacuum installations.

Installation

It has already been stated that most bearings should be installed with a light interference fit between the bore of the bearing and the shaft to preclude the possibility of rotation of the inner race of the bearing with respect to the shaft. Such a condition would result in uneven wear of the bearing elements and early failure. To install the bearing then requires rather heavy forces applied axially. Care must be exercised so that the bearing is not damaged during installation. The installation force should be applied directly to the inner race of the bearing.

If the force were applied through the outer race, the load would be transferred through the rolling elements to the inner race. Because of the small contact area, it is likely that such transfer of forces would overstress some element, exceeding the static load capacity. Brinelling would result, along with the noise and rapid wear that accompany this condition. For large bearings it may be necessary to heat the bearing to expand its diameter in order to keep the installation forces within reason. Removal of bearings intended for reuse requires similar precautions. Bearing pullers are available to facilitate this task.

TABLE 14–8 Types of Greases Used for Bearing Lubrication

Group	Type of grease	Operation temperature range ($^{\circ}F$)
I	General-purpose	-40 to 250
II	High-temperature	0 to 300
III	Medium-temperature	32 to 200
IV	Low-temperature	-67 to 225
V	Extremely high-temperature	Up to 450

Preloading

Some bearings are made with internal clearances that must be taken up in a particular direction to ensure satisfactory operation. In such cases, preloading must be provided, usually in the axial direction. On horizontal shafts, springs are typically used, with axial adjustment of the spring deflection sometimes provided to adjust the amount of preload. When space is limited, the use of Belleville washers is desirable because they provide high forces with small deflections (see Chapter 18). Shims can be used to adjust the actual deflection and preload obtained. On vertical shafts the weight of the shaft assembly itself may be sufficient to provide the required preload.

Bearing Stiffness

Stiffness is the deflection that a given bearing undergoes when carrying a given load. Usually the radial stiffness is most important because the dynamic behavior of the rotating shaft system is affected. The critical speed and the mode of vibration are both functions of the bearing stiffness. Generally, the softer the bearing (low stiffness) is, the lower the critical speed of the shaft assembly will be. Stiffness is measured in the units used for springs, such as pounds per inch or newtons per millimeter. Of course, the stiffness values are quite high, with values of 500 000 to 1 000 000 lb/in reasonable (87.5 to 175 MN/m). The manufacturer should be consulted when such information is needed, because it is rarely included in standard catalogs.

Operation under Varying Loads

The load/life relationships used thus far assume that the load is reasonably constant in magnitude and direction. If the load varies considerably, an effective mean load must be used for determining the expected life of the bearing. (See References 4, 6, and 12.) Oscillating loads also require special analysis because only a few of the rolling elements share the load. See Section 14–16 for additional information about life prediction under varying loads.

Sealing

When the bearing is to operate in dirty or moist environments, special shields and seals are usually specified. They can be provided on either or both sides of the rolling elements. Shields are typically metal and are fixed to the stationary race, but they remain clear of the rotating race. Seals are made of elastomeric materials and do contact the rotating race. Bearings fitted with both seals and shields and precharged at the factory with grease are sometimes called *permanently lubricated*. Although such bearings are likely to give many years of satisfactory service, extreme conditions can produce a degradation of the lubricating properties of the grease. The presence of seals also increases the friction in a bearing. Sealing can be provided outside the bearing in the housing or at the shaft/housing interface. On high-speed shafts, a *labyrinth seal*, consisting of a non-contacting ring around the shaft